

Farkaleet Series

Statistics and Probability (Engineering Statistics)

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PREFACE

This book is the zenith of the experience that I have gained throughout my career as a teacher, teaching in different institutions at various levels – diploma, degree, masters. The book is also valuable for various professional examinations such as PCS, CSS, B. Tech, Engineering services and the readers who want to gain a sound knowledge of the subject.

I have always felt that most of the available textbooks on “Statistics and Probability” take-up the subject in a round-about and confusing manner. In this book, I have tried to explain each topic in a very simple and direct method, avoiding unnecessary confusions.

Each topic is covered in depth from the basic concepts by providing multiple problems. The book is written in accordance with the syllabi of engineering universities in particular and other examinations in general. I would welcome advice and suggestions leading to improving the book.

My sincere thanks are due to all renowned co-authors and publishers whose works and text have been the source of assistance to me in writing this book. I am also indebted to my colleagues who have extended their cooperation while writing, editing and printing the book. I am also thankful to the publisher for printing the book and preparing its title page.

Authors are indebted to the EDITORS who have very much carefully read the material, edited the book and gave their valuable suggestions to improve the standard of the book.

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CHAPTER ONE

INTRODUCTION TO STATISTICS

1.1 INTRODUCTION

The word “*Statistics*” comes from the Latin word “*status*”, which means a political state. This in turn means information valuable to the state. For example, one wishes to collect information regarding the number of people living in whole country, number of I.T industries in certain city, production of wheat in particular province. In today’s world, this word has now acquired different meanings. For example:

- The word *statistics* refers to numerical facts. For instance, statistics of road and railway accidents, statistics of crimes, statistics of births and statistics of newly built educational institutions in a country.
- Statistics* are numerical quantities calculated from sample observation. For example the mean of sample is known as statistic. (This will be made clear in coming chapters.)
- The word *statistics* is defined as discipline that includes procedures and techniques used to collect, summarize process and analyze the numerical data to make inferences and take valid decisions based on this data.

Decisions are often made on the basis of numeric facts, that is; rate of inflation, unemployment figures, number of defective circuits produced during manufacturing process, expected number of accidents in a particular month. Such numeric facts are known as data. Statistics is that branch of mathematics which deals with the collection, presentation, analysis and interpretation of such data. When this data is arranged in some order, it is known as information. We shall consider how this information is used to aid the decision process.

Some Important Definitions and Notations

When referring to data, the following general terms are used:

A list of data collected in a given study is known as *data set*.

A list of data collected without putting it into any order is called *raw data*.

A *variable* is that characteristic of data which is subject to change and the data collected for a variable is called its *value*.

For example, a variable X that represents the number of children in a family chosen at random, and then it may take a value 0, 1, 2, 3. Generally, X is a variable that takes n values $x_1, x_2, \dots, x_n$.

Example 1.1.1 The following data was collected for the 5 applicants who were short listed for a job. Define the data set, the raw data, the variables and the values.

Age	Gender	Aptitude Test Result	Communication Skills Rating	Number of Previous Jobs
18	M	70.3	1	1
23	M	62.6	3	3
25	F	72.5	1	2
19	M	60.1	2	1
22	F	52.4	2	2

Solution: The data set is the information recorder on 5 applications. The raw data is the list of numbers and letters as they were collected. The five variables in the data set are:

$x_1 \equiv$ Age $x_2 \equiv$ Gender $x_3 \equiv$ Test results
 $x_4 \equiv$ Communication skills rating (1: good, 2: average, 3: poor)
 $x_5 \equiv$ Number of previous jobs

The values of different variables in say first row are:

$x_1 \equiv 18$, $x_2 \equiv M$, $x_3 \equiv 70.3$, $x_4 \equiv$ Communication skills rating (1: good),
 $x_5 \equiv$ Number of previous jobs = 1.

1.2 TYPES OF DATA

Any data may be of two types, namely, *qualitative data* and *quantitative data*.

Qualitative Data

When the data is presented in categories, it is known as *qualitative data*. It can be in numeric or letter. For example, in the above example gender is qualitative variable which is categorized as “M” and “F”. The communication skill rating is also qualitative variable which is categorized numerically as 1 (good), 2 (average) and 3 (poor). These numbers are used as labels for categories, and are meaningless as quantities.

Quantitative Data

In quantitative data, the values of the items are always numeric and represent a measurement. We may categorize quantitative data as “*Discrete* and *Continuous*”. Discrete quantitative variable can take only certain values such as integers. For example, in example 1, the number of previous jobs held by an applicant is a discrete quantitative variable since it is not possible to have 1.2 previous jobs. Other examples of discrete quantitative variable are the number of accidents occurred in certain town, number of daily flights from capital city, etc, are the examples of discrete data. It may be noted that in certain cases, discrete quantitative variable may have non-integer values. For example, ladies shoe sizes can be 4, 4 ½, 5, 5 ½. Shoe sizes such as 4.16 are not possible, so the variable is discrete. Continuous type of variables may assume any value over a range or interval. For illustration, in Example 1, the aptitude test result is a continuous quantitative variable because it can take any value ranging from 0% to 100%. Age is also a continuous quantitative variable since it is measured over time. Continuous quantitative variable however, may be rounded to the nearest integer as in case of ages of different candidates appearing in an aptitude test.

Collection of Data

Data can be collected from different sources. The purpose of collecting the data is to keep the records for future use. For example, companies keep records on sales, profits, employees etc. Government keeps record for unemployment ratio, weapons to be manufactured for war, number of education and research institutions to be built in public or private sectors for higher education. Data can be collected using either *experiment* or a *survey*.

An *experiment* is performed under controlled conditions. One or more of the variables are controlled and the effect on the other variables is observed. For example, suppose a computer manufacturing company is interested in how the amount spent on advertising affects the number of sales. They decide to fix some amount on advertising in each 12 consecutive months and to record the sale. This method of collecting the data is called experiment since the amount spent is controlled and the resulting effect on sales is observed.

A *survey* is performed when the variables cannot be controlled. For example, if computer manufacturing company wants to predict the future level of demand for its product, it is

not possible to control the factors which will influence future demand. Instead, the company can market test the product in selected cities and from this estimates the total demand. Questionnaires are often used to collect the data from surveys.

Populations and Samples

A *population* is the set of all possible items, usually relating to a group of people and objects, which is of interest in a particular study. A *sample* is a selection of items taken from a population which is used for examination. It is often very time consuming and expensive to collect information on all items in the population in order to make decisions. For example, if a computer manufacturing company wants to investigate the future demand of a new CPU, it is not possible to ask all potential computer users (the population) whether or not they will purchase the new CPU. In this case, a sample of potential computer users will be asked to establish the demand. This sample will not provide the exact value of the future demand, but it can be used to provide an *estimate*. The accuracy of any decision made using a sample will depend on how representative the sample is of the population as a whole and on the number of items included in the sample. In order for a sample to be representative, the items contained in the sample must be selected at random, so that every item in the population has an equal chance of being included in the sample. This concept is discussed in more detail in Chapter 8. The accuracy of decision can also be improved by increasing the size of the sample. Chapters 8 and 9 illustrate how decisions about populations can be made using samples.

Descriptive and Inferential Statistics

Statistics as a subject, may be divided into two categories namely, *descriptive statistics* and *inferential statistics*.

Descriptive statistics is that branch of statistics which deals with concepts and methods of summarizing and presenting the data in tabular or graphical form. On the other hand, *Inferential statistics* deals with the procedures for making inferences about the characteristics that describes the population from the knowledge derived from a sample. This phase of statistics is based on probability theory as the inference which are made on the basis of sample evidence, cannot be absolutely certain.

1.3 SCOPE OF STATISTICS

Statistics has come to play an important role in almost every field of life and human activity. Statistics is perhaps a subject that is used by everybody. In early stage, Statistics was used by the state to collect information on public affairs for administration. Gradually its use was extended to all scientific experiments where collection, presentation and analysis of data were made for valid conclusion. The significance of hypothesis, prediction of future course on the bases of past experiences can be estimated by this useful tool.

Consciously or unconsciously Statistics has been entered in all branches of knowledge where a study of quantitative phenomenon is required. There is hardly any field whether it is trade, industry, commerce, economics, biology, psychology, sociology, botany, physics, chemistry, education, medicine where statistical tools are not applicable. Statistics can be used in various fields from simple to complex. The various uses and applications of Statistics make statistical procedures important and valuable in a decision making context.

Some applications which illustrate how Statistics is used in Business and Economics are briefly presented.

Accounting Applications

Suppose that a mail order firm is interested in collecting information concerning recent ordering practices of its customers. The concerned manager asks for information about

the amounts billed on recent orders. The manager may be provided with the estimates of the average amount billed per order to enable him to arrive at some conclusion.

Marketing Applications

In an attempt to measure consumer acceptance of a new product, a firm has identified several test market areas and has selected a sample of consumers in each area. The individuals in the sample are asked in the interviews whether or not they prefer the new product over the product they are currently using. An analysis of the sample data will enable management to estimate the percentage of customers who will prefer the new product.

Finance Applications

A Federal agency would like to identify the current practices and trends in house mortgages. Using a sample data from Finance Corporation about the average amount of mortgages issued, an inference could be made about the trend in house mortgages and the future demand for house mortgages loans.

Production Applications

Many manufacturing and production applications of Statistics involve quality control. A manufacturing firm that produces soaps and shampoos is interested to know periodically, if the automatic mechanism to fill the containers is accurate. A quality control inspector takes the sample of the containers from the production line. On the basis of this sample, if under filling or over filling is suspected, the line can be shut down for repair otherwise the production process will be allowed to continue.

Engineering Applications

Almost in every engineering discipline the use of statistics is very much acknowledged. Statistical analysis is used in stochastic process, noise analysis where we have to see the chance that signals received correctly or not. In environmental engineering one has to predict how much pollutant the river water is at certain location. In civil engineering one has to see the chance that concrete fulfills the structure requirements and what is the average breaking strength of steel produced in certain manufacturing company, etc.

The Role of Computer

In last 10 to 15 years, the computer has played an ever – increasing role in Statistical analysis. The data used in Statistical Studies are stored in the computer through input devices and the computer is being used more and more for data summarization tasks. Computer programs have been written to handle the stored data according to requirements. The increasing availability to software packages makes the computer a valuable tool in inferential Statistics, in addition to its use for data summarization. For example, it is extensively used for Regression Analysis, Analysis of Variance, Forecasting and so on.

We have seen that Statistics now holds a central position in almost every field and the importance of Statistics is increasing day by day. So we cannot deny its important and animated role in any way.

1.4 LIMITATIONS OF STATISTICS

The definition of Statistics specifies that it is a subject in its own right. It may therefore be desirable to know its limitations and characteristics in order to appreciate and understand its general nature. Some of its limitations are summarized as follows:

Statistics deals with aggregates and cannot provide information about a particular individual.

Statistics deals with those aspects of things which; are described numerically either by counts or measurements.

- Statistics deals with characteristics which are random in nature.

- Statistical laws are valid on the average or in the long run. There is no guarantee that certain law will hold in all cases. Statistical inference is therefore made in the face of uncertainty.
- Statistical conclusions are drawn under uncertainty and may not be exact or completely valid. That is why we have to use the concepts of probability.
- Statistical results might be misleading and incorrect if sufficient care in collecting, processing and interpreting the data is not taken or if the Statistical data are handled by a person who is not well skilled in the subject matter of Statistics.

WORKSHEET 01

1. Define the terms: data set, raw data, variable and value.
2. How will you represent 4th, 6th, and 10th value of a variable?
3. The following data are recorded for 5 manufacturers producing steel bars in Karachi.

A	B	C	D	E	F
1	26	1000	4	KORANGI	2
2	15	880	2	N. ABAD	1
3	30	2200	3	ORANGI	3
4	11	700	2	SITE	1
5	21	900	2	M. PIR	2

NOTE: A, company; B, turnover (Rs. Million); C, number of employees; D, number of different steel bars made; E, location; F, change in profit from the previous year (1, increase; 2, stable; 3, decrease)

- i. How many variables are recorded in this data set?
 - ii. How many items are recorded for each variable?
 - iii. What is the value of 3rd item of variable C?
4. Define “qualitative” and “quantitative data variables. What is the difference between discrete and continuous quantitative variables?
 5. State whether the following variables are qualitative, discrete quantitative, or continuous quantitative.
 - i. Salary
 - ii. Marital status
 - iii. Number of defective items in a lot
 - iv. Life time of an electronic component
 - v. Number of computers sold in a given year
 - vi. Make of a car
 - vii. Internet users in a town
 - viii. Grade obtain in an aptitude test
 - ix. Color of car
 - x. Profit increase/decrease
 6. Refer to Q. 3, which of the variables are qualitative and which are quantitative? If the variable is quantitative, which are discrete and which are continuous?
 7. Define what do you meant by an experiment and survey. What methods are used to collect data in a survey?
 8. Define what is meant by a “population” and “sample”. Why a sample in comparison to population is investigated?
 9. State two ways in which the accuracy of decision made using a sample can be increased.
 10. Differentiate between descriptive statistics and inferential statistics.
 11. Describe the applications of Statistics in various fields of science?
 12. What are the limitations of statistics?

CHAPTER TWO

PRESENTATION OF DATA

2.1 INTRODUCTION

Raw data is often difficult to understand and interpret. We are often interested in describing and summarizing the whole data rather than the individual data item. Thus raw data needs to be summarized using tables and graphs. For example, it is difficult to visualize an individual educational institutions run by government. However, if they are placed in tabular or graphical form it will be easy to understand that how many educational institutions in different cities are running under the government. Similarly, it is difficult to visualize the number of voters in every city but if the total voters are categorized according to district wise, it will give a clear picture of total voters in a country.

This chapter discusses the most frequently used methods of summarizing data using tables and graphs.

Summarizing Qualitative Data: Frequency Distribution

A frequency distribution is a table showing the frequency or number of items in each category. For example, a computer company employs 30 female and 40 males. This data in frequency distribution can be presented as under:

Gender of Employees	Number of Employees
Male	40
Female	30
Total	70

Relative Frequency Distribution

The relative frequency of a category is the proportion of items in the category. If there are n items in data set, then relative frequency of a category is defined as:

$$\text{Relative frequency of a category} = (\text{frequency of category})/n$$

The relative frequency can also be expressed as a percentage as follows:

$$\text{Percentage of items in a category} = \text{Relative frequency} \times 100$$

The relative frequency table is used to show the proportion or percentage of data item in each category.

Gender of Employees	Number of Employees	Relative Frequency	Percentage
Male	40	0.57	57
Female	30	0.43	43
Total	70	1.00	100

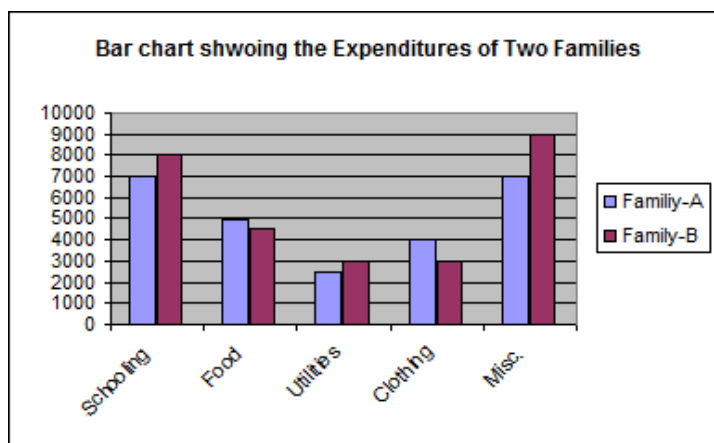
The table shows that in 'Computer Company' approximately 57% male employees and 43% female employees are working.

Bar Chart

A bar chart is a graphical representation of data set. The horizontal axis specifies the label of the category and vertical axis the frequency of respective category. The following table shows the monthly expenditures of two families.

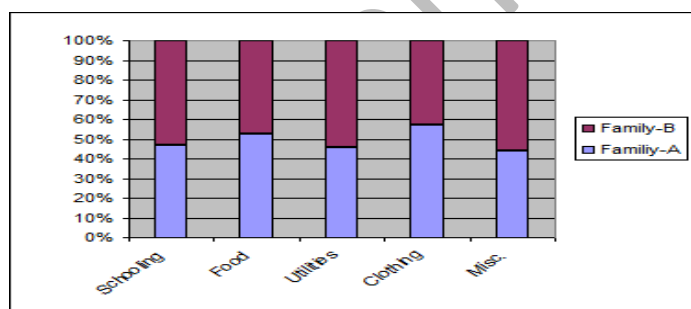
	Schooling	Food	Utilities	Clothing	Misc.
Family 'A'	7000	5000	2500	4000	7000
Family 'B'	8000	4500	3000	3000	9000

The following bar chart represents the expenditures incurred by two families.



Component Bar Chart

Bar charts are also useful when the variable of interest can be subdivided into a number of components which are to be illustrated at the same time. This gives rise to a *component bar chart*. This is shown in the following figure related to above example.

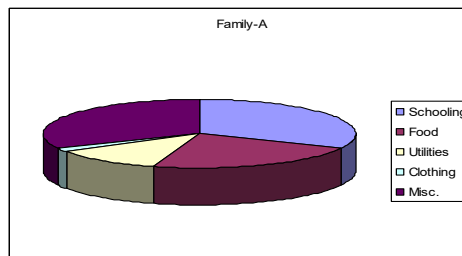
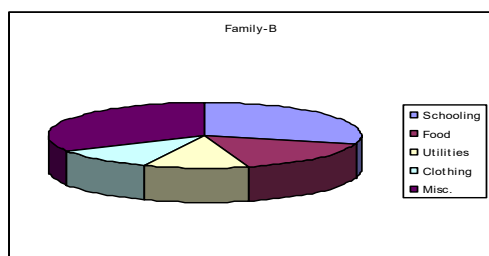


Pie Chart

A pie chart is a circular diagram showing the frequency or relative frequency of data set. The slices of pie usually represent the percentage of items in each category. It may be noted that pie chart is used only when the data is categorized in not more than six categories. To draw a pie chart, first draw a circle. Then divide the circle into slices which correspond to the frequency, relative or percentage frequency of items in each category. The size of each slice is calculated by using the formula:

$$\text{No. of degree for a category} = \text{Relative frequency of a category} \times 360$$

The pie charts showing the monthly expenditures of two families are shown below:



2.2 SUMMARIZING QUANTITATIVE DATA

Grouped Data

We know that there are two types of quantitative data; discrete and continuous. The tables and graphs appropriate to summarize the data depend on its type. The following three sections describe the methods appropriate for summarizing continuous quantitative data whilst the fourth section describes methods which are appropriate for summarizing discrete quantitative data.

Since continuous quantitative data can take any value over a specified range, rarely two or more items have the same value. Thus, counting the frequency of each value is not an appropriate way to summarize the data. Thus, continuous quantitative data are grouped into non-overlapping classes or categories. The grouped data is a frequency distribution showing the frequency of items in each of the category. The following steps are taken to form a grouped data.

- I. Number of classes: There should be at least 5 and at most 20 classes depending on the number of data set.
- II. Width of classes: The width of a class is the range of values covered by the class. The width of a class depends on the number of classes chosen in step I. The class width can be calculated using the formula

$$\text{Class width} = (\text{largest value} - \text{smallest value}) / \text{number of classes}$$
- III. Class limits: The lower class limit is the smallest value assigned to a class. Similarly, the upper class limit is the largest value assigned to the class. The limits must be defined so that no class overlaps; that is, the value of item must fall in one and only one class. There should also be no gap between the classes. The first class should contain the value of the smallest item and the last class should contain the value of the largest item.
- IV. Constructing the table: Count the number of items that fall into e. Use the tally and present the frequency of each class.

Example 2.2.1 The following table shows soil compaction in lb/psi taken from 32 different points from the place where a multistoried building is to be constructed. Construct the frequency distribution.

6.49	8.90	22.95	13.39	18.63	4.44	24.99	4.44	34.98	8.12	8.99	6.99
21.25	24.99	1.26	9.97	9.98	3.99	4.35	12.49	2.19	7.75	11.99	1.69
4.65	4.50	9.85	13.89	10.45	7.49	29.97	2.77				

Solution:

Step I: Since there are 32 data points which is relatively small sample, hence we take say five, classes.

Step II: Largest value = 34.98 : Smallest value = 1.26 : No. of classes = 5

Hence, Width of class = $(34.98 - 1.26)/5 = 6.744 \sim 7$

Step III: Since smallest value is 1.26, hence it is appropriate to consider lower class limit of the first class as 1 and the upper class limit of the first class be therefore taken as $1 + 7 = 8$. Now if we start second class from 8 then the first and second class will overlap. Thus we set the upper limit at 7.99. The second class therefore should start from 8 and end at 14.99, etc.

Step IV: Count the number of individual data item in each class. For this use tally which makes counting process little easier.

Using the same table we may also form grouped relative and percentage frequency distribution.

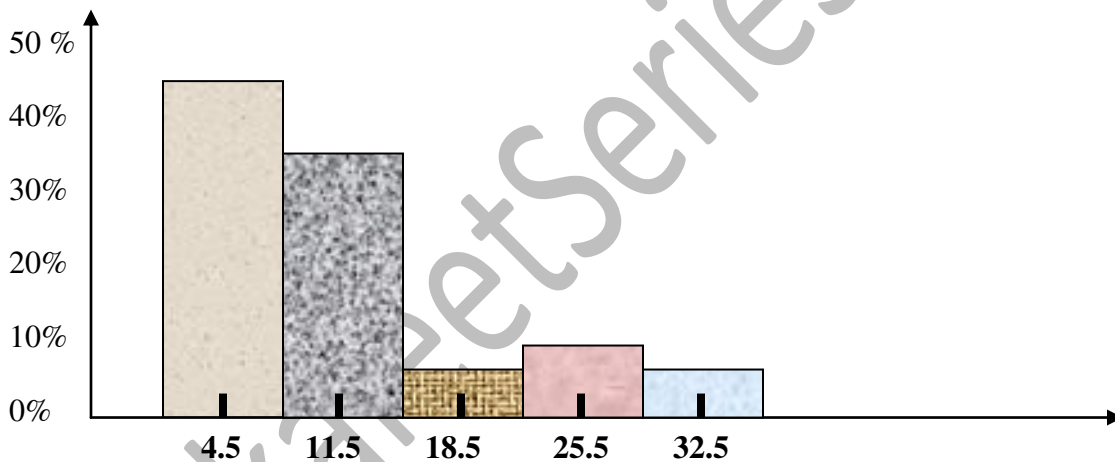
The following is the grouped data for continuous quantitative data set as given above.

Soil Compaction	Tally	Frequency	Relative frequency	Percentage
1.00 – 7.99		14	0.4375	43.75
8.00 – 14.99		11	0.34375	34.375
15.00 – 21.99		02	0.0625	6.25
22.00 – 28.99		03	0.09375	9.375
29.00 – 35.99		02	0.0625	6.25
		TOTAL 32	1.00	100

Histogram

A histogram is a graph (bars) showing the grouped data or grouped relative or percentage data frequency distribution of a set of data. The horizontal axis shows the variable of interest and vertical axis the frequency of respective class. In histogram the rectangles (bars) are not separated since they represent continuous data.

The histogram for the above grouped data is shown as under.



The values shown on the x-axis represent the class mid point which are obtained by taking the mean of lower and upper class limits of each class.

NOTE: It may be noted that rarely it happens that the class sizes of some of the bars are not equal. In this case the heights of the rectangle bars having different class size are adjusted. This is done by proportionally decreasing the height of that rectangle whose size is large.

Example 2.2.2 Construct the histogram for the following frequency distribution relating to ages of telephone operators.

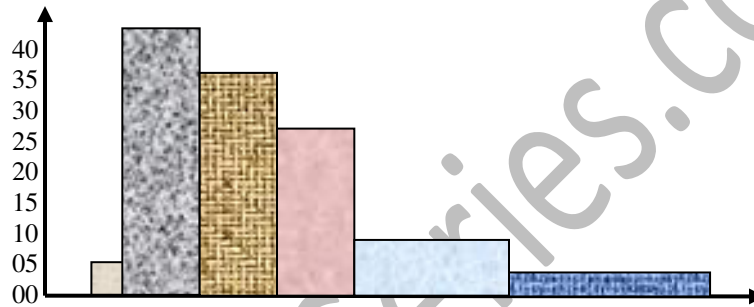
Age (years)	18-19	20-24	25-29	30-34	35-44	45-59
No. of operators	9	188	160	123	84	15

Solution: Here we observe that width of some classes is different. Thus before a histogram is constructed, we draw a new frequency distribution showing class boundaries, size of a class and proportional height of each rectangle. It may be noted that

in constructing the histogram, class boundaries also play a vital role. The class boundary is obtained by taking the mean of upper class limit of any class and the lower class limit of next higher class. This is shown below:

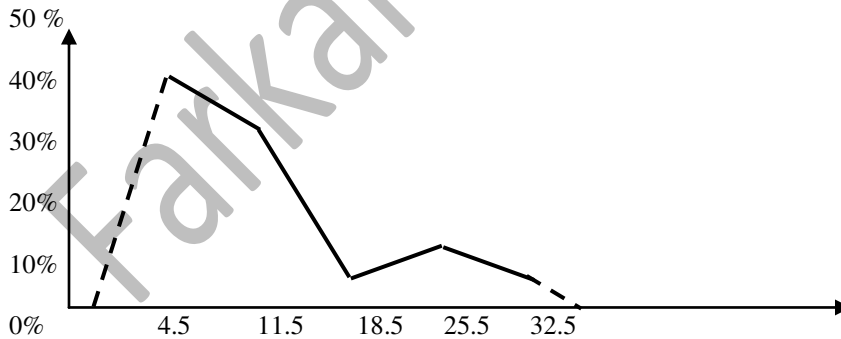
Class boundaries	Width	Frequency	Proportional heights
17.5-19.5	2	9	$9 \div 2 = 4.5$
19.5-24.5	5	188	$188 \div 5 = 37.6$
24.5-29.5	5	160	$160 \div 5 = 32.0$
29.5-34.5	5	123	$123 \div 5 = 24.6$
34.5-44.5	10	84	$84 \div 10 = 8.4$
44.5-59.5	15	15	$15 \div 15 = 1$

The histogram is shown as under:



Frequency Polygon

This is another way to represent grouped data graphically. Frequency polygon is constructed by plotting the point (x_i, f_i) where x_i is the class mid point of the i^{th} class and f_i is the corresponding frequency and then connecting them by straight line segments. Finally extend the first and last line segment to meet the x-axis at the mid points x_{i-1} and x_{i+1} . The following figure shows the related frequency polygon of Example 1.



Cumulative Frequency Distribution: Frequency polygon and Ogive

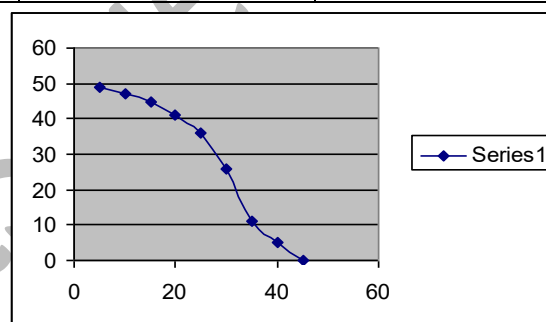
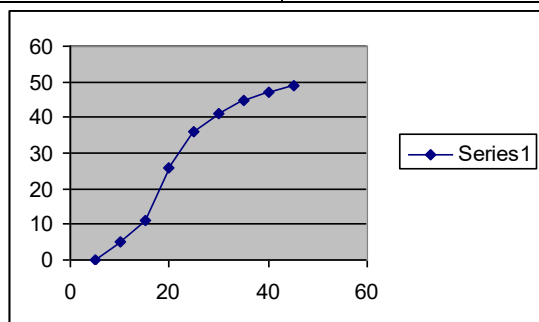
Very often, it is desired to show in a tabular as well as diagrammatic form, not the relative frequencies in the various intervals, but the cumulative frequencies above or below a given value. For example, we may wish to read off from the table or diagram the number of students whose marks are less than or greater than any particular value. Tabular form of this type is called cumulative distribution and graphical form is known as cumulative frequency curve of *ogive*. Let us consider the following example.

Example 2.2.3 Draw the cumulative frequency distribution and ogive “less than” and “more than” from the following distribution of marks obtained by 49 students.

Marks	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45
No. of students	5	6	15	10	5	4	2	2

Solution: The cumulative frequency distribution for “less than” and “more than” is shown as under.

Less than	Cumulative frequency	Greater than	Cumulative frequency
5	0	5	49
10	5	10	47
15	11	15	45
20	26	20	41
25	36	25	36
30	41	30	26
35	45	35	11
40	47	40	5
45	49	45	0



Cumulative frequency curves for “less than” and “greater than” values

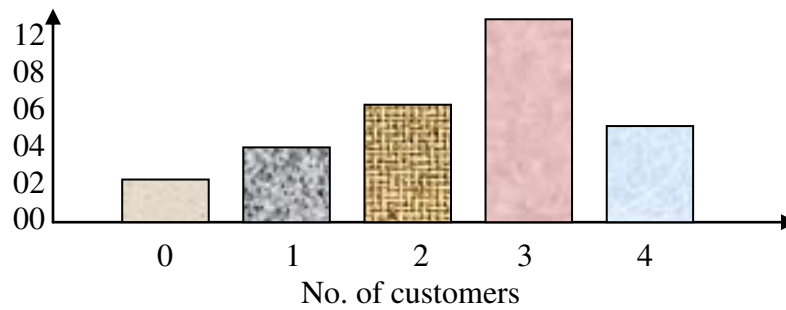
Summarizing Discrete Quantitative Data

If the data is discrete and many different values have occurred, then the data can be summarized using tables and graphs as described for continuous quantitative data. Note that rectangular bars should be separated to indicate the discrete nature of the data.

Example 2.2.4 An automatic cash dispenser is situated outside a bank. A survey of the usage of the dispenser was performed. The numbers of customers using the dispenser during 30 consecutive ten minute period are given by: 2, 1, 3, 3, 0, 2, 4, 2, 3, 3, 3, 3, 2, 2, 3, 1, 3, 4, 3, 1, 4, 3, 0, 1, 2, 4, 2, 3, 4, 3. Summarize the data using table and graph.

Solution: The following table shows the frequency distribution for the data. The histogram is also depicted on the next page.

No. of customers in 10 minute period	Frequency
0	2
1	4
2	7
3	12
4	5
Total:	30



WORKSHEET 02

1. The following data refer to the strengths (psi in 100) of steel bars produced in certain week in a factory.

125, 126, 135, 132, 130, 128, 123, 120, 135, 121, 128, 126, 132, 134, 138, 129, 130, 124, 126, 122, 135, 131, 139, 137, 132.

Prepare a frequency table with class – interval of 4 rupees each, such that

- The lower class – limit of the first class interval is 119.
- The lower class – limit of the last class interval is 135.
- The mid – point of the last class is 137.

2. The following data give the July rainfall (in cm.) at a certain place for 25 years (1950 – 1974):

40.8 45.5 48.7 46.3 44.4 38.4 46.0 39.1 41.6
 54.5 48.0 49.0 51.5 48.9 46.0 49.9 55.0 48.0
 58.7 59.9 53.0 53.1 61.9 54.4 47.7

Prepare a frequency distribution taking equal width of class – intervals i.e. 38 – 42, 42 – 46, ..., 58 – 62. Also draw a histogram and a frequency polygon on the same graph paper.

3. The following table gives the weights recorded to nearest pound of 40 students at a university:

138 164 150 132 144 125 149 136
 148 152 144 150 156 168 135 142
 146 173 142 147 135 157 161 145
 146 158 140 147 126 138 176 163
 119 154 165 153 140 135 145 128

- Tabulate the data into a frequency distribution taking class interval of size 9.
- Draw a histogram for the distribution obtained in (a) above.

4. The data given below show the daily wages paid to the workers in a factory.

200 165 175 185 210 117 100 176 175
 179 214 126 147 169 175 165 145 156
 175 107 172 124 134 140 175 165 176
 103 145 179 104 210 217 225 217 215
 222 217 179 146 157 165 179 145 190

- Classify this data taking a class interval of 18:
- Construct class boundaries.
- Calculate relative frequencies.
- Find less than and more than cumulative frequencies.
- Draw histogram and ogive.

5. The following data represent the population of United States in millions based on the census taken every 10 years from 1950 – 1990.

Year	1950	1960	1970	1980	1990
Population (in millions)	160	199	226	252	278

Construct a simple bar diagram for this data.

6. The following data showing the expenditure of a salaried employee. Represent the data by Pie – diagram:

Items	Expenditure
Food	1200
House Rent	1000
Clothing	500
Education	300

7. The following data are the compressive strength in pounds per square inch (psi) of 48 specimen of a new aluminum alloy under going evaluation as a possible material for air craft structural elements:

90, 88, 102, 100, 99, 104, 90, 97, 92, 91, 100, 98, 97, 104, 105, 92, 95, 96, 102, 97, 98, 105, 88, 94, 93, 100, 101, 99, 104, 106, 104, 91, 95, 101, 103, 98, 106, 105, 96, 91, 103, 101, 96, 105, 106, 98, 100, 106

Construct a frequency distribution table for the above data. Also draw Histogram.

8. Frequency distribution given below shows the hard disks production at some hardware company for 30 days. Draw histogram, frequency polygon and ogive.

Energy consumption	780 – 739	740 – 779	780 – 839	840 – 879	880 – 939
No. of days	6	10	8	4	2

9. The following data represent the length of life, in seconds, of 40 fruit flies subject to a new spray in controlled laboratory experiment:

66, 79, 60, 81, 82, 60, 76, 90, 70, 81, 74, 72, 86, 82, 87, 83, 71, 68, 65, 72, 93, 79, 54, 75, 82, 50, 94, 57, 64, 86, 71, 80, 66, 85, 79, 81, 62, 76, 82, 80

Set up a relative frequency distribution table for the above data. Also draw histogram and frequency polygon.

10. The following data are the number of cycles to failure of aluminum test coupons subject to repeated alternating stress at 21000 psi , 18 cycles per second.

1115	1567	1223	1782	1055	798	1016
1310	1883	1375	1522	1764	1020	1102
1540	1203	2265	792	1330	865	1605
1502	1270	1910	1000	1608	2130	803

- Construct a frequency distribution table for the data.
- Draw histogram, frequency polygon and ogive.

11. The following data are the comprehensive strengths in pounds per square inch (psi) of 28 specimens of a new aluminum lithium alloy under going evaluation as a possible material for aircraft structural elements.

105, 221, 183, 186, 121, 181, 180, 97, 154, 153, 174, 120, 168, 167, 245, 228, 154, 199, 181, 158, 176, 163, 131, 154, 115, 160, 194, 133.

Draw histogram, frequency polygon and ogive.

12. The following data shows number of products of ICs in some factory per day.

10, 3, 5, 6, 3, 7, 9, 2, 8, 6, 11, 20, 15, 13, 13, 19, 1, 12, 12, 8, 5, 17, 3, 12, 16, 10, 9, 2, 1, 11, 10, 5, 13, 8, 18, 7, 21, 4, 4, 11, 16, 19, 3, 6, 17.

(i) Construct a frequency distribution (ii) Construct the cumulative frequency distribution.

13. Following frequency distribution table shows oil production for 100 days.

Oil production	Frequency	Mid point
15-29	12	22
30-39	26	34.5
40-44	21	42
45-49	18	47
50-54	10	52
55-59	13	57

Draw histogram, frequency polygon and ogive.

CHAPTER THREE

MEASURES OF CENTRAL LOCATION AND VARIATION

3.1 INTRODUCTION

In many cases it is difficult to manage the entire data. Rather a numerical value that is representative of the entire data furnishes better information. As this typical value lies in the centre of data hence the word measure of central tendency is used for this numerical value. This numerical value is also known as average of the entire data set. This book discuss five most common averages: Arithmetic mean or simply the mean, Harmonic mean, Geometrical mean, Median and Mode. We shall also discuss limitations of each of these averages and particular situations where these averages are being used.

Arithmetic Mean

Perhaps most widely used average in our daily life is the arithmetic mean. The arithmetic mean which is commonly known as mean, of a set of data is calculated by adding all data values and dividing the sum by total number of items in a data set. Using the raw data:

$$\text{Population mean: } \mu = \frac{\sum_{i=1}^N x_i}{N} \quad \text{and} \quad \text{Sample mean: } \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Here n is a sample size and N is the size of population.

If data value x_i occurs with frequency f_i ($i = 1, 2, 3, \dots$) then mean is calculated as follows:

$$\text{Population mean: } \mu = \frac{\sum_{i=1}^N x_i f_i}{\sum_{i=1}^N f_i} \quad \text{and} \quad \text{Sample mean: } \bar{x} = \frac{\sum_{i=1}^n x_i f_i}{\sum_{i=1}^n f_i}$$

If the data is given in frequency distribution form then above formulae may be used by considering x_i as class mid point and f_i the corresponding class frequency.

Example 3.1.1 The following data shows sample strength in psi of 5 steel bars: 900, 880, 790, 1150 and 1010. Find the average strength of bars.

$$\text{Solution: } \bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{900 + 880 + 790 + 1150 + 1010}{5} = 946 \text{ psi}$$

Example 3.1.2 Twenty candidates appeared in aptitude test of 10 marks. Their marks are shown as under: 2, 2, 2, 3, 3, 3, 3, 5, 5, 5, 5, 6, 6, 6, 7, 7, 8, 9, 9, 10. Find the average marks obtained by each candidate.

Solution: Here we observe that 2 appears 3 times, 3 appears 4 times, 5 appears 4 times, 6 appears 3 times, 7 occurs 2 times, 8, 9 and ten appear 1, 2 and 1 times respectively. Thus using the formula:

$$\bar{x} = \frac{\sum_{i=1}^n x_i f_i}{\sum_{i=1}^n f_i} = \frac{2(3) + 3(4) + 5(4) + 6(3) + 7(2) + 8(1) + 9(2) + 10(1)}{3 + 4 + 4 + 3 + 2 + 1 + 2 + 1} = 5.3$$

Hence the average marks of 20 candidates are 5.3.

Example 3.1.3 The following data shows the number of cracks on 50 km long highway. Find the average number of cracks.

No. of cracks	0-5	5-10	10-15	15-20	20-25	25-30
Frequency	2	6	20	10	8	4

Solution: Calculations are made as under:

No. of cracks	Frequency (f_i)	Mid-point (x_i)	$x_i f_i$
0-5	2	2.5	5
5-10	6	7.5	45
10-15	20	12.5	250
15-20	10	17.5	175
20-25	8	22.5	180
25-30	4	27.5	110
Sum	50		765

$$\bar{x} = \sum_{i=1}^n x_i f_i / \sum_{i=1}^n f_i = \frac{765}{50} = 15.3.$$

Thus there are averagely 15.3 cracks in a km on 50 km long highway.

Properties of Mean

For numeric data:

- The mean can always be calculated and it is always unique for given data set.
- The mean takes account of all the value in a sample. This is a good property if there occurs no extreme value.
- Mean is always affected by extreme value in which case it is not recommended to use it. For example, the mean of the numbers 2, 3, and 4 is 3, where as the mean of 2, 3, 4 and 99 is 27.
- Sum of deviations of each data value from their mean is always zero.

Proof: Consider, $\sum_{i=1}^n (x_i - \bar{x}) = (x_1 - \bar{x}) + (x_2 - \bar{x}) + \dots + (x_n - \bar{x})$

$$= x_1 + x_2 + \dots + x_n - n\bar{x} = \sum_{i=1}^n x_i - n\bar{x} = n\bar{x} - n\bar{x} = 0$$

For example, the mean of numbers 4, 7, 8 and 11 is 7.5, then

$$(4 - 7.5) + (7 - 7.5) + (8 - 7.5) + (11 - 7.5) = -3.5 - 0.5 + 0.5 + 3.5 = 0$$

Geometric Mean and Harmonic Mean

Geometric mean G of n data values $x_1, x_2, \dots, x_n$ is defined as the n^{th} root of the product $(x_1 \cdot x_2 \cdot \dots \cdot x_n)$; that is,

$$G = (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n)^{1/n}.$$

Taking log on both sides, we obtain:

$$\log G = \log (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n)^{1/n} = \frac{\log x_1 + \log x_2 + \dots + \log x_n}{n} = \frac{1}{n} \sum_{i=1}^n \log x_i.$$

The geometric mean G is then found by taking the anti-log on both sides.

If the data values $x_1, x_2, \dots, x_n$ occur with frequencies $f_1, f_2, \dots, f_n$ respectively, then geometric mean is computed as:

$$G = (x_1^{f_1} \cdot x_2^{f_2} \cdot \dots \cdot x_n^{f_n})^{1/n} \text{ or } \log G = \frac{1}{n} \sum f_i \log x_i, \text{ where } n = \sum f_i$$

This formula is also applicable in computing the geometric mean from the grouped data where x_i is considered as class mid point and f_i as the corresponding frequency.

Example 4.1.4 Find the geometric mean from the frequency distribution given below.

Weight(gm)	65-84	85-104	105-124	125-144	145-164	165-184	185-204
f_i	9	10	17	10	5	4	5

Solution: The computation is shown in the following table.

Weight(gm)	x_i	f_i	$\log x_i$	$f_i \log x_i$
65-84	74.5	9	1.8722	16.8498
85-104	94.5	10	1.9754	19.7540
105-124	114.5	17	2.0589	35.0013
125-144	134.5	10	2.1287	21.2870
145-164	154.5	5	2.1889	10.9445
165-184	174.5	4	2.2418	8.9672
185-204	194.5	5	2.2889	11.4445
Sum		60		124.2483

$$\log G = \frac{1}{n} \sum f_i \log x_i = \frac{124.2483}{60} = 2.0708 \quad \text{then } G = \text{anti log}(2.0708) = 117.7 \text{ grams}$$

Harmonic Mean

Harmonic mean H of n data values $x_1, x_2, \dots, x_n$ is defined as the mean of reciprocals of $x_1, x_2, \dots, x_n$; that is,

$$H = \text{Reciprocal of } \left(\frac{(1/x_1 + 1/x_2 + \dots + 1/x_n)}{n} \right) \quad \text{or} \quad H = \frac{n}{\sum \frac{1}{x_i}}$$

If the data values $x_1, x_2, \dots, x_n$ occur with frequencies $f_1, f_2, \dots, f_n$ respectively, then harmonic mean is computed as:

$$H = \text{Reciprocal of } \frac{(f_1/x_1 + f_2/x_2 + \dots + f_n/x_n)}{n} \quad \text{or} \quad H = \frac{n}{\sum f_i/x_i}, \text{ where } n = \sum f_i$$

This formula is also applicable in computing the harmonic mean from the grouped data where x_i is considered as class mid point and f_i as the corresponding frequency.

Example 3.1.5 Find the harmonic mean from the frequency distribution given below.

Weight(gm)	65-84	85-104	105-124	125-144	145-164	165-184	185-204
f_i	9	10	17	10	5	4	5

Solution: The computation is shown in the following table.

Weight(gm)	x_i	f_i	f_i/x_i
65-84	74.5	9	0.12081
85-104	94.5	10	0.10582
105-124	114.5	17	0.14847
125-144	134.5	10	0.07435
145-164	154.5	5	0.03236
165-184	174.5	4	0.02292
185-204	194.5	5	0.02571
Sum		60	0.53044

$$\text{Now, } H = \frac{n}{\sum f_i/x_i} = \frac{60}{0.53044} = 113.11 \text{ gram.}$$

The, arithmetic mean of above frequency distribution is 122.5. Now,

$$A.M = 122.5$$

$$G.M = 117.7$$

$$H.M = 113.11$$

One may observe that:

$$H.M < G.M < A.M$$

It may be noted that harmonic mean is an appropriate type of average to be used when averaging certain kinds of ratios or rates. To illustrate this let us consider a situation where in a car race a racer covers first 50 km at the rate of 120 km/h; at the rate of 130 km/h during second 50 km and finally he covers third 50 km at the rate of 150 km/h. Here we see that distances are constant but speed varies, hence harmonic mean is the appropriate average to be used. Now,

$$H = \frac{3}{\frac{1}{120} + \frac{1}{130} + \frac{1}{150}} = \frac{3}{0.008333 + 0.0076923 + 0.0066667} = 132.21 \text{ km/h}$$

Thumb Rule to Apply Harmonic and Arithmetic Means

When rates are expressed as p per q and p is constant, the harmonic mean is the appropriate mean; but if q is constant then arithmetic mean is to be applied.

The Median

It was mentioned before that main drawback of arithmetic mean is that it is always affected by extreme values. In this situation; the median is the most appropriate average to be used.

The median is the middle value or the mean of two middle values when the data values are arranged in order of magnitude. For example;

The median of data values: 2, 3, 3, 4, 5, 6, 6, 7, 7, 7, 9 is 6.

The median of data values: 2, 3, 3, 4, 4, 4, 5, 6, 7, 8, 8, 9 is $(4 + 5)/2 = 4.5$

The main drawback of median is that before its computation, the data values are to be arranged in order of magnitude. This is cumbersome when number of data values is very large.

Placing the data in order from smallest to largest is known as *ranking the data*. The item with the smallest value has rank 1; the second smallest value has rank 2 and so on. Thus median is the value of the item which has half of the ranked data below it and half of the data above it. This concept is therefore extended to define *quartile, decile and percentile*. The first quartile (lower quartile) is the value of the item which has 25% of the ranked data below it and third quartile is the value of the item which has 75% of ranked data below it.

Properties of Median

- For any data set the median is unique
- Median is not affected by extreme values as it only considers middle values

Method of Finding Median from Discrete Quantitative Data

When it is required to compute the median from “Discrete Quantitative Data” we proceed as under. Let the sum of frequencies of all categories be n find $m = n/2$. If m is odd the median is m^{th} observation of the category. If m is even then median is the average of m^{th} and $(m + 1)^{\text{st}}$ observations.

Example 3.1.6 The following data shows the number of hours worked by part time staff in a software company. Draw the frequency distribution and find the mean and median number of hours.

18	20	20	24	20	18	20	20	24	18	20	18
20	18	24	24	24	20	20	24	18	20	18	20

20 18 20 24 18 20 20 20 20 18 24 20

Solution: We observe that there are three categories, that is; 18, 20, 24. The frequency distribution is shown in the following table.

Value of category	Frequency	$x_i f_i$
18	10	180
20	18	360
24	08	192
Sum	36	732

Mean = $\frac{\sum x_i f_i}{\sum f_i} = \frac{732}{36} = 20.34$. To find median, we proceed as under:

Here $n = 36$ then $n/2 = 18$. This is an even, hence median is average of 18th and 19th observation of the data. Thus median is $(20 + 24)/2 = 22$.

Thus mean hours worked by each part time staff member is 21 hours and median number of hours is 22.

Computation of Median from Grouped Data

The median is computed from grouped data by using the formula:

$$\text{Median} = L + \left(\frac{n/2 - C}{f_m} \right) h$$

Here, L is lower class boundary of median class

n is sum of all class frequencies

C is cumulative frequency before the median class

f_m is the frequency of the median class

h is the size of median class

Example 3.1.7 The following grouped data shows kilowatt hours of electricity used in one month by 72 residential consumers in certain area. Compute the median electricity consumed by each consumer.

Electricity consumed	5-21	25-44	45-64	65-84	85-104	105-124	125-144	145-164
No. of users	4	6	14	22	11	5	7	3

Solution: Here $n = 72 \Rightarrow n/2 = 36$. To find median class i.e; the class that contains median, we proceed as under:

$$4 < 36 \quad 4 + 6 = 10 < 36 \quad 4 + 6 + 14 = 24 < 36 \quad 4 + 6 + 14 + 22 = 46 > 36$$

Hence, median class is 4th class (65 – 84). Now

$$L = 64.5$$

$$C = 4 + 6 + 14 = 24$$

$$f_m = 22$$

$$h = 20$$

Thus, Median = $64.5 + 20.(36 - 24)/22 = 75.4$ KWH

Hence median consumption of electricity is 75.4 KWH.

Quartiles, Deciles and Percentiles

When the complete data is divided into four equal parts each part is called quartile.

Similarly, when the data is divided into ten equal parts each part is called deciles and when it is divided into 100 equal parts then each part is known as percentile.

Look at the following division of a line segment into four equal parts.



$Q_1 \qquad Q_2 \qquad Q_3$

Observe that by giving three cuts the line is divided into four equal parts. Hence there are always three “Quartiles” in the entire data. These are usually denoted by Q_1 , Q_2 and Q_3 .

Similarly, there will be nine deciles and 99 percentiles. These are denoted by:

$D_1, D_2, \dots, D_9$ and $P_1, P_2, \dots, P_{99}$ respectively. It may be noted that

$$\text{Median} = Q_2 = D_5 = P_{50}.$$

Quartiles, Deciles and Percentiles are computed in the same way as median is computed. Here we list the formulae for their computation.

Quartile: $Q_i = L_i + \left(\frac{n \times i / 4 - C}{f_i} \right) h \qquad i = 1, 2, 3.$

Decile: $D_i = L_i + \left(\frac{n \times i / 10 - C}{f_i} \right) h \qquad i = 1, 2, 3, \dots, 9.$

Percentile: $P_i = L_i + \left(\frac{n \times i / 100 - C}{f_i} \right) h \qquad i = 1, 2, 3, \dots, 99.$

Inter Quartile Range and Quartile Deviation

In practice, sometimes we are interested in knowing the range where 50% of observations lies in the mid of entire data. For example, a management of certain software company wishes to increase the salary of part time staff members. Before the salary is increased, management wants to find that “what is the range of salary of mid 50% of people”. In other words, they want to find out $Q_3 - Q_1$. This is known as “Inter Quartile Range”. The half of Inter Quartile Range is called “Quartile Deviation”, that is;

$$Q.D = (Q_3 - Q_1)/2$$

The Mode

The mode of a data is value, category or class that occurs with the highest frequency.

Properties of Mode

- Mode is easy to find and is used as a guide to typical value in a sample, that is; the value which occurs with highest frequency. Hence it is used very much in “Stock Exchange” market where one is only interested in a value with highest price or sale etc.
- The mode can be used in qualitative as well as quantitative data.
- It has a draw back that it may not exist and even if it exists it may not be unique. This is in contrast with mean and median.

For example:

- The mode of data 2, 2, 3, 4, 4, 4, 5, 5, 6, 7, 7 is 4 because 4 occur with highest frequency 3. Such data is called *uni-modal* data.
- The data 2, 3, 3, 3, 4, 4, 5, 5, 6, 6, 6, 7, 7, 8 has two modes namely 3 and 6 as these are the numbers which occur with highest frequencies 3. Such data is called *bi-modal* data.
- The data 1, 3, 4, 6, 7, 9 or 2, 2, 3, 3, 4, 4, 6, 6, 8, 8 has no mode as each number in the both data sets occur with same frequency.

Computation of Mode from Grouped Data

From the grouped data, mode is computed by using the formula:

$$\text{Mode} = L + \frac{\Delta_1}{(\Delta_1 + \Delta_2)} h . \text{ Here;}$$

L : is the lower class boundary of modal class.

Δ_1 : is the difference between the modal frequency and the frequency of class before it.

Δ_2 : is the difference between the modal frequency and the frequency of class after it.

h : is the size of modal class.

Example 3.1.8 The following grouped data shows kilowatt hours of electricity used in one month by 72 residential consumers in certain area. Compute the modal (highest) electricity consumption by each consumer.

Electricity consumed	5-24	25-44	45-64	65-84	85-104	105-124	125-144	145-164
No. of users	4	6	14	22	11	5	7	3

Solution: We observe that highest frequency is 22, hence 65 – 84 is the modal class.

Now, $L = 64.5$, $\Delta_1 = 22 - 14 = 8$ $\Delta_2 = 22 - 11 = 11$ $h = 20$

Thus, $\text{Mode} = L + \frac{\Delta_1}{(\Delta_1 + \Delta_2)} h = 64.5 + \frac{8}{(8+11)} \times 20 = 72.92$

Hence, averagely maximum electricity consumed is 72.92 KWH.

Computation of Mode from Qualitative Data

As mentioned earlier that mode is the only type of average that can be used to compute the average from qualitative data. The following example illustrates this fact.

Example 09: The following data represents the grades given to job applicants in an aptitude test.

A B D A D A D D D B B B
A D C B C B D D

Find the modal grade.

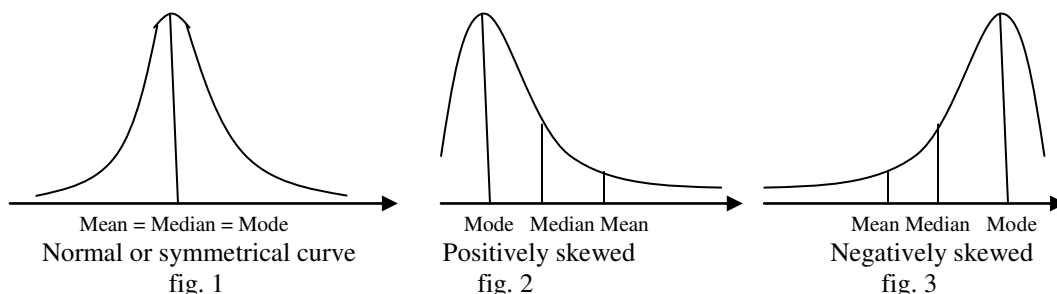
Solution:

Category: A B C D
Frequency 4 6 2 8

Since grade “D” occurs 8 times hence modal grade is D.

Relation between Mean, Median and Mode

Before we discuss the relation between mean, median and mode readers may notice that a data that is discrete or continuous and follows the shape as shown figure1 is called normal or symmetric data otherwise it is skewed or asymmetrical data as shown in figures 2 and 3.



- If the data type is symmetrical then mean, median and mode are equal and lie at the centre of the curve.
- If the data type is asymmetrical or skewed then median lies in between the mean and mode where mode is the point of maxima.

For moderate skewed curve, mean, median and mode are related by equation:

$$\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median}) \quad \text{or} \quad \text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

This empirical formula does not hold for the data that is highly skewed.

3.2 MEASURES OF DISPERSION OR VARIATION

Measures of variation provide information about the variability or spread of a set of data. For example variation in lives of electric bulbs produced by two different companies. When summarizing a set of data, a measure of central location (Average) alone does not provide enough information. A measure of variation must also be used since two sets of data may have the same mean but not the same variability. The following example illustrates this fact.

Two machines are used to manufacture the steel sheets. The samples are taken from both machines and are recorded in mm as under:

Machine	Thickness				
A	7	8	9	10	11
B	15	4	5	20	1

The mean thickness for each machine is 9 mm but variability for machine B is greater than that for machine A. This is very much clear from the above data. We shall show you very soon how to compute the variability.

There are many measures of variations. We will focus on the four most commonly used measures: the range, the mean deviation, the standard deviation and the variance.

The Range

The range is simply the difference between the largest and smallest values in the data set. The range is very quick and simple to calculate. It is used as general indication of the variability of the data. Since range only considers the smallest and largest values in a set of data, it will give a misleading impression of variability of the data. Hence, it is not very much useful in practice.

Example 3.2.1 During manufacturing process of motherboards it was found that in a particular week the maximum and minimum number of defective motherboards produce were 1000 and 880. Find the range.

Solution: Range = 1000 - 880 = 120.

You may realize that value 120 does not give further information about the data other than the variability of the data is 120. This may not be same for the other week's production.

Mean Deviation

Mean deviation which, is also known as average deviation or mean absolute deviation is defined as:

$$\text{M.D} = \frac{\sum |x_i - \mu|}{N} \text{ for population data.} \quad \text{M.D} = \frac{\sum |x_i - \bar{x}|}{n} \text{ for sample data}$$

Remarks:

- ✓ We may place median in place of mean $\bar{x}$ in computation of M.D.

- ✓ The absolute value $|x_i - \bar{x}|$ are taken in computation of M.D because we know that sum of the deviations of each data value from its mean is zero.
- ✓ It is found that M.D taken from median is less than if it is taken from the mean. However; it is usually taken from the mean.
- ✓ M.D is unsatisfactory for making statistical inferences; hence it is not very much used. However, it is better than range.

Example 3.2.2 Marks obtained by 9 students in mathematics are given as under. Compute the M.D from the mean and median.

45 32 37 39 36 46 41 48 36

Solution: The mean marks are $\bar{x} = (45 + 32 + 37 + 39 + 36 + 46 + 41 + 48 + 36)/9 = 40$ marks. Putting the data in ascending order, we get:

32 36 36 37 39 41 45 46 48

You may observe that median marks are: 39

Now necessary calculations are made in the following table.

x_i	$x_i - \bar{x}$	$ x_i - \bar{x} $	$ x_i - \text{median} $
32	-8	8	7
36	-4	4	4
36	-4	4	3
37	-3	3	2
39	-1	1	0
41	1	1	2
45	5	5	6
46	6	6	7
48	8	8	9
Sum: 360		40	39

$$\text{M.D (from mean)} = \frac{\sum |x_i - \bar{x}|}{n} = \frac{40}{9} = 4.4 \text{ marks}$$

$$\text{M.D (from median)} = \frac{\sum |x_i - \text{median}|}{n} = \frac{39}{9} = 4.3 \text{ marks}$$

The Variance

The variance of a set of data values is defined as the mean of the squares of deviations of all data values (observations) from their mean. If this variance is calculated from the entire population, it is called “Population Variance” and is denoted by σ^2 . If sample data is used, the variance is called “Sample Variance” and is denoted by s^2 . Thus

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N} \text{ for population data,} \quad s^2 = \frac{\sum (x_i - \bar{x})^2}{n} \text{ for sample data}$$

It may be noted that variance is in square of units. It is extremely important for making statistical inferences because of its mathematical properties. These will be made clear in coming chapters.

Standard Deviation

The positive square root of the variance is called “Standard Deviation”. Symbolically,

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}} \text{ for population data,} \quad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} \text{ for sample data}$$

Standard deviation is expressed in the same unit as that of observation.

When the data are grouped into frequency distribution having k classes, where x_i is the class mid points and f_i the corresponding class frequency, then sample variance and sample standard deviations are computed as:

$$s^2 = \frac{\sum (x_i - \bar{x})^2 f_i}{n}, s = \sqrt{\frac{\sum (x_i - \bar{x})^2 f_i}{n}} \text{ respectively, where } n = \sum_{i=1}^k f_i.$$

Points to note:

- It may be noted that some authors use following formulas for computing variance and standard deviation:

$$s^2 = \frac{\sum (x_i - \bar{x})^2 f_i}{n-1} \text{ or } s = \sqrt{\frac{\sum (x_i - \bar{x})^2 f_i}{n-1}} \quad i = 1, 2, \dots, (n-1).$$

This is due to the fact that for frequency distribution the number of observations (sum of all frequencies) is usually large hence, dividing the sum by n or $(n-1)$ is practically equivalent.

- The other reason is that knowledge of $(n-1)$ deviations routinely determine the remaining deviation, as the sum of n deviations is zero.
- The expression $\sqrt{\frac{\sum (x_i - a)^2}{n}}$, where 'a' is some arbitrary constant, is called the *root-mean-square-deviation* which becomes standard deviation when this arbitrary constant is replaced by the mean $\bar{x}$.

Another Method of Computing Variance

$$\begin{aligned} \text{Since } s^2 &= \frac{\sum (x_i - \bar{x})^2}{n} = \frac{\sum (x_i^2 - 2\bar{x}x_i + \bar{x}^2)}{n} = \frac{\sum x_i^2}{n} - 2\bar{x} \frac{\sum x_i}{n} + \frac{\sum \bar{x}^2}{n} \\ &= \frac{\sum x_i^2}{n} - 2\bar{x} \bar{x} + \frac{n\bar{x}^2}{n} = \frac{\sum x_i^2}{n} - 2\bar{x}^2 + \bar{x}^2 = \frac{\sum x_i^2}{n} - \bar{x}^2 \end{aligned}$$

$$\text{Thus, } s^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2.$$

This shows that sample variance is the mean of squares minus square of the means. In case of frequency distribution, standard deviation is computed as follows:

$$s^2 = \frac{\sum f_i x_i^2}{n} - \left(\frac{\sum f_i x_i}{n} \right)^2$$

Example 3.2.3 Marks obtained by 9 students in mathematics are given as under. Compute the standard deviation.

45 32 37 39 36 46 41 48 36

Solution: The necessary calculations are made in the following table.

x_i	32	36	36	37	39	41	45	46	48	360(sum)
x_i^2	1024	1296	1296	1369	1521	1681	2025	2116	3204	14632(sum)

$$\text{Now, } s^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2 = \frac{14632}{9} - \left(\frac{360}{9} \right)^2 = 1625.78 - 1600 = 25.78 \text{ (Marks)}^2$$

Thus, $s = 5.07$ marks.

Example 3.2.4 The following data shows weight in (1000 kg) of steel bars produced in a steel factory. Find the standard deviation.

Weight	65-84	85-104	105-124	125-144	145-164	165-184	185-204
f_i	9	10	17	10	5	4	5

Solution: The necessary calculations are shown in the following table.

Weight (1000 kg)	x_i	f_i	$f_i x_i$	$f_i x_i^2$
65-84	74.5	9	670.5	49952.25
85-104	94.5	10	945.0	89302.5
105-124	114.5	17	1946.5	222874.25
125-144	134.5	10	1345.0	180633.6
145-164	154.5	5	772.5	119351.25
165-184	174.5	4	698.0	121801
185-204	194.5	5	972.5	189151.25
Sum		60	7350	973335

$$\text{Thus, } s^2 = \frac{\sum f_i x_i^2}{n} - \left(\frac{\sum f_i x_i}{n} \right)^2 = \frac{973335}{60} - \left(\frac{7350}{60} \right)^2 = 1216 (1000 \text{ kg})^2$$

Or, $s = 34.87$ (1000 kg)

Significance of Standard Deviation

We know that average or mean is the number that lies in the centre of data set. Hence it is also known as balancing point of the entire data. The standard deviation is a very important concept of statistical analysis that serves as a basic variability measure. Thus its interpretation or significance is not so straight forward. However, it may be noted that small value of standard deviation indicates that data values are close to their mean and its large value indicates that data values are scattered widely about their mean. Russian mathematician P.I. Chebyshev stated a rule known as “Chebyshev Rule” which states that “For any data set, the interval $(\bar{x} - ks, \bar{x} + ks)$, where k is a number greater than 1, contains at least $(1 - 1/k^2)$ % of the entire data. For example, if $k = 2$ the interval $(\bar{x} - 2s, \bar{x} + 2s)$ contains at least $(1 - 1/4)$ % = 75 % of the entire data.

Coefficient of Variation

The standard deviation has a definite mathematical meaning and it utilizes all the data values and is helpful to mathematical treatment but is very much affected by extreme values. In such situations, its relative measure called coefficient of variation is used. It is denoted by (C.V) and is defined as:

$$\text{C.V} = \frac{\text{STANDARD DEVIATION}}{\text{MEAN}} \times 100\% = \frac{s}{\bar{x}} \times 100\%$$

Coefficient of variation is almost used in comparing the variability of two sample data. As the coefficient of variable is a pure number without units, it is therefore used to compare the variation in two or more data sets that are measured in different units. For example, one may be measured in number of inhabitants living in certain area and other may be consumption of electricity by the people living in that area.

Example 3.2.5 Two plants were installed by a certain textile mills on the same date and were in operation under identical conditions. The first six months output of both plants is given as under. State which of the two plants is more consistent?

Month	Plant – I	Plant – II
-------	-----------	------------

January	10	20
February	30	25
March	10	25
April	30	25
May	45	30
June	25	10
Total	150	135

Solution: To check the consistency, we find coefficient of variation of output of each plant.

Plant – I :

x	10	30	10	30	45	25	Total = 150
x ²	100	900	100	900	2025	625	Total = 4650

$$\bar{x} = \frac{\sum x}{n} = \frac{150}{6} = 25, \quad \sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2 = \frac{4650}{6} - \left(\frac{150}{6} \right)^2 = 150 \text{ or } \sigma = \sqrt{150} = 12.25.$$

$$\text{C.V. of Plant – I} = \frac{\sigma}{\bar{x}} \times 100\% = \frac{12.25}{25} \times 100\% = 49\%.$$

Plant–II:

x	20	25	25	25	30	10	Total = 135
x ²	400	625	625	625	900	100	Total = 3275

$$\bar{x} = \frac{\sum x}{n} = \frac{135}{6} = 22.5, \quad \sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2 = \frac{3275}{6} - \left(\frac{135}{6} \right)^2 = 39.58 \text{ or } \sigma = \sqrt{39.58} = 6.29.$$

$$\text{C.V. of Plant – II} = \frac{\sigma}{\bar{x}} \times 100\% = \frac{6.29}{22.5} \times 100\% = 27.96\%.$$

Since Coefficient of Variation of Plant – II is less than the Coefficient of Variation of Plant – I, therefore Plant – II is said to be more consistent.

Example 3.2.6 In a survey of domestic consumption of electricity during a month, two samples of 100 residents each in two cities were selected and the current consumed was as follows:

Number of units	Consumers of City – A	
	Consumers of City – A	Consumers of City – B
0 – 49	8	6
50 – 99	12	12
100 – 149	25	30
150 – 199	30	40
200 – 249	15	7
250 – 299	10	5

Compare the coefficient of variation (C.V.) for both cities and comment.

Solution:

For City – A:

No. of units	f	x	fx	x^2	fx^2
0 – 49	8	24.5	196	600.25	4802
50 – 99	12	74.5	894	5550.25	66603
100 – 149	25	124.5	3112.5	15500.25	387506.25
150 – 199	30	174.5	5235	30450.25	913507.50
200 – 249	15	224.5	3367.5	50400.25	756003.75
250 – 299	10	274.5	2745	75350.25	753502.50
Total	100		15550	177851.5	2881925

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{15550}{100} = 155.5$$

$$\sigma^2 = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2 = \frac{2881925}{100} - \left(\frac{15550}{100} \right)^2 = 4638.9721 \text{ or } \sigma = \sqrt{4638.9721} = 68.11$$

$$\text{C.V. for City – A} = \frac{\sigma}{\bar{x}} \times 100\% = \frac{68.11}{155.5} \times 100\% = 43.8\%$$

For City – B:

No. of units	f	x	fx	x^2	fx^2
0 – 49	6	24.5	147	600.25	3601.5
50 – 99	12	74.5	894	5550.25	66603
100 – 149	30	124.5	3735	15500.25	465007.5
150 – 199	40	174.5	6980	30450.25	1218010
200 – 249	7	224.5	1571.5	50400.25	352801.75
250 – 299	5	274.5	1372.5	75350.25	376751.25
Total	100		14700	177851.5	2482775

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{14700}{100} = 147$$

$$\sigma^2 = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2 = \frac{2482775}{100} - \left(\frac{14700}{100} \right)^2 = 3218.2929 \text{ or } \sigma = \sqrt{3218.2929} = 56.73$$

$$\text{C.V. for City – B} = \frac{\sigma}{\bar{x}} \times 100\% = \frac{56.73}{147} \times 100\% = 38.59\%$$

Since C.V.(B) < C.V.(A), therefore consumption of electricity in city – B is more consistent.

WORKSHEET 03

1. The following table contains data which represent the lives recorded to the nearest tenth of a year, of 40 cars batteries of a certain brand:

C.I.	1.5 – 1.9	2.0 – 2.4	2.5 – 2.9	3.0 – 3.4	3.5 – 3.9	4.0 – 4.4
F	2	1	4	15	10	5

Calculate (a) Mean (b) Median

2. The following table contains weekly wages of 10 employees recorded in the last week of the month of the December 2007:

Weekly wages	61-70	71-80	81-90	91-100	101-110
No. of employees	1	2	4	2	1

Compute from the above data:

(a) Mean (b) Median (c) Mode

3. In a factory, the time during working hours in which a machine is not operating as a result of breakage or failure is called a downtime. The following distribution shows a sample of the length of the down times of a certain machine:

Downtime (minutes)	0 – 9	10 – 19	20 – 29	30 – 39	40 – 49
Frequency	2	15	17	13	3

Find (a) the mean and the mode (b) the standard deviation (c) 3rd Quartile

4. The following distribution gives the number of defective pieces in 100 uniform size samples of a certain kind of bolt:

No. of defective bolts	0-4	5-9	10-14	15-19	20-24	24-29
frequency	12	20	48	10	7	3

Compute: (a) Mean (b) Median (c) Mode

5. The following table shows the lives of batteries of a certain brand:

Life in months	11-15	16-20	21-25	26-30	31-35
No. of batteries	1	2	5	1	1

Calculate: (a) Mean deviation from median (b) standard deviation (c) coefficient of variation (d) variance.

6. A manufacturing company owns two plants Plant – I and Plant – II which manufacture the same product. The weekly output of two plants for the past 5 years as follows:

Weekly output (in 000 units)	Number of weeks	
	Plant – I	Plant – II
11 – 15	10	15
16 – 20	15	20
21 – 25	135	60
26 – 30	65	150
31 – 35	35	15

Which of the two plants gave more stable production during the period?

7. Production of two machines are given as under, select a machine which is more profitable? (Hint: One should have to choose a machine with higher consistency level output).

Output	11-15	16-20	21-25	26-30	31-35
Machine-A	25	40	90	55	50
Machine-B	15	20	60	150	15

8. A manufactured product consists of various components. Each of the components has a limited life time. The company has tested the product to determine the reliability of the component. Following table shows frequency distribution of components, which fail during first 100 hours of operation? Compute mean, mode, mean deviation, standard deviation and 37th percentile:

No. of components failing	01 – 10	11 – 20	21 – 30	31 – 40	41 – 50	51 – 60
Life in hours	05	09	22	32	20	12

9. Contamination on the silicon wafer can seriously affect the yield of good integrated circuits produced. A sample of 42 wafers resulted in the following oxygen concentrations:

C-I	2.0-2.4	2.5-2.9	3.0-3.4	3.5-3.9	4.0-4.4
Frequency	4	8	16	10	4

Compute the median, mode & standard deviation.

10. A manufacturing company own 2 plants, which manufacture the same product. The weekly output of the two plants for the past 5 years was as follows:

Weekly output of 1000 units	Plant-I	Plant-II
11-15	25	15
16-20	40	20
21-25	90	60
26-30	55	150
31-35	50	15

Which of these two plants give more stable production during the period.

11. Fifty fuses were tested and the following currents were recorded for each experiment. The table is shown below:

C.I	6.5 - 7.5	7.5 - 8.5	8.5 - 9.5	9.5 - 10.5	10.5 - 11.5	11.5 - 12.5	12.5 - 13.5
Frequency	02	08	11	12	08	05	04

Compute the median, mode and standard deviation.

12. An aptitude test was given to 80 students of university, and the distribution of marks is given in the following data. Calculate the mean and standard deviation. What conclusion about the variability in data would you draw?

C-I	0-4	5-9	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49	50-54
'f'	1	3	8	9	9	23	13	7	4	2	1

13. Two samples of size each were under observation. The mean for the first sample was 1.16 with standard deviation of 0.21, the mean for the second sample was 1.75 with standard deviation of 0.35, which sample show greater relative dispersion.

14. In a factory, the time during working hours in which a machine is not operating as a result of breakage or failure is called a down time. The following time distribution shows a sample of the length of the down times of a certain machine.

C-I	0-9	10-19	20-29	30-39	40-49
Frequency	2	15	17	13	3

Find the mean and median and standard deviation.

CHAPTER FOUR

INTRODUCTION TO PROBABILITY

4.1 INTRODUCTION

Probabilities are associated with situations involving doubts or uncertainties about the outcomes. For example, which horse to bet on in a race, whether to buy shares in a company, how much orders a firm will get this week, how many defective circuit boards will a lot contain, what is the chance that a newly learnt driver would make an accident. All these examples involve decisions to be made in situation with uncertainty.

Probability theory attempts to make some measure of certainty and uncertainty to the outcome. Probability is a numerical measure of the likelihood of an event occurring. It is measured on a scale from 0 to 1, with zero indicating the impossibility of the event and unity representing certainty. In many applications, values zero and unity are rarely encountered – outcomes are uncertain to a greater or lesser degree. Hence, a managerial decision making always base on some assessment of the chance of possible outcomes.

Experiments, Sample Space

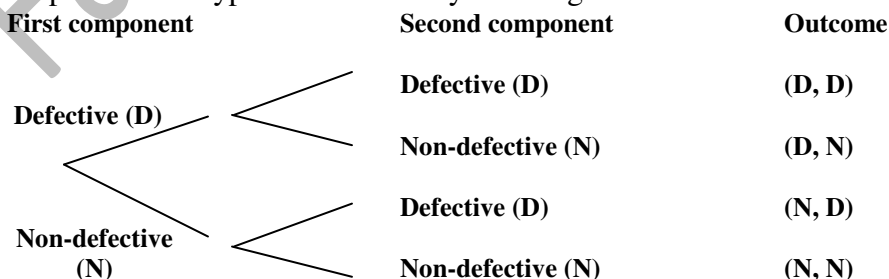
In probability term experiment is defined as a process which generates a well-defined outcome that are not predictable in advance, but where we do know all of the possible outcomes. The set of all possible outcomes of any experiment is known as sample space.

<u>Experiment</u>	<u>Possible outcomes</u>
Rolling a fair die	$S = \{1, 2, 3, 4, 5, 6\}$
Selecting a component from a production line	$S = \{D, N\}$ (defective, non-defective)
A bid for contract	$S = \{S, F\}$ (success, failure)
Grade a candidate acquires	$S = \{A, B, C, D, F\}$
Demand for a new product	$S = \{0, 1, 2, 3, \dots, n\}$

Suppose we wish to choose two items from a production line for inspection. The sample space will be:

$$S = \{(D, D), (D, N), (N, D), (N, N)\}$$

It may be noted that the outcomes (D, N) and (N, D) are different in terms of probability. We can represent this type of outcomes by tree diagram as shown below.



Similarly, if three items are selected from the production line, the sample space will be

$$S = \{(D, D, D), (D, D, N), (D, N, D), (N, D, D), (D, N, N), (N, D, N), (N, N, D), (N, N, N)\}$$

Let us consider another experiment. Suppose a coin is flipped. It is flipped again if a head occurs. If a tail occurs then a die is rolled. The sample space is:

$$S = \{HH, HT, T1, T2, T3, T4, T5, T6\}$$

Basic Rules of Counting

If an experiment consists of several steps, the first having n_1 outcomes, the second having n_2 outcomes and so on, then the total number of possible outcomes from this experiment will be:

$$n_1 \times n_2 \times n_3 \times \dots$$

For the above experiment there are two steps (selecting the first component and then the second component) each step has two possible outcomes – the component being defective or non-defective. Therefore, $n_1 = 2$ and $n_2 = 2$. Thus by above rule, the possible outcomes are $2 \times 2 = 4$. Similarly, if three items are selected from the production line, the possible ways are $2 \times 2 \times 2 = 8$.

Let us consider another experiment of tossing a fair coin and rolling a die. A coin shows two outcomes H or T and die can show up with 1, 2, 3, 4, 5, or 6. Hence, total possible outcomes are $2 \times 6 = 12$. The sample space for this experiment is:

$$S = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$$

Permutations

Above rule of counting may be extended to n distinct elements. If n increases it becomes very difficult to write down the sample space for all possible outcome of the experiment. Permutation is the process by which one can select say r elements out of n elements ($r \leq n$). It is defined as any ordered subset from a set of n distinct elements. The number of permutations of r objects, selected in a definite order from n distinct objects is denoted by

$$P_r^n \text{ and is defined as: } P_r^n = \frac{n!}{(n-r)!}$$

For example,

$$P_2^2 = \frac{2!}{(2-2)!} = 2, P_3^3 = \frac{3!}{(3-3)!} = 6, P_n^n = \frac{n!}{(n-n)!} = n! \text{ and } P_0^n = \frac{n!}{(n-0)!} = 1$$

Example 4.1.1 A president, secretary and treasurer are to be selected from five persons for a club. In how many ways we can make this selection?

Solution: As there are five persons hence a president can be selected in 5 ways. Once the president is selected then a secretary can be selected in four ways and then treasurer can be selected in three ways. Hence total number of ways the President, The secretary and

the treasurer are selected is $5 \times 4 \times 3 = 60$ or $P_3^5 = \frac{5!}{(5-3)!} = 120 / 2 = 60$.

Extension of Permutation Rule

The numbers of permutation of n objects of which n_1 are of one kind, n_2 are of second kind, ..., n_k are of k^{th} kind, where $n = n_1 + n_2 + n_3 + \dots + n_k$ is defined as:

$$\frac{n!}{n_1! \cdot n_2! \cdot n_3! \cdot \dots \cdot n_k!}$$

Example 4.1.2 How many different ways can 3 red, 4 yellow, and 2 blue bulbs be arranged in a string of Christmas tree lights with 9 sockets

Solution: Using the extended rule of permutation the number of ways this experiment is done is:

$$\frac{9!}{3!.4!.2!} = 1260 \text{ ways.}$$

Example 4.1.3 In how many ways can seven doctors be assigned one triple and two double rooms in a hotel?

Solution: The number of possible partitions would be: $\frac{7!}{3!.2!.2!} = 210 \text{ ways.}$

Rule of Combination

A combination is any subset of r objects, selected without regard to their order, from a set of n distinct objects. This is denoted by C_r^n and is defined as:

$$C_r^n = \frac{n!}{r!(n-r)!}$$

It may be observed that: $C_0^n = C_n^n = 1$.

The basic difference between permutation and combination is that in permutation the order in which the objects are selected is important whereas in combination, the selection of elements or objects is done without considering the order.

Let us consider a simple example. Let there be a set of three elements $\{a, b, c\}$. If all elements are considered with order, the number of permutations will be:

“abc”, “acb”, “bac”, “bca”, “cab” and “cba”. Thus total number of permutations is 6. This may be confirmed by using the formula $P_3^3 = \frac{3!}{(3-3)!} = 6$.

The combination of 3 objects taking all at a time is $C_3^3 = \frac{3!}{3!(3-3)!} = 1$.

You may observe that combination of a, b and c is any one of above six values say ‘abc’ or ‘cab’, etc.

Example 4.1.4 A committee of three persons is to be formed out of five people. In how many ways this committee can be formed?

Solution: In this problem a committee of three persons is required hence, the order is not desired so, we use combination rule to get:

$$C_3^5 = \frac{5!}{3!(5-3)!} = 120/12 = 10.$$

Sample Space and Events

The set of all possible outcomes of any random experiment is known as sample space. Any subset of a sample space is called an event. For example, if S is a sample space when a fair die is rolled then the event may be the occurrence of: an even number, an odd number, a number greater than 5, a number less than 4, a number divisible by 3 or a number greater than 6. Events are usually denoted by capital letters. For the sample space $S = \{1, 2, 3, 4, 5, 6\}$, the events defined as above are:

$A = \{2, 4, 6\}$, $B = \{1, 3, 5\}$, $C = \{6\}$, $D = \{1, 2, 3\}$, $E = \{3, 6\}$ and $F = \{\}$. Here F is a null event as the occurrence of 7 is not possible when a fair die is rolled.

4.2 PROBABILITY

A probability is a numeric measure of the chance that an event occurs. If A is an event then probability of A is denoted by $P(A)$. The axioms of probabilities are:

1. Probability of an event always lie in the range 0 to 1, that is, $0 \leq P(A) \leq 1$.
2. $P(S) = 1$
3. $P(\phi) = 0$

Interpretation of Probability

If A is the event then the following interpretations can be made.

- If $P(A)$ is close to 1 then the event A is very likely to occur.
- If $P(A)$ is close to 0 then the event A is unlikely to occur.
- If $P(A) = 1$ then the event A is certain to occur.
- Other values of $P(A)$ indicate on an increasing scale how likely event A is to occur.

Assigning Probabilities

There are three general methods for assigning probabilities to events.

1. The classical method
2. The relative frequency method
3. The subjective method

The Classical Method

If the sample space S for an experiment consists of n outcomes which are equally likely, (that is, the chance of occurrence of each of its element is same), and the event of interest A consists of m of these outcomes, then this method defines the probability that event A will occur as:

$$P(A) = N(A)/N(S) = m/n$$

Here $N(A)$ stands for number of sample points in the event A.

Example 4.2.1 When a fair die is rolled, what is the probability that (i) 6 occurs (ii) an even number occurs.

Solution: Here the sample space $S = \{1, 2, 3, 4, 5, 6\}$. Let A be the event that 6 occurs and B be the event that an even number occurs then $A = \{6\}$ and $B = \{2, 4, 6\}$. Now by definition, since every sample point of sample space S has same chance of occurrence, hence,

$$P(A) = N(A)/N(S) = 1/6 \text{ and } P(B) = N(B)/N(S) = 3/6 = 1/2.$$

Example 4.2.2 A sample of fifty electric components of same type is chosen at random from a lot. It was observed that 5 of them were defective. What is the probability that a non-defective component was chosen?

Solution: Let A be the event that a component chosen is non-defective. Since the number of non-defective components is 45 and every component has the same chance of being chosen, hence $P(N) = N(A)/N(S) = 45/50 = 0.9$.

Example 4.2.3 Six resistors of 50Ω (Type A) and four resistors of 100Ω (Type B) are placed in a box. If six resistors are chosen at random what is the probability that three of type A and three are of type B resistors.

Solution: Total number of possible out comes in the sample space S is

$$N(S) = {}^{10}C_6 = \frac{10!}{6!(10-6)!} = 210.$$

Let E represent the event that three resistors of type A and three of type B are selected.

Then number of sample points in E will be $N(E) = {}^6C_3 \cdot {}^4C_3 = 80$.

Thus, $P(E) = N(E)/N(S) = 80/210 = 0.38$.

Note that in computation of sample points in E we have used multiplication rule of counting.

Relative Frequency Method

The classical method of finding the probability of an event has a drawback that each sample point of sample space is equally likely, that is, the chance of occurrence of each sample point is same. There are cases where the outcomes of an experiment are not equally likely. In such situations, an alternative method known as “Relative Frequency” method is employed to find the probability of an event. To understand the method, let us consider an example.

Example 4.2.4 The following data shows the sale of hard disks of different size in a week.

HD Make	40 GB	80 GB	100 GB	160 GB	200 GB
No. of Sales	4	9	12	20	05

Draw the table showing the probability of each HD to be sold on any particular day.

Solution: We observe that sample space contains 50 HD and on different days there is a different number of sales. Hence each HD to be sold is not equally likely. Thus we use “Relative Frequency” method to find the probability of each category. This is shown in the following table.

HD Make	Frequency	Relative frequency (probability)
40 GB	4	$4/50 = 0.08$
80 GB	9	$9/50 = 0.18$
100 GB	12	$12/50 = 0.24$
160 GB	20	$20/50 = 0.4$
200 GB	5	$5/50 = 0.1$
Sum	50	1.0

The Subjective Method

In many situations it is even not possible to use above two methods to find the probability when no data is given. In such cases probabilities have to be computed on the basis of person’s belief. The degree of belief will be based on the scale from 0 to 1 or 0% to 100%. If the person is an expert in the given situation then he can give good estimates of probabilities. Probabilities found using this method may not be same for other experts as this method of finding the probabilities is based on personal belief.

For example, a management of Mobile Phone Company wishes to launch a new mobile in the market with more features than the mobiles that are available in the market. The Managing Director is asking the chance that this new product will get a place in the market from the three Directors of the company.

Reply of the 1st Director may be that there is a 90% chance that this new product will be liked by the people. The 2nd Director is only 80% confident and the 3rd is 85% confident about the sale of this new product.

Here we see that there is no formula under which these probabilities are calculated but it is the personal belief of three Directors. However, it may be noted that their personal belief depends on their experience. Thus without having any experience, the subjective method of probability is misleading as well as meaningless.

Laws of Probability

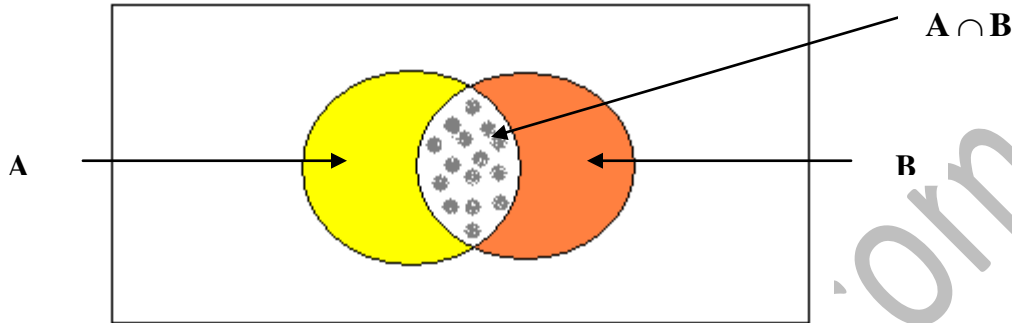
There are situations where simple definition of probability may not help to find the probability of an event. Hence, some laws have been defined to compute the probability of such events.

Additional Law of Probability

Let A and B are two events of sample space S, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof: Consider the Venn diagram as shown in figure below.



Let $N(A) = m_1$, $N(B) = m_2$ and $N(A \cap B) = m_3$. Then $N(A \cup B) = m_1 + m_2 - m_3$. Thus,

$$\begin{aligned} P(A \cup B) &= \frac{N(A \cup B)}{N(S)} = \frac{m_1 + m_2 - m_3}{n} = \frac{m_1}{n} + \frac{m_2}{n} - \frac{m_3}{n} \\ &= P(A) + P(B) - P(A \cap B) \end{aligned}$$

Corollary: If events A and B are disjoint then, $(A \cap B) = \phi$ so that $P(A \cap B) = P(\phi) = 0$

In this case,

$$P(A \cup B) = P(A) + P(B)$$

Example 4.2.5 Eight male and seven female applicants possessing the similar qualification apply for a job. Management wishes to select three of them at random. What is the probability that (i) all male candidates will be selected (ii) at least one female candidate will be selected.

Solution: The total number of applicants is 15. If three of them are to be selected then

possible numbers of ways are: $C_3^{15} = \frac{15!}{3!(15-3)!}$

Thus $N(S) = 455$ sample points.

(i) If A is the event that all three selected candidates are male then A contains

$$C_3^8 = \frac{8!}{3!(8-3)!} = 56 \text{ sample points. Hence } P(A) = 56/455 = 0.123.$$

(ii) Let B be the event that at least one female candidate is selected. This event is the union of three sub-events, that is;

1 female, 2 male OR 2 females, 1 male OR 3 female, no male

The number of ways this can happen is computed by using both multiplication law of permutation and addition law of probability as:

$$N(B) = C_1^7 \cdot C_2^8 + C_2^7 \cdot C_1^8 + C_3^7 \cdot C_0^8 = 196 + 168 + 35 = 399 \text{ sample points}$$

Thus,

$$P(B) = N(B)/N(S) = 399/455 = 0.877.$$

Example 4.2.6 Three manufacturers produce HD in three types. The weekly production level is shown as under.

Manufacture	Type of HD			Total
	100 GB	200 GB	300 GB	
M1	1000	1500	2000	4500
M2	1500	1000	2500	5000
M3	2500	2500	1000	6000
Total	5000	5000	5500	15,500

A hard disk is chose at random. What is the probability that either

- (i) it was manufactured by M1 or M3
- (ii) it was manufactured by M2 or it was 300 GB hard disk
- (iii) it was 100 GB or 200 GB hard disk.

Solution: The required probabilities are:

$$(i) P(M_1 \cup M_3) = P(M_1) + P(M_3) = (4500 + 6000) / 15,500 = 0.678$$

$$(ii) P(M_2 \cup D_{300}) = P(M_2) + P(D_{300}) - P(M_2 \cap D_{300}) = (5000 + 5500 - 2500) / 15,500 = 0.516$$

$$(iii) P(D_{100} \cup D_{200}) = P(D_{100}) + P(D_{200}) = (5000 + 5000) / 15,500 = 0.645$$

Example 4.2.7 An integer is chosen from first 200 integers 1, 2, ..., 200. An integer is chosen at random. What is the probability that the integer chosen at random is either divisible by 6 or by 8 or by both.

Solution: Let S be the sample space that contains integers from 1 to 200. Let A be the event that integer is divisible by 6 and B be the event that it is divisible by 8. Then

$$S = \{1, 2, \dots, 200\}, A = \{6, 12, \dots, 198\}, B = \{8, 16, \dots, 200\}.$$

The set of integer which are divisible by 6 as well as 8 is

$$(A \cap B) = \{24, 48, 72, 96, 120, 144, 168, 192\}$$

$$\text{Now } N(A) = 200/6 = 33, N(B) = 200/8 = 25, N(A \cap B) = 8.$$

Hence the probability that an integer

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = (33 + 25 - 8) / 200 = 0.25$$

Example 4.2.8 Three types of bulbs A, B, and C are produced in an electric company. Due to their lives, bulb A is twice as likely to be purchased as bulb B and bulb B is twice as likely to be purchased as bulb C. A bulb is chosen at random, what is the probability that either bulb A or bulb B will be purchased by a person.

Solution: Let p be the probability that bulb C is bought. Then as given

$$P(B) = 2p \text{ and } P(A) = 2(2p) = 4p. \text{ Since sum of all probabilities is equal to 1, hence}$$

$$p + 2p + 4p = 1 \quad \text{or} \quad 7p = 1 \text{ or } p = 1/7$$

$$\text{Thus, } P(A) = 4/7, P(B) = 2/7 \text{ and } P(C) = 1/7.$$

Hence the probability that either bulb A or B will be purchased is:

$$P(A \cup B) = P(A) + P(B) = \frac{4+2}{7} = 6/7$$

Complement Law

The complement of an event A is the event that A does not occur. It is usually denoted by A^c and contains all experimental outcomes of sample space which are not in A.

$$\text{Since } (A \cup A^c) = S, \text{ therefore, } P(A \cup A^c) = P(S) = 1$$

$$\text{Or, } P(A) + P(A^c) = 1 \text{ thus, } P(A^c) = 1 - P(A)$$

Example 4.2.9 The probability that a sample chosen from a population that is going through inspection process will be passed is $1/3$. What is the probability that sample fails to pass through inspection team?

Solution: Let A be the event that inspection team passes the sample, then $P(A) = 1/3$.

Thus, the probability that it will be failed to pass through the inspection team is

$$P(A^c) = 1 - P(A) = 1 - 1/3 = 2/3$$

Conditional Probability

In many circumstances the probability of an event A occurring may be influenced by the information that another event B has already occurred. For example, the probability that 2 occurs when a fair die is rolled is $1/6$ keeping in mind that sample space $S = \{1, 2, 3, 4, 5, 6\}$. Now if the die is rolled and we are told that an even number has occurred then immediately our mind would accept that new sample space $S = \{2, 4, 6\}$. At this instant if we are asked: What is the probability that 2 has occurred? Our reply would certainly be $1/3$. Such probability of an event is known as conditional probability. Thus conditional probability of an event occurs in a reduced sample space and is usually denoted by $P(A|B)$ and is stated as “Probability of an event A given B or probability of an event A given that an event B has occurred”. In the above example, we write this as:

$$P(2 | \text{given that an even number has occurred}) = 1/3$$

Here A is the event that number 2 occurs and B is the event that an even number has occurred, that is, $P(A|B) = 1/3$. Let us look at this more precisely:

According to the given conditions: $A = \{2\}$, $B = \{2, 4, 6\}$, $A \cap B = \{2\}$. Now with regard to sample space

$$S = \{1, 2, 3, 4, 5, 6\}: P(A) = 1/6, P(B) = 3/6 \text{ and } P(A \cap B) = 1/6.$$

$$\text{Now, } \frac{P(A \cap B)}{P(B)} = \frac{1}{6} \div \frac{3}{6} = \frac{1}{6} \times \frac{6}{3} = \frac{1}{3} = P(A|B)$$

Thus conditional probability of an event A given that event B has occurred (or simply given B) is:

$$P(A|B) = P(A \cap B) / P(B)$$

The idea of conditional probability provides the power of re-evaluating the probability of an event in the light of additional information, that is, when it is known that another event has occurred. The probability $P(A|B)$ is an *updating* of $P(A)$ based on the knowledge the given event B has occurred.

Example 4.2.10 The following contingency table shows the population of adults in a small village who have completed their graduation. The table is categorized according to sex and status as follows:

	Employed	Unemployed	Total
Boys	460	40	500
Girls	140	260	400
Total	600	300	900

An individual is selected for country's tour, what is the probability that unemployed girl is selected.

Solution: Let G be the event that girl is selected and U be the event that unemployed person is selected. Now we have to find $P(G|U)$. By the definition of conditional

$$\text{probability: } P(G|U) = \frac{P(G \cap U)}{P(U)} = \frac{260}{900} \div \frac{300}{900} = \frac{260}{300} = \frac{13}{15}$$

Example 4.2.11 The probability that a regular scheduled flight departs on time is 0.83; the probability that it arrives on time is 0.82; and the probability that it departs and arrives on time is 0.78. Find the probability that on a given day the regular flight (i) arrives on time given that it departs on time (ii) departs on time given that it has arrived on time.

Solution: Let D be the event that flight departs on time, A be the event that it arrives on time. Then as given: $P(D) = 0.83$, $P(A) = 0.82$ and $P(D \cap A) = 0.78$.

(i) Here we require $P(A | D) = P(A \cap D) / P(D) = 0.78 / 0.83 = 0.94$.

(ii) Here we require $P(D | A) = \frac{P(A \cap D)}{P(A)} = \frac{0.78}{0.82} = 0.95$

Example 4.2.12 In a box there are 100 resistors. They are categorized as under:

	Tolerance		
	5%	10%	Total
Resistance			
22 Ω	10	14	24
47 Ω	28	16	44
100 Ω	24	08	32
Total	62	38	100

A resistor is chosen at random where each resistor is equally likely. Let 'A' be the event that "47 Ω " resistor is chosen, 'B' be the event that "5% tolerance" resistor is chose and 'C' be the event that a "100 Ω " resistor is chosen. Find (i) $P(A|B)$ (ii) $P(A|C)$ and (iii) $P(B|C)$.

Solution: By definition (i) $P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{28 / 100}{62 / 100} = \frac{28}{62} = \frac{14}{31}$

(ii) $P(A | C) = \frac{P(A \cap C)}{P(C)} = \frac{0 / 100}{32 / 100} = \frac{0}{32} = 0$ (iii) $P(B | C) = \frac{P(B \cap C)}{P(C)} = \frac{24 / 100}{32 / 100} = \frac{24}{32} = \frac{3}{4}$

Note: $A \cap C = \phi$ or $P(A \cap C) = P(\phi) = 0$

Multiplication Law for Probability

The multiplication rule is based on the definition of conditional probability and is used to find $P(A \cap B)$. Rearranging the equation of conditional probability, we have:

$$P(A \cap B) = P(A | B).P(B) \quad \text{or} \quad P(A \cap B) = P(B | A).P(A)$$

Example 4.2.13 A box of 200 resistors contains 12 that are defective. Two resistors are chosen at random without replacement. What is the probability that both are defective?

Solution: Let A be the event that 1st resistor is defective and B be the event that 2nd resistor is also defective. Then,

$P(A) = 12/200$. Now when the 2nd resistor is chosen at random then the box contains 199 resistors with only 11 defectives because the 1st resistor is taken away or not replaced.

Hence the 2nd draw depends on the 1st draw. Thus $P(B|A) = 11/199$. Hence the probability that both resistors are defective is:

$$P(A \cap B) = P(A)P(B|A) = (12/200).(11/199) = 0.0033.$$

Example 4.2.14 A student takes a train to get to a school and to a class. The probability that student will arrive at class on time is 0.95 provided the train is on time. If the train is known to be on schedule 70% of the time, what is the probability that student will be on time to the class?

Solution: Let C be the event that student arrives in class and T be the event that train arrives on time. We require to find the “probability that student will be on time to the class and train is on schedule”. This is given by

$$P(C \cap T) = P(C|T).P(T) = 0.95(0.7) = 0.665$$

This means there is 66.5% chance that “Train is on schedule and student reaches the class on time”.

Independent Events

Conditional probability not only allows finding the probability of an event in a reduced sample space but it also helps to see whether or not two events occur independently.

The two events A and B are independent if and only if:

$$P(A|B) = P(A) \quad \text{or} \quad P(B|A) = P(B)$$

Multiplication Law of Probabilities for Independent Events

If events A and B are independent then multiplication law of probability can be extended to another law known as multiplication law of probabilities for independent events which, is given by:

$$P(A \cap B) = P(A)P(B)$$

Example 4.2.15 A fair coin is tossed twice. Let A be the event that tail occur on the first toss and B be the event that head occur on the second toss. Show that events A and B are independent.

Solution: Here $A = \{(T, H), (T, T)\}$, $B = \{(H, H), (T, H)\}$ and $P(A \cap B) = \{(T, H)\}$

Now $S = \{(H, H), (H, T), (T, H), (T, T)\}$. Hence,

$$P(A) = 2/4 = 1/2, P(B) = 2/4 = 1/2 \text{ and } P(A \cap B) = 1/4$$

We see that $P(A \cap B) = P(A)P(B)$, that is, $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Note: If events A and B are independent events then so are the events: A and B^c, A^c and B, A^c and B^c.

Example 4.2.16 The probability that an ambulance is available when called is 0.80 and that fire engine is available when needed is 0.9. If both ambulance and fire engine work independently, what is the probability that

- both of them are available when called
- both are not available when called
- ambulance is available but not the fire engine
- fire engine is available but not the ambulance

Solution: Let A be the event that “Ambulance” is available and F be the event that “Fire engine” is available when needed. As given,

$$P(A) = 0.8 \text{ and } P(F) = 0.9 \text{ then } P(A^c) = 1 - P(A) = 1 - 0.8 = 0.2$$

$$\text{Also, } P(F^c) = 1 - P(F) = 1 - 0.9 = 0.1. \text{ We are required to find:}$$

- (i) $P(A \cap F) = P(A)P(F) = 0.8 \times 0.9 = 0.72$
(ii) $P(A \cap F^c) = P(A)P(F^c) = 0.2 \times 0.1 = 0.02$
(iii) $P(A \cap F^c) = P(A)P(F^c) = 0.8 \times 0.1 = 0.08$
(iv) $P(A^c \cap F) = P(A^c)P(F) = 0.2 \times 0.9 = 0.18$

Example 4.2.17 A bag contains 4 white balls and 3 black balls. A second bag contains 3 white balls and 5 black balls. A ball is drawn at random from the 1st bag and placed in the 2nd bag. Now a ball is drawn from the 2nd bag what is the probability that it is a black ball?

Solution: Let B_1 be the event that Black ball is drawn from the 1st bag.

Let W_1 be the event that Black ball is drawn from the 1st bag, and

Let B_2 be the event that Black ball is drawn from the 2nd bag.

We are interested to find:

$$P\{(B_1 \cap B_2) \cup (W_1 \cap B_2)\} = P(B_1 \cap B_2) + P(W_1 \cap B_2) \quad [\text{By addition law of probability}]$$

Using multiplication law of probability,

$$P\{(B_1 \cap B_2) \cup (W_1 \cap B_2)\} = P(B_1).P(B_2 | B_1) + P(W_1).P(B_2 | W_1) = \frac{3}{7} \cdot \frac{6}{9} + \frac{4}{7} \cdot \frac{5}{9} = \frac{38}{63}.$$

Total Theorem (Rule of Elimination)

If the events $A_1, A_2, A_3, \dots, A_n$, constitute the partition of sample space S where $A_i \neq \phi$ for $i = 1, 2, 3, \dots, n$, then for any other event B of S :

$$P(B) = \sum_{i=1}^n P(A_i \cap B) = \sum_{i=1}^n P(A_i).P(B | A_i)$$

Proof: Consider Venn diagram as shown in figure.

We observe that event B has some elements in common with each A_i . Moreover, each part is mutually exclusive (disjoint). Thus event B is union of:

$$B = (A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_n \cap B). \text{ Thus}$$

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_n \cap B) = \sum_{i=1}^n P(A_i \cap B) = \sum_{i=1}^n P(B | A_i).P(A_i).$$

This proves the result.

Example 4.2.18 Three members of a club nominated for the office of president. The probability that Mr. Ali, will be elected is 0.3; the probability that Mr. Bilal will be selected is 0.5; and the probability that Mr. Chohan will be selected is 0.2. Given that Mr. Ali or Mr. Bilal or Mr. Chohan is selected, the probability that they will increase the membership fee is 0.8, 0.2 and 0.4 respectively. What is the probability that there will be increase in membership fee?

Solution:

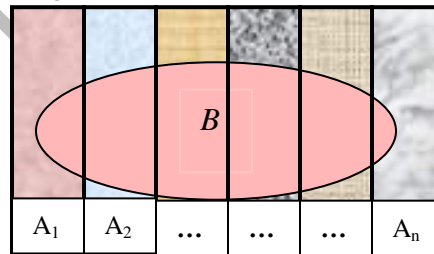
Let A_1 be the event that Mr. Ali is selected or $P(A_1) = 0.3$

Let A_2 be the event that Mr. Bilal is selected or $P(A_2) = 0.5$

Let A_3 be the event that Mr. Chohan is selected or $P(A_3) = 0.2$

Let B be the event that membership fee is increased no matter who so ever is selected.

We are required to find $P(B)$. Now, by total law of probability,



$$P(B) = \sum_{i=1}^3 P(B | A_i) \cdot P(A_i) = P(B | A_1) \cdot P(A_1) + P(B | A_2) \cdot P(A_2) + P(B | A_3) \cdot P(A_3)$$

Here, $P(B|A_1) = 0.8$, $P(B|A_2) = 0.2$, $P(B|A_3) = 0.4$. Thus,

$$P(B) = 0.8(0.3) + 0.2(0.5) + 0.4(0.2) = 0.42.$$

It may be noted that $P(A_1) + P(A_2) + P(A_3) = 1$

Now instead of asking for $P(B)$ by the rule of elimination, suppose that we now consider the problem of finding the conditional probability $P(A_1|B)$, that is, what is the probability that Mr. Ali is selected given that he would increase the membership fee. Question of this type can be answered by using the Bay's theorem.

Example 4.2.19 A binary symmetric channel (BSC) error probability is P_e . The probability of transmitting 1 is Q , and that of transmitting 0 is $1 - Q$. Determine the probability of receiving 1 and 0.

Solution: The probability error means if 1 is transmitted but is received as 0 and vice versa. This probability is denoted by P_e .

Now let x and y be the transmitted bits and received bits, respectively. As given:

$$P(0_y|1_x) = P(1_y|0_x) = P_e \text{ and } P(0_y|0_x) = P(1_y|1_x) = 1 - P_e. \text{ Also, } P_x(1) = Q \text{ and } P_x(0) = 1 - Q$$

We need to find $P_y(1)$ and $P_y(0)$.

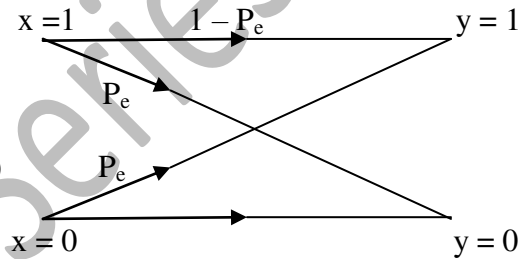
Now, bit 1 can be received in two ways:

(i) when 1 is transmitted & 1 is received

(ii) when 0 is transmitted & 1 is received

$$\text{Thus, } P_y(1) = P(1_y|1_x) \cdot P_x(1) + P(1_y|0_x) \cdot P_x(0)$$

$$= Q \cdot (1 - P_e) + (1 - Q) \cdot P_e = Q + P_e - 2QP_e.$$



$$\text{Similarly, } P_y(0) = P(0_y|1_x) \cdot P_x(1) + P(0_y|0_x) \cdot P_x(0) = P_e \cdot Q + (1 - Q)(1 - P_e) \\ = 1 - Q - P_e + 2QP_e$$

You may observe that $P_y(0) \neq P_y(1)$ because transmission of bits occur with an error.

Example 4.2.20 Over a certain Binary Communication Channel, the symbol 0 is transmitted with probability 0.4 and that of 1 is 0.6. Given that $P(\epsilon|0) = 10^{-6}$ and $P(\epsilon|1) = 10^{-4}$ where ϵ stands for error. Determine $P(\epsilon)$.

Solution: $P(\epsilon)$ means that the error occurs in transmitting the bits 0 and 1. Now the error occurs when 0 is transmitted but 1 is received and when 1 is transmitted but 0 is received. Now, $P(\epsilon) = P(\epsilon|0) \cdot P(0) + P(\epsilon|1) \cdot P(1) = 0.4 \times 10^{-6} + 0.6 \times 10^{-4} = 0.604 \times 10^{-4}$.

Note: $P(\epsilon|0) = 10^{-6}$ means that on the average one of 1 million received 0s will be detected erroneously. Similarly, $P(\epsilon|1) = 10^{-4}$ means that on the average one of ten thousand received 1s will be detected erroneously. $P(\epsilon) = 0.604 \times 10^{-4}$ means on the average, one of $1/(0.604 \times 10^{-4}) = 16,556$ bits (0's or 1's) are received in error.

Bay's Theorem

If the events $A_1, A_2, A_3, \dots, A_n$, constitute the partition of sample space S where $A_i \neq \phi$ for $i = 1, 2, 3, \dots, n$, then for any other event B of S :

$$P(A_i | B) = \frac{P(A_i) \cdot P(B | A_i)}{P(B)} = \frac{P(A_i) \cdot P(B | A_i)}{\sum_{k=1}^n P(A_k) \cdot P(B | A_k)} \quad i = 1, 2, 3, \dots, n$$

Proof: By multiplication law of probability:

$$P(A_i \cap B) = P(A_i | B) \cdot P(B) \quad (i) \quad \text{and} \quad P(B \cap A_i) = P(B | A_i) \cdot P(A_i) \quad (ii)$$

Equating (i) and (ii), we obtain:

$$P(A_i | B) \cdot P(B) = P(B | A_i) \cdot P(A_i)$$

$$\text{Thus, } P(A_i | B) = \frac{P(B | A_i) \cdot P(A_i)}{P(B)} = \frac{P(B | A_i) \cdot P(A_i)}{\sum_{k=1}^n P(B | A_k) \cdot P(A_k)}. \text{ This proves the theorem.}$$

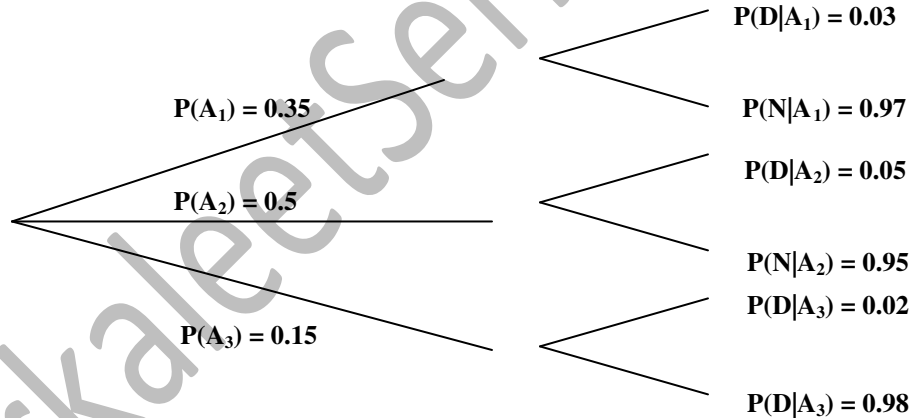
Example 4.2.21 With reference to Example 20, if someone is considering joining the club but delays his/her decision to find out that fees have been increased, what is the probability that Mr. Ali was elected as president of the club?

Solution: We are required to find:

$$P(A_1 | B) = \frac{P(B | A_1) \cdot P(A_1)}{P(B)} = \frac{0.8(0.3)}{0.42} = 0.57$$

Example 4.2.22 In a HD manufacturing factory, machines A_1 , A_2 , and A_3 manufacture 35%, 50% and 15% of the total output. Due to some mechanical problem, Machines A_1 , A_2 and A_3 produce 3%, 5% and 2% defective HDs. All HDs are mixed and then one of them is selected at random and found defective. What is the probability that it was produced by machine A.

Solution: The problem may be well understood by using the tree diagram.



Here D is the event that defective HD is chosen and N is the event that non-defective bolt is chosen. Now by Bay's theorem:

$$\begin{aligned} P(A_1 | D) &= \frac{P(D | A_1) \cdot P(A_1)}{P(D | A_1) \cdot P(A_1) + P(D | A_2) \cdot P(A_2) + P(D | A_3) \cdot P(A_3)} \\ &= \frac{0.03(0.35)}{0.03(0.35) + 0.05(0.5) + 0.02(0.15)} = 0.24 \end{aligned}$$

Example 4.2.23 Consider a binary communication system which utilizes two pulses P_1 and P_0 to represent "1" and "0". At the end of noisy channel, there is a receiver that prints the symbol "p" and "q". An experiment is performed and it was observed that:

- (i) When P_0 is transmitted 80% of the time, output is "y"
- (ii) When P_1 is transmitted 60% of the time, output is "x"
- (iii) The typical input sequences are composed of 80% P_0 pulses and 20% P_1 pulses.

Find the probability of error.

Solution: The above information can be summarized as under:

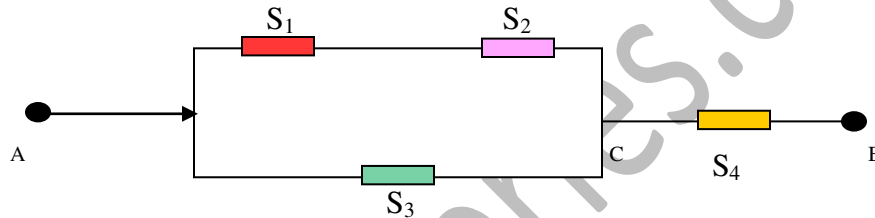
- | | |
|-----------------------|------------------|
| i. $P(y P_0) = 0.8,$ | $P(x P_0) = 0.2$ |
| ii. $P(x P_1) = 0.6,$ | $P(y P_1) = 0.6$ |
| iii. $P(P_0) = 0.8,$ | $P(P_1) = 0.2$ |

If we denote P_e as the probability of error, then: $P_e = P(P_0 \cap p) + P(P_1 \cap q)$.

This means that when P_0 is transmitted then instead of “y”, “x” becomes the output. Similarly, when P_1 is transmitted then instead of “x”, “y” becomes the output. Hence,

$$P_e = P(P_0 \cap x) + P(P_1 \cap y) = P(P_0).P(x | P_0) + P(P_1).P(y | P_1) = 0.8 \times 0.2 + 0.2 \times 0.6 = 0.28.$$

Example 4.2.24 A switching circuit is given below. Switches S_1 , S_2 , S_3 and S_4 open and close at random independently. The probability that any switch is closed at a given time is “p”. What is the probability that at any time there is at least one closed switch from the path A to path B.



Solution: Since we are asked to find the probability that at least one switch is closed, we use the complement law of probability as under:

$$P(\text{path closed}) = 1 - P(\text{path open})$$

Now let us find the probability that path is closed from A to C.

$P(\text{path closed from A to C})$

$$= 1 - P(\text{path open from A to C})$$

$$= 1 - P(\text{path open in the top branch}) \cdot P(\text{path open in the bottom branch})$$

$$= 1 - [1 - P(\text{path closed in the top branch})] \cdot [1 - P(\text{path closed in the bottom branch})]$$

$$= 1 - (1 - p^2) \cdot (1 - p) = 1 - [1 - p^2 - p + p^3] = p + p^2 - p^3. \text{ Thus,}$$

$$P(\text{path closed from A to B}) = P(\text{path closed from A to C}) \cdot P(\text{path closed from C to B}) \\ = (p + p^2 - p^3) \cdot p = p^2 + p^3 - p^4.$$

Example 4.2.25 An elementary binary communication system (BCS) consists of a transmitter that sends one of two possible symbols 1 or 0 over a channel to a receiver. The channel occasionally causes errors to occur so that a 1 shows up at the receiver as a 0 and vice versa. Let us define the following events:

B_1 : The symbol before the channel is 1; with $P(B_1) = 0.6$.

B_2 : The symbol before the channel is 0; with $P(B_2) = 0.4$.

A_1 : The symbol after the channel is 1; and

A_2 : The symbol after the channel is 0

Also let $P(A_1|B_1) = 0.9$; [the probability that 1 is received given that 1 is transmitted].

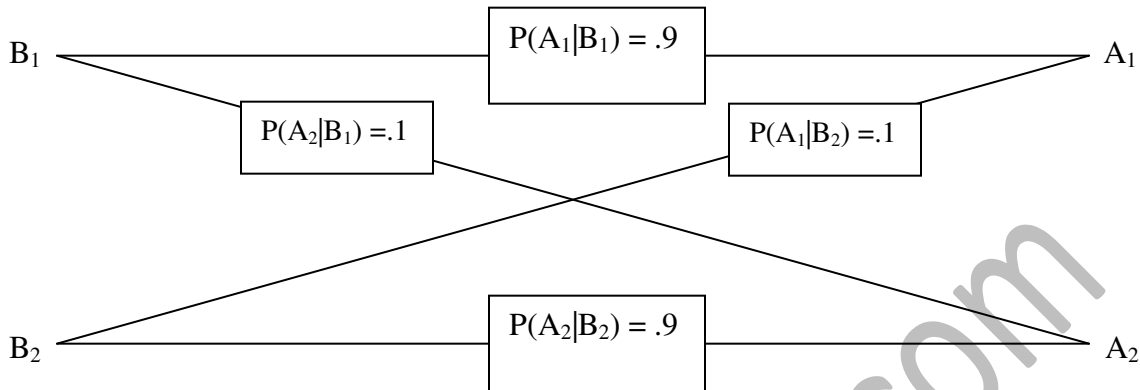
$P(A_2|B_1) = 0.1$; [the probability that 0 is received given that 1 is transmitted].

$P(A_1|B_2) = 0.1$; [the probability that 1 is received given that 0 is transmitted].

$P(A_2|B_2) = 0.9$; [the probability that 0 is received given that 0 is transmitted].

Find $P(A_1)$, $P(A_2)$, $P(B_1|A_1)$, $P(B_1|A_2)$, $P(B_2|A_1)$ and $P(B_2|A_2)$.

Solution: Refer the diagram below:



Using “Total Theorem”:

$$P(A_1) = P(A_1|B_1).P(B_1) + P(A_1|B_2).P(B_2) = 0.9(0.6) + 0.1(0.4) = 0.58 \quad (1)$$

This is the probability that received digit is 1 that is transmitted through channel B_1 or B_2 .

Similarly,

$$P(A_2) = P(A_2|B_1).P(B_1) + P(A_2|B_2).P(B_2) = 0.1(0.6) + 0.9(0.4) = 0.42 \quad (2)$$

This is the probability that received digit is 0 that is transmitted through channel B_1 or B_2 .

Now, using Bay’s Theorem

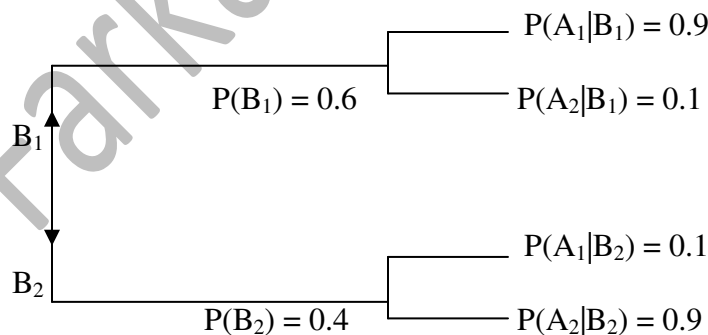
$$P(B_1 | A_1) = \frac{P(A_1 | B_1).P(B_1)}{P(A_1 | B_1).P(B_1) + P(A_1 | B_2).P(B_2)} = \frac{0.9(0.6)}{0.58} = 0.931 \quad [\text{Using (1)}]$$

$$\text{Here, } P(A_1 | B_1).P(B_1) + P(A_1 | B_2).P(B_2) = P(A_1) = 0.58$$

This is the probability that symbol ‘1’ was released given that same was received.

Note: $P(A_1|B_1)$ means the probability that digit 1 is received given that same was transmitted which is 0.9. On the other hand, $P(B_1|A_1)$ means the probability that digit 1 is transmitted given that same was received which is 0.931.

The tree diagram is shown below:



Similarly,

$$P(B_2 | A_2) = \frac{P(A_2 | B_2).P(B_2)}{P(A_2 | B_1).P(B_1) + P(A_2 | B_2).P(B_2)} = \frac{0.9(0.4)}{0.42} = 0.857 \quad [\text{Using (2)}]$$

$$\text{Here, } P(A_2 | B_1).P(B_1) + P(A_2 | B_2).P(B_2) = P(A_2) = 0.42$$

This is the probability that symbol '0' was released given that same was received.

$$\text{Now, } P(B_1 | A_2) = \frac{P(A_2 | B_1).P(B_1)}{P(A_2 | B_1).P(B_1) + P(A_2 | B_2).P(B_2)} = \frac{0.1(0.6)}{0.42} = 0.143 \quad [\text{Using (2)}]$$

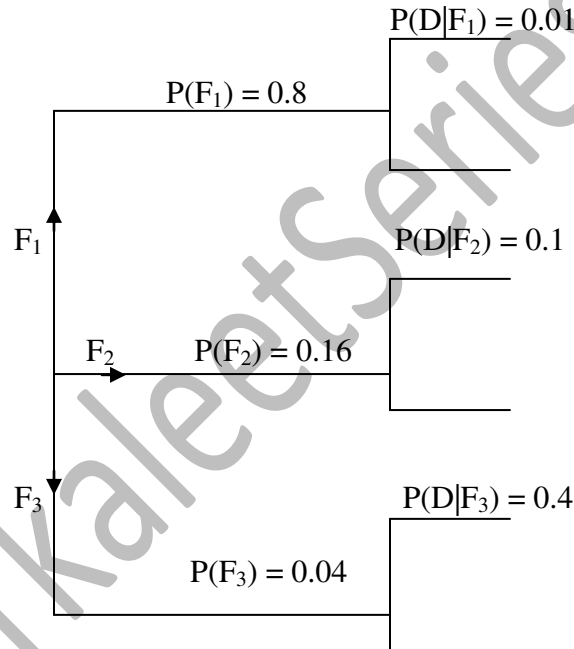
This is the probability that symbol '1' was released given that symbol '0' was received.

$$\text{Now, } P(B_2 | A_1) = \frac{P(A_1 | B_2).P(B_2)}{P(A_1 | B_1).P(B_1) + P(A_2 | B_2).P(B_2)} = \frac{0.1(0.4)}{0.58} = 0.069 \quad [\text{Using (1)}]$$

This is the probability that symbol '0' was released given that symbol '1' was received.

Example 4.2.26 A communication system always encounters one of three possible interference waveforms F_1 , F_2 and F_3 . The probabilities of encountering interference are 0.8, 0.16 and 0.04, respectively. The communication fails with probabilities 0.01, 0.1 and 0.4 when encounters F_1 , F_2 and F_3 , respectively. Given that the system has failed, find the probability that the failure is a result of F_1 , F_2 and F_3 , respectively.

Solution: Let D be the event of failure of interference F_1 , F_2 , and F_3 .



Using the tree diagram and using Bay's theorem

$$\begin{aligned} P(F_1 | D) &= \frac{P(D | F_1).P(F_1)}{P(D | F_1).P(F_1) + P(D | F_2).P(F_2) + P(D | F_3).P(F_3)} \\ &= \frac{0.1(0.8)}{0.1(0.8) + 0.16(0.1) + 0.04(0.4)} = 0.27 \end{aligned}$$

$$\begin{aligned} P(F_2 | D) &= \frac{P(D | F_2).P(F_2)}{P(D | F_1).P(F_1) + P(D | F_2).P(F_2) + P(D | F_3).P(F_3)} \\ &= \frac{(0.16)(0.1)}{(0.1)(0.8) + (0.16)(0.1) + (0.04)(0.4)} = 0.4 \end{aligned}$$

$$\begin{aligned}
 P(F_3 | D) &= \frac{P(D | F_3) \cdot P(F_3)}{P(D | F_1) \cdot P(F_1) + P(D | F_2) \cdot P(F_2) + P(D | F_3) \cdot P(F_3)} \\
 &= \frac{(0.04)(0.4)}{(0.1)(0.8) + (0.16)(0.1) + (0.04)(0.4)} = 0.4
 \end{aligned}$$

WORKSHEET 04

1. In a medical study, patients are classified in 8 ways according to whether they have blood type AB^+ , AB^- , A^+ , A^- , B^+ , B^- , O^+ , O^- , and also according to whether their blood pressure is low, normal, or high. Find the number ways in which a patient can be classified?
2. A college freshman must take a science course, a humanities course, and a mathematics course. If she may select any of 6 science courses, any of 4 humanities, and any of 4 mathematics courses, in how many ways can she arrange her program?
3. A developer of a new subdivision offers a prospective home buyer a choice of 4 designs, 3 different heating systems, a garage or carport, and a patio or screened porch. How many different plans are available to this buyer?
4. A contractor wishes to build 9 houses, each different in design. In how many ways can he place these homes on a street if 6 lots are on one side of the street and 3 lots are on the opposite sides?
5. Four married couples have bought 8 seats in a row for a concert. In how many different ways can they be seated? (a) With no restrictions? (b) If each couple is to sit together? (c) if all the men sit together to the right of all the women?
6. In how many ways can 3 oaks, 4 pines, and 2 maples be arranged along a property line if one does not distinguish between trees of the same kind?
7. A shipment of 12 television sets contains 3 defective sets. In how many ways can a hotel purchase 5 of these sets and receive at least 2 of the defective sets?
8. Two cards are drawn in succession from a deck without replacement. What is the probability that both cards are greater than 2 and less than 8?
9. The probability that an American industry will locate in Munich is 0.7, the probability that it will locate in Brussels is 0.4, and the probability that it will locate in either Munich or Brussels or both is 0.8. What is the probability that the industry will locate (a) In both cities? (b) In neither city?
10. The probabilities that a service station will pump gas into 0, 1, 2, 3, 4, or 5 or more cars during a certain 30 – minute period are 0.03, 0.18, 0.24, 0.28, 0.10, and 0.17. Find the probability that in this 30 – minute period (a) More than 2 cars receive gas? (b) At most 4 cars receive gas? (c) 4 or more cars receive gas?
11. In the senior year of a high school graduating class of 100 students, 42 studied mathematics, 68 studied psychology, 54 studied history, 22 studied both mathematics and history, 25 studied both mathematics and psychology, 7 studied history but neither

mathematics nor psychology, 10 studied all three subjects, and 8 did not take any of the three. If a student is selected at random, find the probability that

- (a) A person enrolled in psychology takes all three subjects;
- (b) A person not taking psychology is taking both history and mathematics.

12. The probability that an automobile being filled with gasoline will also need an oil change is 0.25; the probability that it needs a new oil filter is 0.40; and the probability that both the oil and filter need changing is 0.14.

- (a) If the oil had to be changed, what is the probability that a new oil filter is needed?
- (b) If a new oil filter is needed, what is the probability that the oil has to be changed?

13. The probability that a doctor correctly diagnoses a particular illness is 0.7. Given that the doctor makes an incorrect diagnosis, the probability that the patient enters a law suit is 0.9. What is the probability that the doctor makes an incorrect diagnosis and the patient sues?

14. A town has two fire engines operating independently. The probability that a specific fire engine is available when needed is 0.96.

- (a) What is the probability that neither is available when needed?
- (b) What is the probability that a fire engine is available when needed?

15. The probability that a person visiting his dentist will have an X – ray is 0.6; the probability that a person who has an X – ray will also have a cavity filled is 0.3; and the probability that a person who has had an X – ray and a cavity filled will also have a tooth extracted is 0.1. What is the probability that a person visiting his dentist will have an X – ray, a cavity filled, and a tooth extracted?

16. An allergist claims that 50% of the patients she tests are allergic to some type of weed. What is the probability that

- (a) Exactly 3 of her next 4 patients are allergic to weeds?
- (b) None of her 4 patients are allergic to weeds?

17. Police plan to enforce speed limits by using radar traps at 4 different locations within the city limits. The radar traps at each of the locations L_1, L_2, L_3 and L_4 are operated 40%, 30%, 20% and 30% of the time, and if a person who is speeding on his way to work has probabilities of 0.2, 0.1, 0.5, and 0.2, respectively, of passing through these locations, what is the probability that he will receive a speeding ticket?

18. A large industrial firm uses 3 local motels to provide overnight accommodations for its clients. From past experience it is known that 20% of the clients are assigned rooms at the Ramada Inn, 50% at the Sheraton, and 30% at the Lakeview Motor Lodge. If the plumbing is faulty in 5% of the rooms at the Ramada Inn, in 4% of the rooms at the Sheraton, and in 8% of the rooms at the Lakeview Motor Lodge, what is the probability that (a) a client will be assigned a room with faulty plumbing? (b) a person with a room having faulty plumbing was assigned accommodations at the Lakeview Motor Lodge?

19. A computer owns 2 cars, 1 a compact and 1 a standard model. About $\frac{3}{4}$ of the time he uses the compact to travel to work, and about $\frac{1}{4}$ of the time the larger car is used. When he uses the compact car, he usually gets home by 5:30 P.M. about 75% of the time; if he uses the standard – sized car, he gets home by 5:30. P.M. about 60% of the time (but he enjoys the air conditioner in the larger car). If he gets home at 5:35 P.M., what is the probability that he used the compact car?

20. There are 100 resistors in a box. The 10 resistors are of 100 ohm's and marked red. Twenty resistors are of 200 ohm's and marked black. Find the probability of drawing either of 100 or 200 ohm's resistors.

21. A lot of 100 semiconductor chips contain 20 that are defective. Three chips are selected at random without replacement. (i) What is the probability that first one is defective? (ii) What is the probability that the 3rd one selected is non-defective given that first and second are defectives.

22. In a semiconductor manufacturing process the contamination on high, medium and low levels is 30%, 20% and 50% respectively. Given that the probability of the chip produced on high-level contamination being defective is 0.10, on medium level is 0.02 and on low level is 0.04, a chip is selected at random and found to be non-defective. What is the probability that it is produced on medium level contamination?

23. The following contingency table provides a cross classification of age group of drivers and accident severity.

(i) Find the marginal probability of total group.

(ii) Find the probability that a person selected at random got minor injury given that his age group is in (30-50).

Show that the event defined in part (ii) is independent or dependent.

Accident Severity		Age Group of Drivers			
		18 -30	30-50	Above 50	Total
	Fatal Accident	12	8	16	36
	Serious Injury	25	14	41	80
	Minor	48	35	70	153
	Total	85	57	127	269

24. A steel plate contains 20 bolts. Assume that the five bolts are not torque to the proper limit. Four bolts are selected at random without replacement to be checked for torque.

i. What is the probability that all four of the selected bolts are torque to the proper limit?

ii. What is the probability that at least one of the selected bolts is not torque to the proper limit?

25. Two methods A and B are available for teaching a certain industrial skill. The failure rate is 20% for A and 10% for B. However B is more expensive and hence is used only 30% of the time. A worker is taught the skill by one of the methods, but fails to learn it correctly. What is the probability that he was taught by method A.

26. In a primary election, there are four candidates for mayor, five candidates for city treasure and two candidates for country attorney.

(a) In how many ways can a voter mark his ballot for all of three of these offices?

(b) In how many ways can a person vote if he exercises his option of not voting for a candidate for any or all of these offices?

26. A multiple choice test contains 15 questions, each permitting a choice of three alternatives. In how many different ways can a student check off her answers to these questions
27. In how many ways can five persons line up to get on a bus? In how many ways can they line up if two of the persons refuse to follow each other?
28. A shipment of 10 television sets includes three that are defective. In how many ways can a hotel purchase four of these sets and receive at least two of the defective sets?
29. Near a certain exit of I-17, the probabilities are 0.23 and 0.24 that a truck stopped at a roadblock will have faulty brakes or badly worn tires. Also the probability is 0.38 that a truck stopped at a roadblock will have faulty brakes and/or badly worn tires. What is the probability that a truck stopped at this roadblock will have faulty brakes as well as badly worn tires?
30. If a person visits his dentist, suppose that the probability that he will have his teeth cleaned is 0.44, the probability that he will have a cavity filled is 0.24, the probability that he will have a tooth extracted is 0.21, the probability that he will have his teeth cleaned and a cavity filled is 0.08, the probability that he will have his teeth cleaned and tooth extracted is 0.11, the probability that he will have cavity filled and tooth extracted is 0.07, and the probability that he will have his teeth cleaned, a cavity filled, and a tooth extracted is 0.03. What is the probability that a person visiting his dentist will have at least one of these things done to him?
31. The probability that the serviceability of a new X-ray machine will be related very difficult, difficult, average, easy, or very easy are, respectively, 0.12, 0.17, 0.34, 0.29, and 0.08. Find the probabilities that the serviceability of the machine will be rated
- (a) difficult or very difficult (b) neither very difficult nor very easy
(c) Average or worse (d) average or better
32. A police department needs new tires for its patrol cars and the probabilities are 0.15, 0.24, 0.03, 0.28, 0.22, and 0.08 that it will buy Uniroyal tires, Goodyear tires, Michelin tires, Goodrich tires, or Armstrong tires. Find the probabilities that it will buy
- (a) Goodyear tires or Goodrich tires (b) Uniroyal, Michelin, or Goodrich tires
(c) Michelin or Armstrong tires (d) Uniroyal, Michelin, General or Goodrich tires
33. Four candidates are seeking a vacancy on a school board. If A is twice as likely to be elected as B, and B and C are given about the same chance of being elected, while C is twice as likely to be elected as D, what are the probabilities that
- (a) C will win (b) A will not win?
34. Two cards are randomly drawn from a deck of 52 playing cards. Find the probability that both cards will be greater than three and less than 8.
35. Given that $P(A) = 0.59$, $P(A) = 0.30$ and $P(A \cap B) = 0.21$, find
- (a) $P(A \cup B)$ (b) $P(A \cap B')$ (c) $P(A' \cup B')$ (d) $P(A' \cap B')$
36. Four married couples living in a certain suburb, the probability that the husband will vote in a school board election is 0.21, the probability that the wife will vote in the election is 0.2, and the probability they will both vote is 0.51. What is the probability that at least one of them will vote?
37. A biology professor has two graduate assistants helping her with her research. The probability that the older of the two assistants will be absent on any given day is 0.08, the probability that the younger of the two assistants will be absent on any given day 0.05,

and the probability that the both will be absent on any given day is 0.02. Find the probabilities that

- a) Either or both of the graduate assistants will be absent on any given day
 - b) at least one of the two graduate assistants will not be absent on any given day
 - c) Only one of the two graduate assistants will be absent on any given day
38. A manufacturer of airplane parts knows from past experience that the probability is 0.8 that an order will be ready for shipment on time, and it is 0.72 that an order will be ready for shipment on time and will also be delivered on time. What is the probability that such an order will be delivered on time given that it was ready for shipment on time?
39. A box of fuses contains 20 fuses, of which five are defective. If three of the fuses are selected and random and removed from the box in succession without replacement, what is the probability that all three fuses are defective?
40. The members of consulting firm rent cars from three rental agencies: 60% from agency 1, 30% from agency 2, and 10% from agency 3. If 9% of cars from agency 1 need a tune-up, 20% of the cars from agency 2 need a tune-up, and 6% of the cars from agency 3 need a tune-up, what is the probability that a rental car delivered to the firm will need a tune-up?
41. In a certain state .25% of all cars emit excessive amounts of pollutants. If the probability is 0.99 that a car emitting excessive amounts of pollutants will fail the states vehicular emission test, and the probability is 0.17 that a car not emitting excessive amounts of pollutants will nevertheless fail the test, what is the probability that a car that fails the test actually emits excessive amounts of pollutants?
42. There are 90 applicants for a job with the news department of television station. Some of them are college graduate and some are not, some of them have at least three years experience and some have not, with exact breakdown being

	college graduates	non college graduates
at least three years experience	18	9
less than three years experience	36	27

If the order in which applicants are interviewed by the station manager is random and G is the event that the first applicant interviewed is college graduate, T is the event that first applicant interviewed has at least three years experience, determine each of the following probabilities:

- (a) $P(G)$ (b) $P(T')$ (c) $P(G \cap T)$ (d) $P(G' \cap T')$ (e) $P(T/G)$ (f) $P(G' | T')$
43. Use the results of problem 18 to verify that

$$(a) P(T/G) = \frac{P(G \cap T)}{P(G)} \quad (b) P(G' | T') = \frac{P(G' \cap T')}{P(T')}$$

44. A bin contains 100 balls, of which 25 are red, 40 are white, and 35 are black. If two balls are selected from the bin, what is the probability that one will be red and one will be white (a) if first ball is replaced before the second is drawn (b) if the second ball is drawn without replacing the first.

45. The probability of surviving a certain transplant operation is 0.55. If a patient survives the operation; the probability that his or her body will reject the transplant within a month is 0.20. What is the probability of surviving both of these critical stages?

46. A mail-order house employs three stock clerks U, V and W. who pull items from shelves and assemble them for subsequent verification and packaging makes a mistake in an order (gets a wrong item or the wrong quantity) one item in hundred makes a mistake in an order five times in a hundred, and W makes a mistake in an order three times in a hundred. What are the probabilities that (a) a mistake will be made in order (b) if a mistake is made in order, the order was filled by U (c) if a mistake is made in an order, the order was filled by V.

47. An explosion at a construction site could have occurred as the result of static electricity, malfunctioning of equipment, carelessness, or sabotage. Interviews with construction engineers analyzing the risk involved led to the estimates that such an explosion would occur with probability 0.25 as a result of static electricity, 0.20 as result of malfunctioning of equipment, 0.40 as a result of carelessness, and 0.75 as a result of sabotage. It is also felt that the prior probabilities of the four causes of the explosion are 0.20, 0.40, 0.25 and 0.15. Based in all this information, what is (a) the most likely cause of the explosion (b) the least likely cause of the explosion.

CHAPTER FIVE

RANDOM VARIABLES

5.1 INTRODUCTION

In most applied problems involving probabilities we are interested in a particular aspect of the outcomes of experiments. For example, when we roll a pair of dice we are interested in the total, and not in the outcome for each die; when we interview a randomly chosen couple we may be interested in the size of their family and their joint income, but not in the number of years they have been married; and when we buy bulbs we may be interested in their durability to their brightness, but in their price.

In each of these examples we are interested in numbers that are associated with the outcomes of sample space that is, in the values taken on by random variables. In the language of probability and statistics, the total we roll with a pair of dice is a random variable, the size of the family and the joint income of a married couple are the random variables, and so are the durability and brightness of a light bulb are the random variables.

Random Variables

An experiment is usually denoted by a capital letter, such as X , and the outcomes of the experiment by a corresponding lower case letter, x . Here, X is called a random variable because the value it takes in a given experiment is a chance or random outcome. For example, the random variable X may be the number of sales in a single day and it could take the value $x = 0, 1, 2, \dots, n$. Some authors use the other names for random variable, for example, 'chance variable', 'stochastic variable', and 'variate'. There are two types of random variables:

1. When the outcomes of the experiment or the values of random variable can only be a countable number of values, i.e., discrete outcomes, the random variable is called a discrete random variable.
2. When the random variable can take any value over a range of values it is called continuous random variable. Continuous random variable is discussed in the next section.

Example 5.1.1 The following are discrete random variables, identify the values the random variables can take:

- i. the number of customers entering a shop during a day
- ii. the shoe sizes of a sample of 100 families
- iii. the number of positive answers given by a candidate in a questionnaire containing 20 questions.
- iv. the number of telephone calls to a company in a given hours

Solution: (i) The values of this random variable could be, 0, 1, 2, ...
(ii) The values of this random variable could be 3, 3½, 4½, 5, 5½, ...
(iii) The value of this random variable could be 0, 1, 2, ..., 20
(iv) The value of this random variable could be 0, 1, 2, 3, ..., n

Discrete Probability Distributions

The probability distribution of a discrete random variable X is a table or formula showing the possible values of X along with their associated probabilities. These probabilities are written as $P(X = x)$ or just $P(x)$ where x is an element of X .

Example 5.1.2 When two fair dice are rolled write down the probability distribution showing the sum of the numbers occur on both dice.

Solution: The sample space when two dice are rolled is:

$$S = \left\{ \begin{array}{cccccc} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{array} \right\}$$

If X is a random variable that sum of numbers occur on both dice then,

$$X = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

Thus probability distribution is:

x	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

Observe that there are two pairs whose sum is 3, that is, $A = \{(1, 2), (2, 1)\}$. Hence the probability that sum of 3 occurs is $2/36$ as there are 36 sample points in S .

We have mentioned above that probability distribution of a random variable X is a table or formula showing the possible values of X along with their associated probabilities. For example, the probability distribution function in example 02 may be expressed as:

$$P(x) = \frac{6 - |x - 7|}{36} \text{ for } x = 2, 3, \dots, 12$$

This formula can easily be verified by substitution. Thus,

$$P(2) = [6 - |2 - 7|]/36 = (6 - 5)/36 = 1/36,$$

$$P(3) = [6 - |3 - 7|]/36 = (6 - 4)/36 = 2/36, \text{ etc.}$$

Example 5.1.3 Find the formula for the probability distribution of total heads obtained in four tosses of a fair coin.

Solution: The sample space when four fair coins are tossed is:

$$S = \{HHHH, HHHT, HHTH, HTHH, THHH, HHTT, HTHT, HTTH, THHT, THTH, TTHH, HTTT, THTT, TTHT, TTTH, TTTT\}$$

Now the $P(0) = 1/16$, $P(1) = 4/16$, $P(2) = 6/16$, $P(3) = 4/16$, $P(4) = 1/16$. Observe that

$$1 = {}^4C_0, \quad 4 = {}^4C_1, \quad 6 = {}^4C_2, \quad 4 = {}^4C_3, \quad 1 = {}^4C_4$$

Hence formula for probability distribution is: $P(x) = \frac{{}^4C_x}{16}$, $x = 0, 1, 2, 3, 4$.

Example 5.1.4 A shipment of 8 similar PCs contains 3 defective PCs. If a college makes a random purchase of 2 PCs, write down the probability distribution of this distribution for the purchase of PCs.

Solution: The sample space contains ${}^8C_2 = 28$ sample points. Now probability distribution

for this problem is $P(x) = \frac{{}^3C_x \cdot {}^5C_{2-x}}{{}^8C_2}$, for $x = 0, 1, 2$. Here x is number of defective

PCs.

Explanation: Total number of PCs is 8 in which 3 are defective and therefore non-defective PCs are 5. If x is number of defective PCs, then x may assume 0, 1, 2 values. Now, if a PC is chosen and found non-defective, it will from 5 and therefore, no PC will be selected from 3 defectives PCs. In this case using the formula as mentioned above, $P(0) = 10/28$. The probability distribution is shown below:

x	0	1	2
$P(X = x) = P(x)$	10/28	15/28	3/28

Properties of Discrete Probability Distribution

There are three important properties of discrete random variable X . They are:

1. Probabilities associated with all possible outcome of X lie between 0 and 1, that is, $0 \leq P(X) \leq 1$.
2. The sum of all probabilities is 1, that is, $\sum P(x) = 1$.
3. To find the probabilities of several outcomes of X , we add the probabilities. For instance, $P(1 \leq X \leq 4) = P(1) + P(2) + P(3) + P(4)$. Similarly, $P(1 < X < 4) = P(2) + P(3)$.

Remark: Some of the authors use symbol $f(x)$ instead of $P(x)$ for the discrete probability distribution.

Cumulative Distribution Function for Discrete Random Variable

There are many problems in which it is of interest to know the probability that the value of a random variable is less than or equal to some real number x . This we write as:

$$F(x) = P(X \leq x) = \sum_{t \leq x} P(t) = \sum_{t \leq x} f(t)$$

Such a function refers to as cumulative distributive function or simply the distribution function. From now on we shall also use $f(x)$ instead of $P(x)$. It may be noted that:

1. $F(-\infty) = 0$
2. $F(\infty) = 1$
3. If $a < b$ then $F(a) \leq F(b)$.

Example 5.1.5 Find the distribution function of the total number of heads obtained in four tosses of fair coin.

Solution: The probability distribution is shown below. Then by definition,

x	0	1	2	3	4
$P(X = x) = f(x)$	1/16	2/16	6/16	2/16	1/16

$$F(0) = f(0) = 1/16$$

$$F(1) = P(X \leq 1) = f(0) + f(1) = 1/16 + 2/16 = 3/16$$

$$F(2) = P(X \leq 2) = f(0) + f(1) + f(2) = 1/16 + 2/16 + 6/16 = 9/16$$

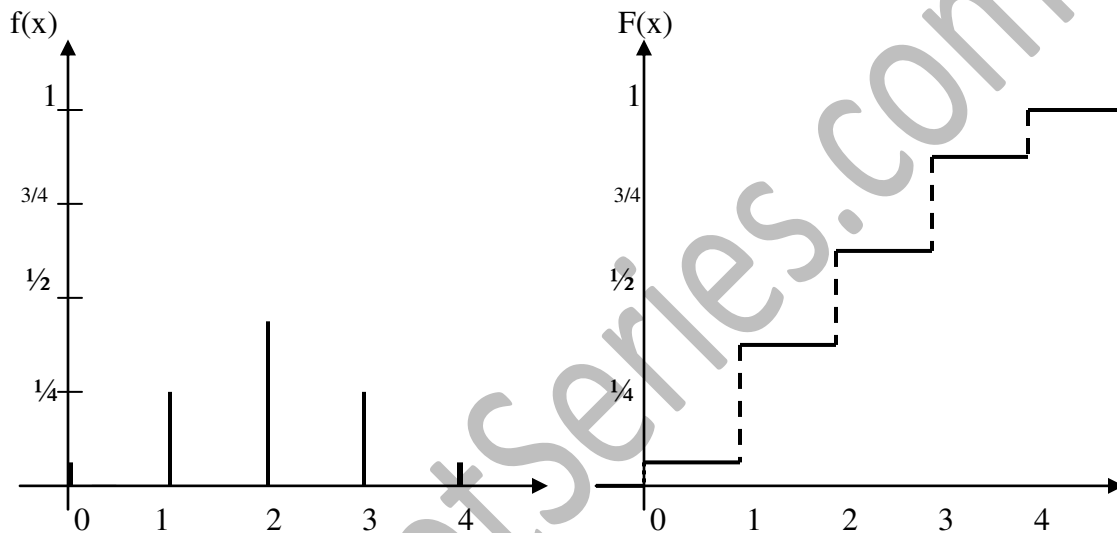
$$F(3) = P(X \leq 3) = f(0) + f(1) + f(2) + f(3) = 1/16 + 2/16 + 6/16 + 2/16 = 15/16$$

$$F(4) = P(X \leq 4) = f(0) + f(1) + f(2) + f(3) + f(4) = 1/16 + 2/16 + 6/16 + 2/16 + 1/16 = 1$$

Hence the distribution function is given by:

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1/16 & \text{for } 0 \leq x < 1 \\ 5/16 & \text{for } 1 \leq x < 2 \\ 11/16 & \text{for } 2 \leq x < 3 \\ 15/16 & \text{for } 3 \leq x < 4 \\ 1 & \text{for } x \geq 4 \end{cases}$$

It is interesting to see the graphs of both discrete probability distribution and associated cumulative distribution function. They are presented in the following figure.



Continuous Random Variable and Probability Density Function

In the last paragraphs we introduced the concept of discrete random variable as a single-valued function defined over the points of a sample space with a probability measure. In the continuous case, where random variables can assume values on a continuous scale, the procedure is very much same. The outcomes of experiments are represented by the points on line segment.

As for discrete variables, probabilities associated with continuous random variables can often be represented by a formula known as probability density function (p.d.f) for a continuous random variable and is denoted by $f(x)$. It does not, however, directly provide probabilities; the probability that the continuous random variable takes on values over a given interval is represented by the area under the graph of $f(x)$ over the given interval as shown in the figure shown below. Thus we have:

A function $f(x)$ defined over the set of all real numbers, is called probability density function of the continuous random variable X if and only if:

$$1. f(x) \geq 0 \quad 2. \int_{-\infty}^{\infty} f(x) dx = 1 \quad 3. P(a < X < b) = \int_a^b f(x) dx$$

Here a and b where $a < b$ are real.

(See the figure on the next page)

Example 5.1.6 Let X has the probability density

$$f(x) = \begin{cases} k.e^{-3x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

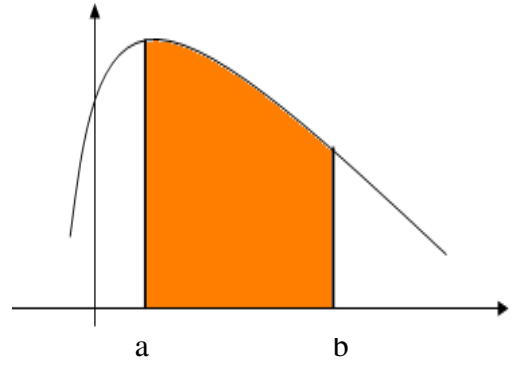
Find k and $P(0.5 < X < 1)$.

Solution: We know that if $f(x)$ is a p.d.f then

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= 1 \text{ or } \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx = 1 \\ &= 0 + k \int_0^{\infty} e^{-3x} dx = 1 \end{aligned}$$

$$= k \left[\frac{e^{-3x}}{-3} \right]_0^{\infty} = k \left[\frac{e^{-\infty} - e^0}{-3} \right] = 1 \text{ or } k \left[\frac{0-1}{-3} \right] = 1 \text{ or } \frac{k}{3} = 1 \text{ or } k = 3$$

$$\text{Now, } P(0.5 < X < 1) = 3 \int_{0.5}^1 e^{-3x} dx = 3(e^{-3} - e^{-1.5}) / (-3) = -e^{-3} + e^{-1.5} = 0.173.$$



Example 5.1.7 The number of minutes that a flight from Karachi to Islamabad is early or late is a random variable with p.d.f given by

$$f(x) = \begin{cases} (36 - x^2) / 288, & -6 < x < 6 \\ 0 & \text{elsewhere} \end{cases}$$

The negative values show that flight is early and the positive values show that it is being late. Find the probability that one of these flights will be

- i. at least 2 minutes early
- ii. at least 1 minute early
- iii. from 1 to 3 minutes early
- iv. exactly 5 minutes late

Solution: (i) Here we have to compute $P(-6 \leq X \leq -2)$. Now by definition:

$$\begin{aligned} P(-6 \leq X \leq -2) &= \frac{1}{288} \int_{-6}^{-2} (36 - x^2) dx = \frac{1}{288} \left[36x - x^3 / 3 \right]_{-6}^{-2} \\ &= \frac{1}{288} [36(-2+6) - (-8+216)/3] = 0.26 \end{aligned}$$

$$(ii) \text{ We require } P(-6 \leq X \leq -1) = \frac{1}{288} \int_{-6}^{-1} (36 - x^2) dx = 0.38$$

$$(iii) \text{ Here we require } P(-3 \leq X \leq -1) = \frac{1}{288} \int_{-3}^{-1} (36 - x^2) dx = 0.22$$

$$(iv) \text{ Finally, we require } P(X = 5) = \frac{1}{288} \int_5^5 (36 - x^2) dx = 0$$

Distributive Function

Distributive function or Cumulative Distributive Function of a random variable X with p.d.f $f(x)$ is denoted by $F(x)$ and is defined as:

$$F(X) = P(X \leq x) = \int_{-\infty}^x f(x) dx, \text{ for any real } x.$$

If $f(x)$ and $F(x)$ are the values of the probability density and the distribution function of X at x, then:

(i) $P(a \leq X \leq b) = F(b) - F(a)$.

(ii) $f(x) = dF/dx$ (This is the relation between p.d.f and distribution function).

Example 5.1.8 The p.d.f of continuous random variable is defined as

$$f(x) = \begin{cases} x^2/3, & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find the distribution function and $P(0.5 \leq X \leq 1)$. Also draw the graphs of $f(x)$ and $F(x)$.

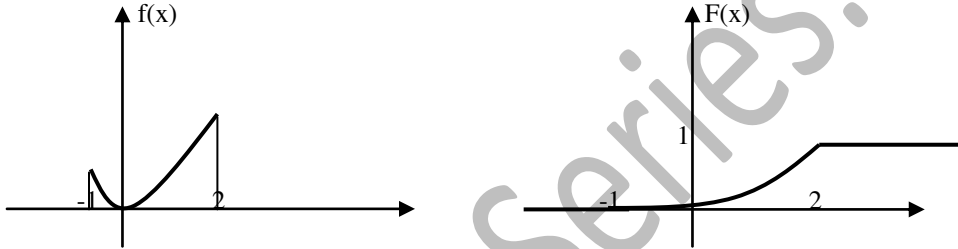
Solution: By definition

$$F(X) = P(X \leq x) = \int_{-\infty}^x f(x) dx = \int_{-1}^x x^2/3 dx = x^3/9 \Big|_{-1}^x = (x^3 + 1)/9$$

$$\text{Thus, } F(x) = \begin{cases} 0, & x < -1 \\ (x^3 + 1)/9, & -1 \leq x \leq 2 \\ 1, & x > 2 \end{cases}$$

Now $P(0.5 \leq X \leq 1) = F(1) - F(0.5) = (2 - 1.125)/9 = 0.097$.

The graphs of $f(x)$ and $F(x)$ are shown under.



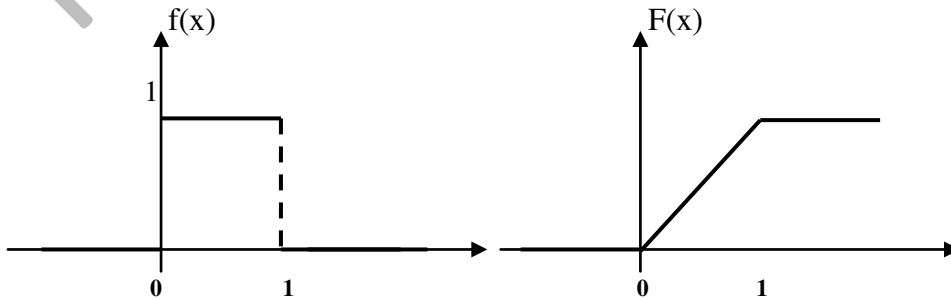
Example 5.1.9 Find a probability density function for the random variable whose distribution function is given by:

$$F(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ x & \text{for } 0 < x < 1 \\ 1 & \text{for } x \geq 1 \end{cases}$$

Also plot the graphs of $f(x)$ and $F(x)$.

Solution: Since $F(x)$ is continuous as well as derivable in each of the subintervals mentioned above, and $f(x) = dF/dx$, hence

$$f(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ 1 & \text{for } 0 < x < 1 \\ 0 & \text{for } x \geq 1 \end{cases}$$



Multivariate Distributions

A random variable, as we know is a real valued function defined over a sample space with a probability measure. It may happen that over the same sample space, we may define more than single random variables. For example, when two dice are rolled, one random variable was sum of the points appeared on both dice, the second random variable may be the product of the numbers and third may be the “Even Number” appears on the second throw, etc. As a second illustration, let us consider a class of 50 students. The class doctor may be interested in their weights, the teacher may be interested in their grades and their principal may be interested in their IQs. Thus over one sample space, we may define many random variables.

In this section we shall be concerned with bivariate case that is a situation where we are interested in a pair of random variables at the same time.

Joint Probability Distribution for Discrete Random Variable

If X and Y are discrete random variables, we write the probability that X will take on the value x and Y will take the value y as $P(X = x, Y = y)$. Thus $P(X = x, Y = y)$ is the probability of the intersection of the values $X = x$, and $Y = y$. Analogous to univariate random variable, we can now, in bivariate case, display the probabilities associated with all pairs of values of X and Y by means of a tables.

Definition: If X and Y are discrete random variables, the function given by

$$f(x, y) = P(X = x, Y = y)$$

for each pair of values (x, y) within the range of X and Y is called the “Joint Probability Distribution”.

Marginal Probability Distributions for Discrete Random Variable

From the joint probability distribution for two random variables X and Y , we can obtain the individual probability distributions for X and Y . Such individual probability distribution is called marginal probability distribution.

Let $f(x_i, y_j)$ be the joint probability distribution of two discrete random variables X and Y . Then the marginal probability distribution of X is defined as:

$$g(x_i) = \sum_{j=1}^n f(x_i, y_j) = f(x_i, y_1) + f(x_i, y_2) + \dots + f(x_i, y_n), \quad i = 1, 2, 3, \dots, m.$$

Similarly, the marginal probability distribution of Y is defined as:

$$h(y_j) = \sum_{i=1}^m f(x_i, y_j) = f(x_1, y_j) + f(x_2, y_j) + \dots + f(x_m, y_j), \quad j = 1, 2, 3, \dots, n.$$

The values of marginal probabilities are often written in the margins of joint probability table as they are the row and column totals in the table. The probabilities in each marginal probability distribution are equal to 1.

$Y \backslash X$	x_1	x_2	x_3	---	x_m	$h(y_j)$
y_1	$f(x_1, y_1)$	$f(x_2, y_1)$	$f(x_3, y_1)$	---	$f(x_m, y_1)$	$h(y_1)$
y_2	$f(x_1, y_2)$	$f(x_2, y_2)$	$f(x_3, y_2)$	---	$f(x_m, y_2)$	$h(y_2)$
y_3	$f(x_1, y_3)$	$f(x_2, y_3)$	$f(x_3, y_3)$	---	$f(x_m, y_3)$	$h(y_3)$
---	---	---	---	---	---	---
y_n	$f(x_1, y_n)$	$f(x_2, y_n)$	$f(x_3, y_n)$	---	$f(x_m, y_n)$	$h(y_n)$
$g(x_i)$	$g(x_1)$	$g(x_2)$	$g(x_3)$	---	$g(x_m)$	1

The Joint Probability Distribution of two discrete random variables X and Y

Thus marginal probability distribution for the random variable X is:

X	x_1	x_2	---	x_m
$P(X = x) = g(x)$	$g(x_1)$	$g(x_2)$	---	$g(x_m)$

Similarly, the marginal probability distribution for the random variable Y is:

Y	y_1	y_2	---	y_n
$P(Y = y) = h(y)$	$h(y_1)$	$h(y_2)$	---	$h(y_n)$

Bivariate probability distribution function satisfies the following properties:

1. $f(x, y) \geq 0$ for each pair (x, y) .
2. $\sum_x \sum_y f(x, y) = 1$, where double summation extends over all possible pairs (x, y) within the domain.
3. $\sum g(x_i) = 1$ and $\sum h(y_j) = 1$

Conditional Probability Distribution

Let X and Y be two discrete random variables with joint probability distribution $f(x_i, y_j)$. Then conditional probability of X given $Y = y_j$ denoted as $f(x_i | y_j)$ is defined as:

$$f(x_i | y_j) = P(X = x_i | Y = y_j) = \frac{P(X = x_i \cap Y = y_j)}{P(Y = y_j)} = \frac{f(x_i, y_j)}{h(y_j)}. \text{ Similarly}$$

$$f(y_j | x_i) = P(Y = y_j | X = x_i) = \frac{P(X = x_i \cap Y = y_j)}{P(X = x_i)} = \frac{f(x_i, y_j)}{g(x_i)}$$

Independent Events for Discrete Probability Distribution

Two discrete random variables X and Y are said to be independent, if and only if, for all pairs (x_i, y_j) the joint probability $f(x_i, y_j)$ can be expressed as the product of the marginal probabilities of X and Y, that is, $f(x_i, y_j) = g(x_i) h(y_j)$

Example 5.1.10 Two capsules are selected at random from a bottle containing three “aspirin”, two “sedative” (tranquilizers) and four “laxative” capsules. If X and Y are, respectively, the number of aspirin and sedative capsules included among the two capsules drawn from the bottle, find the probabilities associated with all possible pairs of X and Y. Compute the marginal probability distributions for the random variables X and Y and $P(X = 0 | Y = 1)$. Are the random variables X and Y independent?

Solution: The total number of capsules in a bottle is 9 and the number of ways in which two capsules are selected from it is $C_2^9 = 36$. The joint probability distribution is therefore, given by:

$$f(x, y) = \frac{C_x^3 C_y^2 C_{2-x-y}^4}{C_2^9}, \text{ where } x = 0, 1, 2; y = 0, 1, 2; \text{ and } 0 \leq x + y \leq 2. \quad (1)$$

It may be noted that if “aspirin” and “sedative” capsules are not drawn then certainly “laxative” type of capsule will be drawn. This means if $x = 0$ and $y = 0$ then two capsules are taken from four “laxative” type of capsules. This can be done in $C_2^4 = 6$ ways. Hence $f(0, 0) = 6/36 = 1/6$. The following joint probability table depicts the entire situation.

		X			
Y		0	1	2	Total
	0	1/6	1/3	1/12	7/12
	1	2/9	1/6	-----	7/18
	2	1/36	-----	-----	1/36
	Total	15/36	1/2	1/12	1

Thus marginal probability distribution for the random variable X is:

x	0	1	2	Sum
$P(X = x) = g(x)$	15/36	1/2	1/12	1

Similarly, the marginal probability distribution for the random variable y is:

y	0	1	2	Sum
$P(Y = y) = h(y)$	7/12	7/18	1/36	1

Now consider $P(X = 0 | Y = 1) = f(0, 1)/h(1) = [(2/9)/(7/18)] = 4/7$.

To see that X and Y are independent let us consider

$$f(1, 0) = g(1) \cdot h(0) \quad \text{or} \quad 1/3 = (1/2) \cdot (1/36) \quad \text{or} \quad 1/3 \neq 1/72.$$

This is not true hence two random variables X and Y are not independent.

Example 5.1.11 Determine the value of k for which the function given by

$$f(x, y) = kxy \quad \text{for } x = 1, 2, 3; \quad y = 1, 2, 3.$$

is a joint probability distribution. Are X and Y independent?

Solution: Substituting the various values of x and y, we get:

$$f(1, 1) = k, f(1, 2) = 2k, f(1, 3) = 3k, f(2, 1) = 2k, f(2, 2) = 4k, f(3, 1) = 3k,$$

$$f(3, 2) = 6k, f(3, 3) = 9k. \text{ Now as per condition 2,}$$

$$k + 2k + 3k + 2k + 4k + 6k + 3k + 6k + 9k = 1 \quad \text{or} \quad 36k = 1 \quad \text{or} \quad k = 1/36$$

Thus,

$$f(x, y) = xy/36$$

$$\text{Now, } g(x) = \sum_y f(x, y) = \sum_{y=1}^3 xy/36 = x/36 + 2x/36 + 3x/36 = x/6.$$

Similarly, the marginal probability distribution of Y is defined as:

$$h(y) = \sum_x f(x, y) = \sum_{x=1}^3 xy/36 = y/36 + 2y/36 + 3y/36 = y/6$$

Observe that, $f(x, y) = g(x) \cdot h(y)$. Hence the random variables X and Y are independent.

Cumulative Distribution Function for Discrete Random Variable

As in the univariate case, there are many problems in which it is of interest to know the probability that the values of two variables are less than or equal to some real numbers x and y.

Definition: If X and Y are discrete random variables then their Joint Cumulative Distribution Function is defined as:

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} f(s, t)$$

Example 5.1.12 Refer to example 10, find $F(1, 1)$.

$$\begin{aligned} \text{Solution: } F(1, 1) &= P(X \leq 1, Y \leq 1) = f(0, 0) + f(0, 1) + f(1, 0) + f(1, 1) \\ &= 1/6 + 2/9 + 1/3 + 1/6 = 8/9. \end{aligned}$$

Joint Probability Density Function

As in the univariate case, the joint probability density function of continuous random variables X and Y is defined as:

$$P[(X, Y) \in A] = \iint_A f(x, y) \, dx \, dy$$

where A is the region in xy-plane. It may be noted that analogous to discrete random variable, f(x, y) satisfies the following two properties:

$$1. \quad f(x, y) \geq 0 \quad 2. \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1$$

Marginal Probability Density Functions

The marginal p.d.fs for the continuous random variable X and Y are respectively defined as

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) \, dx$$

Conditional Probability Distribution

Let X and Y be two continuous random variables with joint probability distribution $f(x, y)$. Then conditional probability of X given $Y = y$ is denoted by $f(x|y)$ and is defined as:

$$f(x|y) = P(X = x | Y = y) = \frac{f(x, y)}{h(y)}. \quad \text{Similarly,} \quad f(y|x) = P(Y = y | X = x) = \frac{f(x, y)}{g(x)}$$

Independent Events for Continuous Random Variable

Analogous to discrete random variables two continuous random variables are said to be independent, if and only if, for all pairs (x, y) the joint probability $f(x, y)$ can be expressed as the product of the marginal probabilities of X and Y, that is: $f(x, y) = g(x)h(y)$

Example 5.1.13 Given the joint probability density function

$$f(x, y) = \begin{cases} 3x(y+x)/5 & \text{for } 0 < x < 1, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

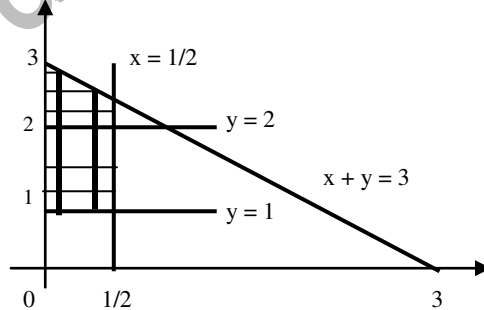
of two random variables X and Y, find $P[(X, Y) \in A]$, where A is the region $\{(x, y) | 0 < x < 1/2, 1 < y < 2\}$. Also compute $P(X + Y < 3)$ under these conditions. Determine the marginal probability density functions for the continuous random variables X and Y. Are X and Y independent? Also find the conditional probability $f(x|y)$.

Solution: $P[(X, Y) \in A] = P(0 < X < 1/2, 1 < Y < 2)$

$$\int_1^{21/2} \int_0^{1/2} 3x(y+x)/5 \, dx \, dy = \int_1^{21/2} \left(\frac{3x^2y}{10} + \frac{3x^3}{15} \right) \Big|_0^{1/2} dy = \int_1^{21/2} \left(\frac{3y}{40} + \frac{1}{40} \right) dy = \left(\frac{3y^2}{80} + \frac{y}{40} \right) \Big|_1^{21/2} = 11/80$$

Under the conditions given in the above part, the region whose area/probability is to be computed is shown below. From the figure, we see that:

$$P(X + Y < 3) = \int_0^{1/2} \int_1^{3-x} 3x(y+x)/5 \, dy \, dx = 1/4$$



The marginal probability of random variable X is:

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy = \int_0^2 3x(y+x)/5 \, dy = 6x(1+x)/5, \quad 0 < x < 1 \quad (\text{Keeping } x \text{ constant})$$

The marginal probability of random variable Y is:

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 3x(y+x)/5 dx = (3y+2)/10, 0 < y < 2 \quad (\text{Keeping } y \text{ constant})$$

Note: Readers are advised to check that:

$$\int_0^1 g(x) dx = 1 = \int_0^2 h(y) dy$$

To see the independence, consider

$$g(x).h(y) = \frac{6x(1+x)}{5} \cdot \frac{(2+3y)}{10} = \frac{6x(1+x)(2+3y)}{50} \neq f(x, y)$$

Hence, X and Y are not independent continuous random variables. Finally the conditional probability of x given y is given by,

$$f(x|y) = \frac{f(x, y)}{h(y)} = \frac{3x(y+x)}{5} \div \frac{(3y+2)}{10} = \frac{6x(y+x)}{(3y+2)}$$

Joint Distribution Function for Continuous Random Variable

Analogous to joint distribution of cumulative distribution function for discrete random variables, we have Joint Distribution Function or Joint Cumulative Distribution Function

defined as: $F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(s, t) ds dt$ for $-\infty < x < \infty, -\infty < y < \infty$

Analogous to relationship between distribution function $F(x)$ and p.d.f $f(x)$, that is, $F(x) = \int f(x) dx$, there exists a relationship between 'Joint Probability Distribution Function' $F(x, y)$ and 'Joint Probability Density Function' $f(x, y)$. This relation is given by:

$$\partial^2 F / \partial x \partial y = f(x, y).$$

5.2 EXPECTED VALUE OF DISCRETE RANDOM VARIABLE

One of the most important and most fundamental concepts associated with statistical methods is that of finding average values of random variables or functions of random variables.

Definition: If X is discrete random variable with possible values $x_1, x_2, x_3, \dots, x_n$, and corresponding probabilities $f_1, f_2, f_3, \dots, f_n$ then expected value of X is defined as:

$$E(X) = \sum_{i=1}^n x_i f_i$$

Other symbols used to denote expectation value of random variable are μ and $\bar{X}$. The expected value of a random variable is also known as its average value because according to definition of mean:

$$\bar{X} = \sum_{i=1}^n x_i f_i / \sum_{i=1}^n f_i$$

Now if f_i is considered as probability then $\sum_{i=1}^n f_i = 1$, in which case $\bar{X} = \sum_{i=1}^n x_i f_i$. This shows that expected value of random variable X is its mean.

Some Important Results

$$1. E(k) = \sum_{i=1}^n k f_i = k f_1 + k f_2 + \dots + k f_n = k(f_1 + f_2 + \dots + f_n) = k.1 = k$$

$$2. E(ax + b) = \sum_{i=1}^n (ax_i + b)f_i = \sum_{i=1}^n (a x_i f_i + b f_i) = a \sum_{i=1}^n x_i f_i + b \sum_{i=1}^n f_i = aE(X) + b$$

Example 5.2.1 A man wins Rs. 5 if he gets three heads or three tails and loses Rs. 1 otherwise. What is the average gain?

Solution: The sample space $S = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$.

The random variable associated with this sample space is: $X = \{5, -1\}$.

Thus probability distribution is:

X	5	-1
P(X = x)	2/8	6/8

Thus expected gain is $E(X) = 5(2/8) - 1(6/8) = (10 - 6)/8 = 4/8 = 0.5$

Hence the expected gain is Rs. 1/2 toss if the player plays a long game.

Example 5.2.2 A committee of three persons is to be selected from four engineers and three doctors. Find the expected number of doctors on this committee.

Solution: Total persons are $4 + 3 = 7$. The number of ways in which three persons are selected is $C_3^7 = 35$. The probability distribution function is, therefore,

$$f(x) = \frac{C_x^3 \cdot C_{3-x}^4}{C_3^7}, \quad x = 0, 1, 2, 3.$$

Here x is number of doctors on the committee. The probability distribution table is:

X	0	1	2	3
f(x) = P(X = x)	4/35	18/35	12/35	1/35

Now, $E(X) = 0(4/35) + 1(18/35) + 2(12/35) + 3(1/35) = 45/35 = 1.2 \sim 1$.

Thus it is expected that there will be one doctor on this committee.

Expected Value of Continuous Random Variable

If X is continuous random variable its expected value is defined as: $E(x) = \int_{-\infty}^{\infty} x f(x) dx$

Example 5.2.3 Let X be a random variable that denote the life of certain electronic circuit board in hours. The p.d.f for this random variable is given by:

$$f(x) = \begin{cases} 20,000/x^3, & \text{for } x > 100 \\ 0, & \text{elsewhere} \end{cases}$$

Find the expected life of the circuit board.

Solution: By definition,

$$E(X) = \int_{100}^{\infty} x \cdot \frac{20,000}{x^3} dx = 20,000 \int_{100}^{\infty} x^{-2} dx = -20,000 \cdot [1/x]_{100}^{\infty} = -20,000 \left[\frac{1}{\infty} - \frac{1}{100} \right] = 200$$

Thus, it should be expected that average life of circuit board is 200 hours.

Expected Value of Function of Random Variable

Let $Y = G(X)$ then expected value of Y is defined as:

$$E(Y) = E[G(X)] = \sum G(x_i) f(x_i). \text{ Here } X \text{ is a discrete random variable.}$$

$$E(Y) = E[G(X)] = \int_{-\infty}^{\infty} G(x) f(x) dx. \text{ Here } X \text{ is a continuous random variable.}$$

Example 5.2.4 Let X be the number of cars that go through a car wash process during any day from 12.00 noon to 13.00 p.m. The probability distribution for this random variable is given by:

X	4	5	6	7	8	9
P(X = x)	1/12	1/12	3/12	3/12	2/12	2/12

If Y is an other random variable that represents the amount in hundreds of rupees earned during this time and is defined as $Y = 2X - 1$, what is the average amount received during this hour?

Solution: The average value of X is:

$$E(X) = 4(1/12) + 5(1/12) + 6(3/12) + 7(3/12) + 8(2/12) + 9(2/12) = 82/12 = 6.84$$

$$\text{Now } Y = 2X - 1, \text{ then } E(Y) = E(2X - 1) = 2E(X) - 1 = 2(6.84) - 1 = 12.38.$$

Thus expected or average amount received is Rs. 1238 during 12.00 and 13.00.

Expected Value of Two Random Variables

If two random variables X and Y occur simultaneously and are related by $g(x, y)$ with joint probability distribution $f(x, y)$ then their expected value is defined as:

$$E[g(x, y)] = \sum_x \sum_y g(x, y) \cdot f(x, y) \quad [X \text{ and } Y \text{ are discrete random variables}]$$

$$E[g(x, y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) \cdot f(x, y) \, dx \, dy \quad [X \text{ and } Y \text{ are continuous random variables}]$$

Two Important Properties

If X and Y are random variables then (i) $E(X + Y) = E(X) + E(Y)$ (ii) $E(XY) = E(X)E(Y)$ when X and Y are independent.

Proof: By definition

$$\begin{aligned} \text{(i) } E(X + Y) &= \sum_i \sum_j (x_i + y_j) f(x_i, y_j) = \sum_i \sum_j x_i f(x_i, y_j) + \sum_i \sum_j y_j f(x_i, y_j) \\ &= \sum_i x_i \sum_j f(x_i, y_j) + \sum_j y_j \sum_i f(x_i, y_j) = \sum_i x_i g(x_i) + \sum_j y_j h(y_j) = E(X) + E(Y) \end{aligned}$$

(ii) Since X and Y are independent hence $f(x_i, y_j) = g(x_i) \cdot h(y_j)$. Therefore,

$$E(XY) = \sum_i \sum_j x_i y_j f(x_i, y_j) = \sum_i \sum_j x_i y_j g(x_i) \cdot h(y_j) = \sum_i x_i g(x_i) \cdot \sum_j y_j h(y_j) = E(X) \cdot E(Y)$$

Example 5.2.5 Let X and Y be the two random variables with joint probability distribution given as under:

$\begin{matrix} \backslash x \\ y \end{matrix}$	2	4
1	0.10	0.15
3	0.20	0.30
5	0.10	0.15

Find: $E(X)$, $E(Y)$, $E(X + Y)$, $E(2X - 3Y)$ and $E(XY)$. Are X and Y independent?

Solution: Before the required entities are to be evaluated, we have to compute the marginal probabilities.

$\begin{matrix} \backslash x \\ y \end{matrix}$	2	4	$h(y)$
1	0.10	0.15	0.25
3	0.20	0.30	0.50
5	0.10	0.15	0.25
$G(x)$	0.40	0.60	1

$$E(X) = \sum x_i g(x_i) = 2(0.40) + 4(0.6) = 3.2,$$

$$E(Y) = \sum y_j h(y_j) = 1(0.25) + 3(0.5) + 5(0.25) = 3$$

$$E(X + Y) = \sum \sum (x_i + y_j) f(x_i, y_j)$$

$$= (2+1)(0.1) + (2+3)(0.2) + (2+5)(0.1) + (4+1)(0.15) + (4+3)(0.3) + (4+5)(0.15) = 6.20$$

You may observe that: $E(X + Y) = E(X) + E(Y)$

$$\text{Also } E(2X - 3Y) = 2 E(X) - 3 E(Y) = 2(3.2) - 3(3) = -2.6$$

Finally, $E(X).E(Y) = (3.2)(3) = 9.6$, and

$$E(XY) = (2)(0.1) + (6)(0.2) + (10)(0.1) + (4)(0.15) + (12)(0.3) + (20)(0.15) = 9.6$$

We see that $E(XY) = E(X) E(Y)$, hence X and Y are independent random variables.

Example 5.2.6 Let X and Y be two continuous random variables with joint p.d.f defined as:

$$f(x, y) = \begin{cases} x(1+3y^2)/4, & 0 < x < 2, \quad 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find $E(X)$, $E(Y)$, $E(X + Y)$ and $E(XY)$. Show that X and Y are independent.

Solution: To determine $E(X)$, $E(Y)$, we must find out the marginal probabilities of X and Y .

$$\text{Marginal probability of } X: g(x) = \int_0^1 x(1+3y^2)/4 \, dy = \left[(xy + xy^3)/4 \right]_0^1 = x/2$$

Marginal probability of Y :

$$h(y) = \int_0^2 x(1+3y^2)/4 \, dx = \left[(x^2/2 + 3x^2y^2)/4 \right]_0^2 = (1+3y^2)/2$$

$$\text{Thus, } E(X) = \int_0^2 x g(x) \, dx = \int_0^2 x^2/2 \, dx = \left[x^3/6 \right]_0^2 = 4/3, \text{ and}$$

$$E(Y) = \int_0^1 y h(y) \, dy = \int_0^1 y(1+3y^2)/2 \, dy = \frac{1}{2} \left[\frac{y^2}{2} + \frac{3y^4}{4} \right]_0^1 = 5/8$$

$$E(X + Y) = E(X) + E(Y) = 4/3 + 5/8 = 47/24. \text{ Finally,}$$

$$E(XY) = \int_0^2 \int_0^1 x y f(x, y) \, dx \, dy = \int_0^2 \int_0^1 x y \frac{x(1+3y^2)}{4} \, dx \, dy = 5/6 \text{ [Readers are advised to evaluate this]}$$

$$\text{Now, } E(X) \cdot E(Y) = (4/3) \cdot (5/8) = 5/6 = E(XY).$$

This shows that X and Y are independent random variables.

5.3 VARIANCE OF RANDOM VARIABLE

If X is a random variable and $\mu = E(X)$ is its mean, we denote the variance of X by $\text{Var}(X)$ or σ^2 and define it as:

$$\text{Var}(X) = \sigma^2 = E(X - \mu)^2$$

Simplifying the right side, we obtain

$$\text{Var}(X) = \sigma^2 = E(X^2 - 2\mu X + \mu^2) = E(X^2) - 2\mu E(X) + E(\mu^2) = E(X^2) - 2\mu \cdot \mu + \mu^2.$$

$$\text{Or, } \text{Var}(X) = \sigma^2 = E(X^2) - \mu^2 = E(X^2) - [E(X)]^2.$$

REMARK: The positive square root of variance is called standard deviation and is denoted by σ .

$$\text{If } X \text{ is discrete random variable then: } \text{Var}(X) = \sum x^2 f(x) - \left(\sum x f(x) \right)^2$$

$$\text{If } X \text{ is continuous random variable then: } \text{Var}(X) = \int_{-\infty}^{\infty} x^2 f(x) \, dx - \left(\int_{-\infty}^{\infty} x f(x) \, dx \right)^2$$

Example 5.3.1 The number of ambulances that are available with two companies Alpha and Beta is given as under:

Company Alpha:

x	1	2	3
$P(X = x)$	0.3	0.4	0.3

Company Beta:

y	0	1	2	3	4
$P(Y = y)$	0.2	0.1	0.3	0.3	0.1

Find the average and variance number of ambulances available in both companies and give your opinion that which of these two companies is better.

Solution:

Company Alpha: $E(X) = 1(0.3) + 2(0.4) + 3(0.3) = 2.$

$$E(X^2) = 1(0.3) + 4(0.4) + 9(0.3) = 4.6$$

Therefore, $\text{Var}(\text{Alpha}) = E(X^2) - [E(X)]^2 = 4.6 - 4 = 0.6$

Company Beta: $E(Y) = 0(0.2) + 1(0.1) + 2(0.3) + 3(0.3) + 4(0.1) = 2.$

$$E(Y^2) = 0(0.2) + 1(0.1) + 4(0.3) + 9(0.3) + 16(0.1) = 5.6$$

Therefore, $\text{Var}(\text{Beta}) = E(Y^2) - [E(Y)]^2 = 5.6 - 4 = 1.6$

Now to see which of these two companies is better in the availability of the ambulance, we have to find their C.Vs.

Coefficient of variance of company Alpha: $C.V = (S.D/\text{Mean}) 100\% = 38.73 \%$.

Coefficient of variance of company Beta: $C.V = (S.D/\text{Mean}) 100\% = 63.26 \%$.

We observe that C.V of company Alpha is less than of company Beta; hence the availability of ambulance at company Alpha is better than company Beta.

Example 5.3.2 The weekly demand of a soft drink in 1000s of liters is a continuous variable with p.d.f

$$f(x) = \begin{cases} 2(x-1), & 1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Find the mean and variance.

Solution: The mean demand is $\mu = E(X) = \int_1^2 x f(x) dx = 2 \int_1^2 x(x-1) dx = 5/3$

$$\text{Also, } E(X^2) = \int_1^2 x^2 f(x) dx = 2 \int_1^2 x^2(x-1) dx = 17/6$$

Now, $\text{Var}(X) = E(X^2) - [E(X)]^2 = 17/6 - 25/9 = 1/18.$

Covariance Between Two Random Variables

If X and Y are two random variables the covariance between them is denoted by $\text{COV}(X, Y)$ or σ_{XY} and is defined as:

$$\text{COV}(X, Y) = \sigma_{XY} = E[(X - \mu_X) \cdot (Y - \mu_Y)]$$

Simplifying right side, we obtain:

$$\begin{aligned} \text{COV}(X, Y) &= \sigma_{XY} = E[(XY - Y\mu_X - X\mu_Y + \mu_X\mu_Y)] = E(XY) - \mu_X E(Y) - \mu_Y E(X) + \mu_X\mu_Y \\ &= E(XY) - \mu_X\mu_Y - \mu_X\mu_Y + \mu_X\mu_Y = E(XY) - \mu_X\mu_Y = E(XY) - E(X)E(Y) \end{aligned}$$

Therefore, $\text{COV}(X, Y) = E(XY) - E(X)E(Y)$

If X and Y are independent random variables then $E(XY) = E(X)E(Y)$. In this case

$$\text{COV}(X, Y) = \sigma_{XY} = E(X)E(Y) - E(X)E(Y) = 0$$

Two Important Results

(i) If a and b are constants then $\sigma_{aX+b}^2 = a^2 \sigma_X^2$

Proof: By definition, $\sigma_{aX+b}^2 = E(aX+b)^2 - [E(aX+b)]^2$

$$\begin{aligned}
&= E(a^2X^2 + 2abX + b^2) - \{a E(X) + b\}^2 \\
&= a^2 E(X^2) + 2ab E(X) + b^2 - a^2[E(X)]^2 - 2ab E(X) - b^2 \\
&= a^2 \{E(X^2) - [E(X)]^2\} = a^2 \sigma_X^2.
\end{aligned}$$

(ii) If X and Y are random variables with joint p.d.f $f(x, y)$ then

$$\sigma_{aX+bY}^2 = a^2 \sigma_X^2 + b^2 \sigma_Y^2 + 2ab \sigma_{XY}$$

Proof: By definition,

$$\begin{aligned}
\sigma_{aX+bY}^2 &= E(aX + bY)^2 - [E(aX + bY)]^2 \\
&= E(a^2X^2 + 2abXY + b^2Y^2) - [aE(X) + bE(Y)]^2 \\
&= a^2E(X^2) + 2abE(XY) + b^2E(Y^2) - a^2[E(X)]^2 - 2abE(X)E(Y) - b^2[E(Y)]^2 \\
&= a^2\{E(X^2) - [E(X)]^2\} + b^2\{E(Y^2) - [E(Y)]^2\} + 2ab\{E(XY) - E(X)E(Y)\} \quad (1) \\
&= a^2 \sigma_X^2 + b^2 \sigma_Y^2 + 2ab \sigma_{XY}
\end{aligned}$$

Corollary: If X and Y are independent then $E(XY) = E(X)E(Y)$ or $E(XY) - E(X)E(Y) = 0$.

In this case equation (1) becomes

$$\sigma_{aX+bY}^2 = a^2 \sigma_X^2 + b^2 \sigma_Y^2$$

Example 5.3.3 If X and Y are random variables with variances 2 and 4 respectively, find the variance of random variable $Z = 3X - 5Y + 8$ if the covariance of X and Y is 2.

Solution: By definition, $\sigma_Z^2 = \sigma_{(aX+bY+c)}^2 = a^2 \sigma_X^2 + b^2 \sigma_Y^2 + 2ab \sigma_{XY}$

$$\text{Or, } \sigma_Z^2 = \sigma_{(3X-5Y+8)}^2 = 3^2(2) + (-5)^2(4) + 2(3)(-5)(2) = 18 + 100 - 60 = 58$$

Example 5.3.4 Refer to the following contingency table, find the covariance between X and Y.

$\begin{matrix} \backslash x \backslash \\ y \end{matrix}$	2	4	$h(y)$
1	0.10	0.15	0.25
3	0.20	0.30	0.50
5	0.10	0.15	0.25
$G(x)$	0.40	0.60	1

Solution: By definition, $E(X) = \sum x_i g(x_i) = 2(0.40) + 4(0.6) = 3.2$. Similarly,

$$E(Y) = \sum y_j h(y_j) = 1(0.25) + 3(0.5) + 5(0.25) = 3$$

$$E(XY) = (2)(0.1) + (6)(0.2) + (10)(0.1) + (4)(0.15) + (12)(0.3) + (20)(0.15) = 9.6$$

$$\text{Now, } \text{COV}(X, Y) = \sigma_{XY} = E(XY) - E(X)E(Y) = 9.6 - 3.2(3) = 0.$$

Example 5.3.5 Let X be the male runner and Y be female runner who have completed the marathon races. Their joint p.d.f is given as under:

$$f(x, y) = \begin{cases} 8xy, & 0 < x < 1, 0 < y < x \\ 0, & \text{elsewhere} \end{cases}$$

Find the covariance of X and Y.

Solution: $\text{COV}(X, Y) = E(XY) - E(X)E(Y)$.

To find $E(X)$ and $E(Y)$ we have to compute $g(x)$ and $h(y)$.

$$g(x) = \int_0^x f(x, y) dy = \int_0^x 8xy dy = \left[4xy^2 \right]_0^x = 4x^3. \text{ Thus}$$

$$g(x) = \begin{cases} 4x^3, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$h(y) = \int_y^1 f(x, y) dx = \int_x^1 8xy dx = \left[4yx^2 \right]_x^1 = 4y(1 - y^2). \text{ Thus}$$

$$h(y) = \begin{cases} 4y(1 - y^2), & 0 < y < x \\ 0, & \text{elsewhere} \end{cases}$$

Now,

$$E(X) = \int_0^1 x g(x) dx = \int_0^1 4x^4 dx = 4/5, \quad E(y) = \int_0^1 y h(y) dy = \int_0^1 4y^2(1 - y^2) dy = 8/15$$

$$\text{Finally, } E(XY) = \int_0^1 \int_0^1 xy f(x, y) dy dx = \int_0^1 \int_0^1 8x^2 y^2 dx dy = 4/9$$

$$\text{Thus, } \text{COV}(X, Y) = E(XY) - E(X)E(Y) = 4/9 - (4/5)(8/15) = 4/225.$$

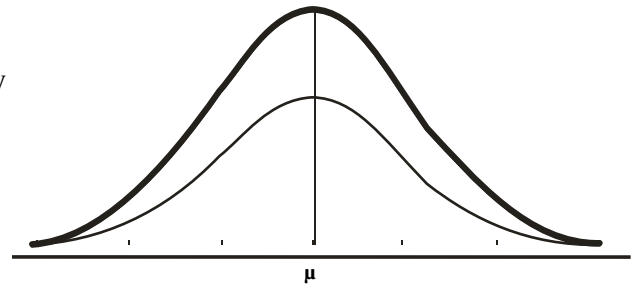
Chebychev's Theorem

In chapter 03, we stated that variance of sample data tells us something about the variability of observations about the mean. If a random variable has small variance or standard deviation, most of the observation gathered around the mean. Therefore, the probability that a random variable assumes a value within a certain interval about the mean is greater than for a similar random with a larger standard deviation.

If we think of probability in terms of area, we expect a continuous distribution with a small standard deviation to have most of its area close to mean μ , as shown in figure with thick curve. The large value of standard deviation σ indicates a greater variability and therefore we expect the area to be more spread out, as shown in figure with thin curve. We can argue the same for discrete probability distribution.

The Russian mathematician P. L. Chebychev (1821 – 1894) discovered that the part of area between any two values symmetric about mean is related to the standard deviation. Since the area under a probability curve or probability histogram is 1, the area between any two numbers is the probability of the random variable assuming the value between these numbers.

The following theorem due to Chebychev gives a traditional estimate of the probability that a random variable assumes a value within k standard deviations of its mean for any real $k > 1$.



Theorem: The probability that any random variable X will assume a value within k standard deviation of the mean is at least $1 - 1/k^2$, that is,

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - 1/k^2$$

Proof: By definition, $\sigma^2 = E[X - \mu]^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$

$$= \int_{-\infty}^{\mu - k\sigma} (x - \mu)^2 f(x) dx + \int_{\mu - k\sigma}^{\mu + k\sigma} (x - \mu)^2 f(x) dx + \int_{\mu + k\sigma}^{\infty} (x - \mu)^2 f(x) dx$$

$$\geq \int_{-\infty}^{\mu-k\sigma} (x-\mu)^2 f(x) dx + \int_{\mu+k\sigma}^{\infty} (x-\mu)^2 f(x) dx$$

NOTE: $\int_{\mu-k\sigma}^{\mu+k\sigma} (x-\mu)^2 f(x) dx$ is positive.

Now, $\mu - k\sigma < x < \mu + k\sigma$ or $(x - \mu)^2 \geq (k\sigma)^2$

$$\sigma^2 \geq \int_{-\infty}^{\mu-k\sigma} (k\sigma)^2 f(x) dx + \int_{\mu+k\sigma}^{\infty} (k\sigma)^2 f(x) dx.$$

Provided Dividing by $(k\sigma)^2$, we obtain

$$1/k^2 \geq \int_{-\infty}^{\mu-k\sigma} f(x) dx + \int_{\mu+k\sigma}^{\infty} f(x) dx \quad (1)$$

Now,

$$\int_{-\infty}^{\mu-k\sigma} f(x) dx + \int_{\mu-k\sigma}^{\mu+k\sigma} f(x) dx + \int_{\mu+k\sigma}^{\infty} f(x) dx = 1$$

or

$$\int_{-\infty}^{\mu-k\sigma} f(x) dx + \int_{\mu+k\sigma}^{\infty} f(x) dx = 1 - \int_{\mu-k\sigma}^{\mu+k\sigma} f(x) dx$$

Using (1), we get $1 - \int_{\mu-k\sigma}^{\mu+k\sigma} f(x) dx \leq 1/k^2$ or $\int_{\mu-k\sigma}^{\mu+k\sigma} f(x) dx \geq 1 - 1/k^2$

or

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}, \quad k > 1$$

If $k = 2$, theorem states that the random variable X has probability of at least $1 - 1/4 = 3/4$ that is about 75% of the area lies in the interval $(\mu - 2\sigma, \mu + 2\sigma)$. If $k = 3$, then there is about 8/9 that is 89% of the area lies in the interval $(\mu - 3\sigma, \mu + 3\sigma)$.

Example 5.3.6 A random variable X has a mean $\mu = 8$, a variance $\sigma^2 = 9$, and unknown probability. Find (a) $P(-4 < X < 20)$ and (b) $P(|X - 8| > 6)$.

Solution: (i) $P(-4 < X < 20) = P(8 - 12 < X < 8 + 12) = P[8 - 3(4) < X < 8 + 3(4)]$. This shows that $\mu = 8$ and $\sigma = 3$ and $k = 4$. Thus,

$$P(-4 < X < 20) \geq 1 - 1/(4)^2 = 15/16.$$

$$(ii) P(|X - 8| > 6) = 1 - P(|X - 8| < 6) \quad (1)$$

Now, $P(|X - 8| < 6) = P(-6 < X - 8 < 6) = P(2 < X < 14) = P[8 - 2(3) < X < 8 + 2(3)]$.

This shows that:

$$\mu = 8 \text{ and } \sigma = 3 \text{ and } k = 2.$$

Thus, $P(|X - 8| > 6) \geq 1 - P(|X - 8| < 6) \geq 1 - [1 - 1/(2^2)] = 1/4$.

It may be noted that Chebychev's theorem hold for every type of distribution hence the results are usually weak. The value given by the theorem is a lower bound only. That is, we know that the probability of random falling within two standard deviations of the mean can be no less than 75%, but we never know that how much more it might actually be?

WORKSHEET 05

1. Classify the following random variables as discrete or continuous:

X: the number of automobile accidents each year in Virginia.

Y: the length of time to play 18 holes of golf.

M: the amount of milk produced yearly by a particular cow.

N: the number of eggs laid each month by 1 hen.

P: the number of building permits issued each month in a certain city.

Q: the weight of grain in pounds produced per acre.

2. A shipment of 8 similar microcomputers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives?

3. If a car agency sells 50% of its inventory of a certain foreign car equipped with airbags, find a formula for the probability distribution of the number of cars with airbags among the next 4 cars sold by the agency?

4. Find the probability distribution for the number of jazz records when 4 records are selected at random from a collection consisting of 5 jazz records, 2 classical records, and 3 polka records. Express your results by means of a formula?

5. Suppose that the error in the reaction temperature, in $^{\circ}\text{C}$, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}.$$

(a) Verify $\int_{-\infty}^{\infty} f(x) dx = 1$ (b) Find $P(0 < X \leq 1)$.

6. The total number of hours, measured in units of 100 hours that a family runs a vacuum cleaner over a period of one year is a continuous random variable X that has the density

function:
$$f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 \leq x < 2 \\ 0, & \text{elsewhere} \end{cases}.$$

Find the probability that over a period of one year, a family runs their vacuum cleaner

(a) less than 120 hours (b) between 50 and 100 hours

7. The proportion of people who respond to a certain mail – order solicitation is a continuous random variable X that has the density function

$$f(x) = \begin{cases} \frac{2(x+2)}{5}, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}.$$

(a) Show that $P(0 < X < 1) = 1$.

(b) Find the probability that more than $1/4$ but less than $1/2$ of the people contacted will respond to this type of solicitation?

8. The life time in hours of a battery cell is known to have the density function:

$$f(x) = \begin{cases} \frac{1}{100} e^{-x/100}, & x \geq 0 \\ 0, & \text{elsewhere} \end{cases}.$$

Find the probability that a battery cell will last more than 150 hours?

9. Two refills for a ball point pen are selected at random from a box that contains 3 blue refills, 2 red refills, and 3 green refills. If X is the number of blue refills and Y is the number of red refills selected, find (a) the joint probability function $f(x, y)$ and (b) $P[(X, Y) \in A]$, where A is the region $\{(x, y) | x + y \leq 1\}$.

10. A candy company distributes boxes of chocolates with a mixture of creams, toffees, and nuts coated in both light and dark chocolate. For a randomly selected box, let X and Y , respectively, be the proportions of the light and dark chocolates that are creams and suppose that the joint density function is:

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}.$$

(a) verify $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$.

(b) Find $P[(X, Y) \in A]$ where A is the region $\{(x, y) | 0 < x < 1/2, 1/4 < y < 1/2\}$.

11. Three resistors are selected at random from a box that contains 3 "Blue resistors" and 2 "Red resistors" and 3 "Black resistors". If X is the number of Red resistors and Y is the number of blue resistors are selected then find the joint probability function $f(x, y)$ and $P(x, y) = \{(x, y) | x + y \leq 1\}$.

12. The weekly demand for soft drinks, in thousands of liters, from a local chain of efficiency stores, is a continuous random variable X having the probability density

function,
$$f(x) = \begin{cases} 2(x-1), & 1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}.$$

Find mean and variance?

13. An accounting firm that does not have its own computing facilities rents time for a consulting company. The firm must plan its computing budget carefully and hence has studied the weekly use of CPU time approximately follows the p.d.f. given by:

$$f(x) = \begin{cases} \frac{3x^2(4-x)}{64}, & 0 \leq x \leq 4 \\ 0, & \text{elsewhere} \end{cases}.$$

Find its expected mean and variance of weekly CPU time?

14. Let X be the random variable denoting the hours of life in an electric bulb. Suppose X is distributed with density function

$$f(x) = \frac{1}{1000} e^{-x/1000}, \quad x > 0.$$

Find the expected life time of such a bulb. Also calculate the variance?

15. An overseas shipment of 5 foreign automobiles contains 2 that have slight paint blemishes. If an agency received 3 of the automobiles at random, list the elements of the sample space S using the letter B and N for “blemished” and “non- blemished” respectively, and then to each sample point a value x of the random variable X representing the number of automobiles purchased by agency with paint blemishes.

16. A grouped data-table for the weekly salaries of the 8 employees is presented in the following table.

Salary(\$) x	f	Relative frequency
240	3	0.375
320	2	0.250
450	1	0.125
600	2	0.250

Determine the mean and the standard deviation of the random variable X .

17. The continuous random variable X has a probability density function that is given by:

$$f(x) = x/4, 1 < x < 3$$

Find $P(1 < x < 2)$

18. A continues random variable Y has a p.d.f of the form $f(y) = A(Y^2 + 4)$, for $0 \leq Y \leq 1$. Find

- The value of A for which $f(y)$ is a probability density function.
- $P(1/2 \leq Y \leq 1)$

19. An accounting firm that does not have its own computing facilities rents time for consulting company. The firm must plan its computing budget carefully and hence has studied the weekly use of CPU time quick thoroughly .The weekly use of CPU time approximately follows the probability density function given by (measure in hours).

$$f(x) = \frac{3}{64} x^2(4-x), 0 < x < 4$$

$$= 0, \text{ elsewhere.}$$

Find expected value and variance of weekly CPU time.

20. In a transformation of digital information, the probability that a bit has a high, moderate and low distribution is 0.01, 0.04, and 0.95 respectively. Suppose that three bits are transmitted and that the amount of distribution each bit is assumed to be independent.

Let X and Y denote the number of bits with high and moderate distribution.

Determine: $P(X = 3, Y = 0)$ and $P(X = 2, Y = 1)$

21. The inner diameter of 100 piston rings from a manufacturing process are measured (in mm) with following results. Find the variance.

Inner dia	69.85	69.90	69.95	70.00	70.05	70.10
No: of rings	4	13	19	30	24	10

22. The PH of water samples from a specific lake is a random variable X with probability density function

$$f(x) = \frac{3}{8}(7 - x)^2, 5 \leq x \leq 7$$

$$= 0, \text{ elsewhere}$$

- Find $E(X)$ and $V(X)$.

- ii. Find an interval shorter than $(5, 7)$ in which at least $3/4$ of the PH measurement must lie.
- iii. Would you expect to see a PH- measurement below 5.5 very often? Why?
23. For each of the following, determine whether the given values can serve as the values of a probability distribution of random variable with the range $x = 1, 2, 3, \text{ and } 4$
- (a) $f(1) = 0.25, f(2) = 0.75, f(3) = 0.25, \text{ and } f(4) = -0.25$
- (b) $f(1) = 0.15, f(2) = 0.27, f(3) = 0.29, \text{ and } f(4) = 0.29$
- (c) $f(1) = 1/19, f(2) = 10/19, f(3) = 2/19, \text{ and } f(4) = 5/19$
24. For each of the following, determine whether the given values can serve as the values of a probability distribution of random variable with the given range.
- (a) $f(x) = \frac{x-2}{5}$ for $x = 1, 2, 3, 4, 5$;
- (b) $f(x) = \frac{x^2}{30}$ for $x = 0, 1, 2, 3, 4$;
- (c) $f(x) = \frac{1}{5}$ for $x = 0, 1, 2, 3, 4, 5$;
25. Verify that $f(x) = \frac{2x}{k(k+1)}$ for $x = 1, 2, 3, \dots, k$ can serve as the probability distribution of random variable with the given range
26. For each of the following, determine 'c' so that the function can serve as the probability distribution of random variable with the given range.
- a. $f(x) = cx$ for $x = 1, 2, 3, 4, 5$
- b. $f(x) = c \binom{5}{x}$ for $x = 0, 1, 2, 3, 4, 5$.
- c. $f(x) = cx^2$ for $x = 1, 2, 3, \dots, k$.
- d. $f(x) = c \left(\frac{1}{4}\right)^x$ for $x = 1, 2, 3, \dots$
27. For what value of k can $f(x) = (1-k)k^x$ serve as the values of the probability distribution of a random variable with the countable infinite range $x = 0, 1, 2, \dots$
28. Show that there are no values of c such that $f(x) = c/x$ can serve as the values of probability distribution of the random variable with the countable infinite range $x = 1, 2, 3, \dots$
29. Construct a probability histogram for each of the following probability distributions

$$(a) f(x) = \frac{\binom{2}{x} \binom{4}{3-x}}{\binom{6}{3}} \text{ for } x = 0, 1, 2$$

$$(b) f(x) = \binom{5}{x} \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{5-x} \text{ for } x = 0, 1, 2, 3, 4, 5$$

30. For each of the following, determine whether the given values can serve as the values of a distribution function of a random variable with the range $x = 1, 2, 3$, and 4 .

- (a) $F(1) = 0.3, F(2) = 0.5, F(3) = 0.8$, and $F(4) = 1.2$;
 (b) $F(1) = 0.5, F(2) = 0.4, F(3) = 0.7$, and $F(4) = 1.0$;
 (c) $F(1) = 0.25, F(2) = 0.61, F(3) = 0.83$, and $F(4) = 1.0$

31. Find the probability distribution of the random variable that has the probability distribution

$$f(x) = x/15 \text{ for } x = 1, 2, 3, 4, 5.$$

32. If X has the distribution function

$$F(x) = \begin{cases} 0 & \text{for } x < 1 \\ \frac{1}{3} & \text{for } 1 \leq x < 4 \\ \frac{1}{2} & \text{for } 4 \leq x < 6 \\ \frac{5}{6} & \text{for } 6 \leq x < 10 \\ 1 & \text{for } x \geq 10 \end{cases}$$

Find (a) $P(2 < X \leq 6)$ (b) $P(X = 4)$ (c) the probability distribution of X

33. The Probability density of the continuous random variable X is given by

$$f(x) = \begin{cases} \frac{1}{5}, & \text{for } 2 < x < 7 \\ 0, & \text{elsewhere} \end{cases}$$

Draw its graph and verify that the total area under the curve is equal to 1. Also find $P(3 < X < 5)$.

34. Show that $f(x) = 3x^2$ for $0 < x < 1$ represents a density function. Sketch a graph of this function, and indicate the area associated with the probability that $0.1 < x < 0.5$. Also calculate the probability that $0.1 < x < 0.5$.

35. Show that $f(x) = e^{-x}$ for $0 < x < \infty$ represents a probability density function. Sketch graph of this function and indicate the area associated with the probability that $x > 1$. Calculate the probability that $x > 1$.

36. The probability density of the random variable Y is given by

$$f(y) = \begin{cases} \frac{1}{8}(y+1) & \text{for } 2 < y < 4 \\ 0 & \text{elsewhere} \end{cases}$$

Find $P(Y < 3.2)$ and $P(2.9 < Y < 3.2)$.

37. The p.d.f of the random variable X is given by

$$f(x) = \begin{cases} \frac{c}{\sqrt{x}} & \text{for } 0 < x < 4 \\ 0 & \text{elsewhere} \end{cases}$$

Find the value of c . Then compute $P(X < 1/4)$ and $P(X > 1)$.

38. The distribution function of the random variable Y is given by

$$Y = \begin{cases} 1 - \frac{9}{y^2} & \text{for } y > 3 \\ 0 & \text{elsewhere} \end{cases}$$

Find $P(Y \leq 5)$ and $P(Y > 8)$.

39. Find the distribution function of the random variable X whose probability density is given by

$$f(x) = \begin{cases} \frac{x}{2} & \text{for } 0 < x \leq 1 \\ \frac{1}{2} & \text{for } 1 < x \leq 2 \\ \frac{3-x}{2} & \text{for } 2 < x < 3 \\ 0 & \text{elsewhere} \end{cases}$$

Also sketch the graph of these probability density and distribution functions.

40. The distribution function of the random variable X is given by

$$F(x) = \begin{cases} 0 & \text{for } x < -1 \\ \frac{x+1}{2} & \text{for } -1 \leq x < 1 \\ 1 & \text{for } x \geq 1 \end{cases}$$

Find $P(-1/2 < X < 1/2)$ and $P(2 < X < 3)$

41. The distribution function of random variable X is given by

$$F(x) = \begin{cases} 1 - (1+x)e^{-x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

Find $P(X \leq 2)$, $P(1 < X < 3)$, and $P(X > 4)$.

42. The number of minutes that a flight from Karachi to Islamabad is early or late is random variable whose probability density function is given by

$$f(x) = \begin{cases} \frac{1}{288}(36-x^2)e^{-x} & \text{for } -6 < x < 6 \\ 0 & \text{elsewhere} \end{cases}$$

Where negative values are indicative of the flights being early and positive values are indicative of its being late. Find the probabilities that one of these flights will be (a) at least 2 minutes early (b) at least 1 minute late (c) anywhere from 1 to 3 minutes early (d) exactly 5 minutes late.

43. The tread wear (in thousands kilometers) that car owners get with a certain kind of tire is a random variable whose probability density is given by:

$$f(x) = \begin{cases} \frac{1}{30}xe^{-x/3} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Find the probabilities that one of these tires will last (a) At most 18,000 kilometers

(b) any where from 27, 000 to 36, 000 kilometers (c) at least 48,000 kilometers.

44. In a certain city the daily consumption of water (in millions of liters) is a random variable whose probability density is given by:

$$f(x) = \begin{cases} \frac{1}{9} x e^{-x/3} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

What are the probabilities that on a given day (a) The water consumption in this city is more than 6 million liters (b) The water supply is inadequate if the daily capacity of this city is 9 million liters?

45. The total life (in years) of five year old dogs in a certain breed is a random variable whose distribution function is given by

$$F(x) = \begin{cases} 0 & \text{for } x \leq 5 \\ 1 - \frac{25}{x^2} & x > 5 \end{cases}$$

Find the probabilities that such five-year old dog will alive (a) Beyond 10 years (b) less than 8 years (c) anywhere from 12 to 15 years.

46. If the joint probability of X and Y is given by

$$F(x, y) = c (x^2 + y^2) \quad \text{for } x = -1, 0, 1, 3; y = -1, 2, 3.$$

Find the value of c. Also find (a) $P(X \leq 1, Y > 2)$ (b) $P(X = 0, Y \leq 2)$ (c) $P(X + Y > 2)$.

47. If the joint probability density of X and Y is given by:

$$f(x, y) = \begin{cases} 2 & \text{for } x > 0, y > 0, x + y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find (a) $P(X \leq \frac{1}{2}, Y \leq \frac{1}{2})$ (b) $P(X + Y > \frac{2}{3})$ (c) $P(X > 2Y)$

Also find an expression for the values of the joint distribution function of X and Y when $x > 0, y > 0$, and $x + y < 1$, and use it to verify result of part (a).

48. Find the joint probability density of the two random variables X and Y whose joint distribution function is given by

$$f(x) = \begin{cases} 1 - e^{-x} - e^{-y} + e^{-x-y} & \text{for } x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Find $P(X + Y > 3)$.

49. A sharpshooter is aiming at a circular target within radius 1. If we draw a rectangular system of coordinates with its origin at the center of the target, the coordinates of the point of impact (X, Y), are random variables having the joint probability density

$$f(x) = \begin{cases} \frac{1}{\pi}, & 0 < x^2 + y^2 < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find (a) $P[(X, Y) \in A]$, where A is the sector of the circle in the first quadrant bounded by the lines $y = 0$ and $y = x$; (b) $P[(X, Y) \in B]$, where $B = \left\{ (x, y) / 0 < x^2 + y^2 < \frac{1}{2} \right\}$.

50. A certain college gives aptitude test in the sciences and the humanities to all entering freshmen. If X and Y are respectively, the proportions of correct answers that a student

gets on the tests in the two subjects, the joint probability distribution of these variables can be approximated with the joint probability density

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & 0 < x < 1, \quad 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

What are the probabilities that a student will get (a) Less than 0.40 on both tests (b) more than 0.80 on the science test and less than 0.50 on the humanities test?

51. Suppose that P, the price (in dollars) of a certain commodity, and S, its total sales (in 10,000 units), are random variables whose joint probability distribution can be approximated closely with the joint probability density

$$f(p, s) = \begin{cases} 5pe^{-ps} \frac{1}{\pi}, & 0.20 < p < 0.40, s > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Find the probabilities that (a) price will be less than 30 cents and sales will exceed 20,000 units (b) the price will be between 25 cents and 30 cents and sales will be less than 10,000 units.

52. If the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} \frac{1}{4}(2x + y), & 0 < x < 1, 0 < y < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Find (a) the marginal density of X (b) the conditional density of Y given Y = 1/4.

Also find (a) Marginal density of Y (b) the conditional density of X given Y = 1

53. If the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} 24y(1 - x - y) & \text{for } x > 0, y > 0, x + y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find (a) marginal probability of X (b) marginal density of Y. Also determine whether the two random variables are independent.

54. If two cards are randomly drawn (without replacement) from an ordinary deck of 52 playing cards is the number of aces obtained in the first draw and W is the total number of aces obtained in both draws, find

(a) The joint probability distribution of Z and W.

(b) The marginal distribution of Z.

(c) The conditional distribution of W given Z = 1.

55. If X is proportion of persons who will respond to one kind of mail-order solicitation is the proportion of persons who will respond to another kind of mail-order solicitation and the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} \frac{2}{5}(x + 4y) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the probabilities that (a) at least 30% will respond to the first kind of mail-order solicitation (b) at most 50% will respond to the second kind of mail-order solicitation given that there has been a 20% response to the first kind of mail-order solicitation

56. The useful life (in hours) of a certain kind of vacuum tube is a random variable having the probability density

$$f(x) = \begin{cases} \frac{20.000}{(x+100)^3} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

If three of these tubes operate independently, find

(a) The joint probability density X_1 , X_2 and X_3 , representing the length of their useful lives (b) The value of $P(X_1 < 100, X_2 < 100, X_3 \geq 200)$.

57. Find the expected value and variance of the discrete random variable X having the probability distribution

$$f(x) = \frac{|x-2|}{7} \quad \text{for } x = -1, 0, 1, 3$$

58. Find the expected value and variance of the discrete random variable Y whose probability density is given by

$$f(x) = \begin{cases} \frac{1}{8}(y+1) & 2 < y < 4 \\ 0 & \text{elsewhere} \end{cases}$$

59. Find the expected value and variance of the discrete random variable Y whose probability density is given by

$$f(x) = \begin{cases} x & 0 < x < 1 \\ 2-x & 1 \leq x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

60. If X takes on the values 0, 1, 2 and 3 with the probabilities $\frac{1}{125}, \frac{12}{125}, \frac{48}{125}$ and $\frac{64}{125}$, find $E(X)$ and $E(X^2)$ and $E[(3X+2)^2]$.

61. If the probability density of X is given by

$$f(x) = \begin{cases} \frac{1}{x(\ln 3)} & 1 < x < 3 \\ 0 & \text{elsewhere} \end{cases}$$

Find $E(X)$, $E(X^2)$, $E(X^3)$, and $E(X^3 + 2X^2 - 3X + 1)$.

62. If the probability density of X is given by

$$f(x) = \begin{cases} \frac{x}{2} & \text{for } 0 < x \leq 1 \\ \frac{1}{2} & \text{for } 1 < x \leq 2 \\ \frac{3-x}{2} & \text{for } 2 < x < 3 \\ 0 & \text{elsewhere} \end{cases}$$

Find the expected value of $g(X) = X^2 - 5X + 3$.

63. The probability that Ms Sarah will sell a piece of property at profit of \$3,000 is $\frac{3}{20}$, the probability that she will sell it at a profit of \$1,500 is $\frac{7}{20}$, the probability that

she will break even is $7/20$, and the probability that she will lose is \$1,500 is $3/20$. What is her expected profit?

64. In a contractor's profit on a construction job can be looked upon as a continuous random variable having the probability density

$$f(x) = \begin{cases} \frac{1}{18}(x+1) & -1 < x < 5 \\ 0 & \text{elsewhere} \end{cases}$$

where the units are in \$1,000, what is her expected profit?

65. If two random variables have the joint density given by

$$f(x, y) = \begin{cases} e^{-x-y} & \text{for } x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Find the value $E(XY)$, $E(X)$, $E(Y)$, and $\text{cov}(X, Y)$.

66. If the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} \frac{1}{3}(x+y) & \text{for } 0 < x < 1, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find the variance of $W = 3X + 4Y - 5$.

CHAPTER SIX

BINOMIAL AND POISSON DISTRIBUTIONS

6.1 INTRODUCTION

As discussed in the previous chapters that discrete probability distribution gives the probability of every possible value of discrete random variable.

Many experiments consist of repeated independent trials with only two possible outcomes each one is complement of the other. For example, when a die is rolled a number may be odd or even, a number divisible by 3 or not. Similarly, when a fair coin is flopped it comes up with head or tail. A horse in the race may win or loose. A candidate appearing in aptitude test may get grade 'A' or not. A part produced in a manufacturing process gets through by quality control department or not, output of digital signal consisting of binary bit 0 or 1 and so forth. If the probability of each outcome remains same throughout the trials then such trials are called the Bernoulli trial and the experiment having n Bernoulli trials is called Bernoulli experiment or binomial experiment. The reason that Bernoulli experiment is also known as binomial experiment will become clear very soon. Thus an experiment is called Bernoulli or binomial experiment if it has the following properties:

- i. The outcomes consist of repeated and independent trial with two possible outcomes traditionally known as "success" or "failure" where the probability of success is denoted by 'p' and that of failure by 'q' with $q = 1 - p$.
- ii. The probability of success/failure remains unchanged throughout the trails.
- iii. The experiment must be repeated a fixed number of times say 'n' and looking for its success with 'k' times.

Bernoulli Trials

The situation considered here is one in which the same experiment is repeated n times and it is desired to find the probability that a particular event occurs exactly k of these times. For example, what is the probability that exactly four '1s' will be observed when a digital sequences consisting of word of 10 bits are released. Such repeated experiments as mentioned above are referred to as Bernoulli trials.

Consider some experiment for which the event A has a probability $P(A) = p$. Hence, the probability that event does not occur is $P(A^c) = q$ where $q = 1 - p$. Now repeat this experiment n time and assume that the trials are independent; that is, the outcome of any one trial does not depend in any way upon the outcome of previous or future trials and let the success of the event A occur k times out of n trials. Because the trials are independent, the probability of this event is:

$$\underbrace{(p.p.p. \dots p)}_{[k\text{-times}]} \cdot \underbrace{(q.q.q\dots q)}_{[(n-k)\text{ times}]} = p^k \cdot q^{n-k}.$$

It may be noted that there are many orders in which exactly k events occur because they can arise in any order. Furthermore, because of the independence, all of these other orders have exactly the same probability as the one specified above.

It is apparent that when one forms a sequence of length n :

The first A can occur in any one of the n ways.

The second A can occur in any one of $(n - 1)$ ways.

The third A can occur in any one of $(n - 2)$ ways.

The fourth A can occur in any one of $(n - 3)$ ways

.....

The k^{th} A can occur in $[n - (k - 1)] = (n - k + 1)$ ways.

Thus, the total number of different sequences of length n containing exactly k 'As' is simply the product of these various possibilities. Thus, since $k!$ orders of k event places are identical:

$$\frac{1}{k!} [n(n-1)(n-2)\dots(n-k+1)] = \frac{n!}{k!(n-k)!} = C_k^n = \binom{n}{k}$$

It is now possible to write the desired probability of A occurring k times as

$$P(A) = C_k^n p^k q^{n-k}$$

Binomial Distribution

Let p be the probability that an event occurs in a single trial (called the probability of success) and let $q = 1 - p$ be the probability that it fails to occur in a single trial (called the probability of failure). Then the probability that the event A occurs exactly k times out of n trials is denoted and defined as

$$P(X = x) = b(n, k; p) = C_k^n p^k q^{n-k}.$$

This is known as binomial or Bernoulli probability distribution. The name binomial distribution is given due to the contribution of binomial coefficients in the above expression.

Example 6.1.1 Assume that message word contains five bits and is transmitted in a channel in which the probability of any one bit being received in error is 0.01. What is the probability that no bits will be received in error?

Solution: Here $p = 0.01$, $q = 0.99$, $n = 5$ and $k = 0$. Using the binomial distribution formula, we obtain,

$$P(X = 0) = b(5, 0, 0.01) = C_0^5 (0.01)^0 (0.99)^5 = 0.951.$$

This shows that in the given channel, there is 95% chance that no bit is received in error.

Example 6.1.2 The probability that a certain kind of electronic device will survive in a given shock is 0.75. Find the probability that out of 5 such devices (a) exactly 2 (b) from 2 to 4 (c) between 2 and 5 (d) at least 3 (e) at most 3 devices will survive the shock.

Solution: Here $p = 0.75$, $q = 0.25$, $n = 5$. It is required to find:

$$(a) P(X = 2) = b(5, 2, 0.75) = C_2^5 (0.75)^2 (0.25)^3 = 0.088$$

$$(b) P(2 \leq X \leq 4) = P(X = 2) + P(X = 3) + P(X = 4)$$

$$= C_2^5 (0.75)^2 (0.25)^3 + C_3^5 (0.75)^3 (0.25)^2 + C_4^5 (0.75)^4 (0.25)^1 = 0.75$$

$$(c) P(2 < X < 5) = P(X = 3) + P(X = 4) = C_3^5 (0.75)^3 (0.25)^2 + C_4^5 (0.75)^4 (0.25)^1 = 0.66$$

$$(d) P(X \geq 3) = 1 - P(X < 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

$$= 1 - \left[C_0^5 (0.75)^0 (0.25)^5 + C_1^5 (0.75)^1 (0.25)^4 + C_2^5 (0.75)^2 (0.25)^3 \right] = 0.896$$

$$(e) P(X \leq 3) = [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)]$$

$$= \left[C_0^5 (0.75)^0 (0.25)^5 + C_1^5 (0.75)^1 (0.25)^4 + C_2^5 (0.75)^2 (0.25)^3 + C_3^5 (0.75)^3 (0.25)^2 \right] = 0.191$$

Theorem: Prove that binomial distribution is probability distribution.

Proof: We have to simply show that sum of all binomial probabilities is equal to 1;

that is, $\sum_{k=0}^n b(n, k, p) = 1$. Now,

$$\sum_{k=0}^n b(n, k, p) = \sum_{k=0}^n C_k^n p^k q^{n-k} = C_0^n p^0 q^n + C_1^n p^1 q^{n-1} + C_2^n p^2 q^{n-2} + \dots + C_n^n p^n q^0$$

$$= q^n + npq^{n-1} + \frac{n(n-1)}{2!} p^2 q^{n-2} + \dots + p^n = (q + p)^n = (1)^n = 1. \text{ Proved.}$$

NOTE: $C_n^n = C_0^n = 1$

Theorem: Prove that mean and variance of binomial distribution respectively are $\mu = np$ and $\sigma^2 = npq$.

Proof: We know that,

$$\text{Mean} = \mu = E(X) = \sum x_i f(x_i) = \sum_{x=0}^n x \cdot b(n, x, p) = \sum_{x=0}^n x \cdot C_x^n p^x q^{n-x}$$

$$= 0C_0^n p^0 q^n + 1C_1^n p^1 q^{n-1} + 2C_2^n p^2 q^{n-2} + 3C_3^n p^3 q^{n-3} + \dots + nC_n^n p^n q^0$$

$$= 0 + npq^{n-1} + 2 \cdot \frac{n(n-1)}{2!} p^2 q^{n-2} + 3 \cdot \frac{n(n-1)(n-2)}{3!} p^3 q^{n-3} + \dots + np^n$$

$$= np \left(q^{n-1} + (n-1)pq^{n-2} + \frac{(n-1)(n-2)}{2!} p^2 q^{n-3} + \dots + p^{n-1} \right)$$

$$= np(q + p)^{n-1} = np(1)^{n-1} = np \quad \text{Thus, mean} = E(X) = np.$$

We know that variance of discrete probability distribution is given by:

$$\sigma^2 = E(X^2) - \{E(X)\}^2 \quad (1)$$

Here, $\{E(X)\}^2 = (np)^2 = n^2 p^2$. Now consider,

$$E(X^2) = \sum x_i^2 f(x_i) = \sum_{x=0}^n x^2 \cdot b(n, x, p) = \sum_{x=0}^n x^2 \cdot C_x^n p^x q^{n-x} = \sum_{x=0}^n (x^2 - x + x) \cdot C_x^n p^x q^{n-x}$$

$$= \sum_{x=0}^n [x(x-1) + x] \cdot C_x^n p^x q^{n-x} = \sum_{x=0}^n [x(x-1)] \cdot C_x^n p^x q^{n-x} + \sum_{x=0}^n x \cdot C_x^n p^x q^{n-x}$$

$$= 0 + 0 + 2 \cdot 1C_2^n p^2 q^{n-2} + 3 \cdot 2C_3^n p^3 q^{n-3} + 4 \cdot 3C_4^n p^4 q^{n-4} + \dots + n(n-1)C_n^n p^n q^0 + E(X)$$

$$= 2 \cdot \frac{n(n-1)}{2!} p^2 q^{n-2} + 3 \cdot 2 \cdot \frac{n(n-1)(n-2)}{3!} p^3 q^{n-3} + 4 \cdot 3 \cdot \frac{n(n-1)(n-2)(n-3)}{4!} p^4 q^{n-4} + \dots$$

$$+ n(n-1)p^n + np$$

$$= n(n-1)p^2 \left[q^{n-2}(n-2)pq^{n-3} + \frac{(n-2)(n-3)}{2!} p^2 q^{n-4} + \dots + p^{n-2} \right] + np$$

$$= n(n-1)p^2(q+p)^{n-2} + np = n(n-1)p^2 + np = n^2p^2 - np^2 + np$$

Thus equation (1) becomes:

$$\sigma^2 = E(X^2) - \{E(X)\}^2 = n^2p^2 - np^2 + np - n^2p^2 = np(1-p) = npq. \text{ [Note: } 1-p=q \text{]}$$

Thus,

Mean of binomial distribution $\mu = np$

Variance of binomial distribution $\sigma^2 = npq$

Standard Deviation of binomial Distribution is $\sigma = \sqrt{npq}$

Example 6.1.3 A consignment of 20, 000 boxes is shipped where each box contains 20 P4-mother boards. The probability that a mother board is being defective is 0.02. Find the mean and standard deviation of number of defective mother boards. Also find the number of boxes containing (a) no defective (b) exactly 1 defective and (c) from 1 to 3 defective mother boards.

Solution: Here $p = 0.02$, $q = 0.98$, $n = 20$.

Mean $= np = 20(0.02) = 0.4$. Standard deviation $= \sigma = \sqrt{npq} = [(20)(0.02)(0.98)]^{.5} = 0.63$

(a) Let us first find the probability that there is no defective mother board in the box.

$$P(X=0) = {}^{20}C_0(0.02)^0(0.98)^{20} = 0.6676.$$

Thus out of 20, 000 boxes there will be $20,000 \times 0.6676 = 13,352$ boxes with no defective mother board.

(b) Now we find the probability that a box contains exactly one defective mother board.

$$P(X=1) = {}^{20}C_1(0.02)^1(0.98)^{19} = 0.2725$$

Thus out of 20, 000 boxes there will be $20,000 \times 0.2725 = 5450$ boxes with exactly one defective mother board.

(c) Let us find the probability that box contains from 1 to 3 defective mother boards.

$$P(1 \leq X \leq 3) = {}^{20}C_1(0.02)^1(0.98)^{19} + {}^{20}C_2(0.02)^2(0.98)^{18} + {}^{20}C_3(0.02)^3(0.98)^{17} = 0.3318$$

Thus out of 20, 000 boxes there will be $20,000 \times 0.3318 = 6636$ boxes that contain from 1 to 3 defective mother boards.

Poisson Distribution

Poisson distribution is named after French mathematician Simon Denis Poisson (1781 – 1840). Poisson published the derivation of this distribution in 1837. Its derivation was based on the assumption that in binomial distribution if n is very large (greater than 30) and p is very small (smaller than 0.01) then binomial distribution takes a new form which he named as Poisson distribution. An event that possesses this property is known as 'Rare Event'. Such event might be number of deaths by electric shocks per year; the number of noisy telephone calls per minute; number of persons born blind per year in some big city; the number of typing errors in a page of a text book; the number of radioactive particles emitted in a given period.

Derivation of Poisson distribution From Binomial distribution

As already has been discussed that Poisson distribution is a limiting case of binomial distribution with the conditions that n is very large and p is very small. Now taking the limit on n to approach to infinity, we have

$$\begin{aligned}\lim_{n \rightarrow \infty} b(n, x, p) &= \lim_{n \rightarrow \infty} C_x^n p^x q^{n-x} = \lim_{n \rightarrow \infty} \left(\frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \right) \quad \text{Note: } q = 1 - p \\ &= \lim_{n \rightarrow \infty} \left(\frac{(n)(n-1)(n-2)\dots(n-x+1)}{x!} p^x (1-p)^{n-x} \right).\end{aligned}$$

Let $np = \mu$ or $p = \mu/n$ and $q = 1 - \mu/n$. Thus

$$\begin{aligned}\lim_{n \rightarrow \infty} b(n, x, p) &= \lim_{n \rightarrow \infty} \left(\frac{(n)(n-1)(n-2)\dots(n-x+1)}{x!} \left(\frac{\mu}{n} \right)^x \left(1 - \frac{\mu}{n} \right)^{n-x} \right) \\ &= \lim_{n \rightarrow \infty} \frac{\mu^x}{x!} \cdot \frac{(n)(n-1)(n-2)\dots(n-x+1)}{n^x} \left(1 - \frac{\mu}{n} \right)^n \left(1 - \frac{\mu}{n} \right)^{-x} \\ &= \lim_{n \rightarrow \infty} \frac{\mu^x}{x!} \cdot n^x \frac{1 \cdot (1-1/n)(1-2/n)\dots[1-(x-1)/n]}{n^x} \left(1 - \frac{\mu}{n} \right)^n \left(1 - \frac{\mu}{n} \right)^{-x} \\ &= \lim_{n \rightarrow \infty} \frac{\mu^x}{x!} \cdot \frac{1 \cdot (1-1/n)(1-2/n)\dots[1-(x-1)/n]}{1} \left[\left(1 - \frac{\mu}{n} \right)^{n/\mu} \right]^\mu \left(1 - \frac{\mu}{n} \right)^{-x}\end{aligned}$$

Now as $n \rightarrow \infty$, $(1-1/n)$, $(1-2/n)$, $(1-3/n)$, and $(1-\mu/n)$ will approach to unity. In this case

$$\lim_{n \rightarrow \infty} b(n, x, p) = \lim_{n \rightarrow \infty} \frac{\mu^x}{x!} \left[\left(1 - \frac{\mu}{n} \right)^{n/\mu} \right]^\mu = \frac{e^{-\mu} \cdot \mu^x}{x!}. \quad \text{Note: } \lim_{n \rightarrow \infty} \left[\left(1 - \frac{\mu}{n} \right)^{n/\mu} \right]^\mu = e^{-\mu}$$

Thus $\lim_{n \rightarrow \infty} b(n, x, p) = \frac{e^{-\mu} \cdot \mu^x}{x!}$. This is Poisson probability distribution.

Definition: If n is large and p is small then binomial distribution takes a new form that is known as Poisson distribution and is defined as:

$$P(X = x) = P(\mu, x) = \frac{e^{-\mu} \cdot \mu^x}{x!}, \quad x = 0, 1, 2, \quad \text{where } \mu = np$$

Example 6.1.4 The probability that a patient recovers from cancer disease is 0.001. If it is known that 30 people are suffering from this disease what is the probability that (a) no person will survive (b) from 2 to 4 persons will survive (c) at least 1 person will survive and (d) at most 1 persons will survive from this disease.

Solution: Here $n = 30$ and $p = 0.001$. Thus $\mu = np = 30(0.001) = 0.03$. Since p is small and n is large, hence we may use Poisson distribution:

$$P(X = x) = P(\mu, x) = \frac{e^{-\mu} \cdot \mu^x}{x!}$$

$$(a) P(X = 0) = P(0.03, 0) = \frac{e^{-0.03} \cdot (0.03)^0}{0!} = 0.97.$$

This means that out of 30 persons there is 97% chance that no one will survive from this disease.

$$(b) P(2 \leq X \leq 4) = P(X = 2) + P(X = 3) + P(X = 4)$$

$$= \frac{e^{-0.03} (0.03)^2}{2!} + \frac{e^{-0.03} (0.03)^3}{3!} + \frac{e^{-0.03} (0.03)^4}{4!} = 0.00044$$

$$(c) P(X \geq 1) = 1 - P(X < 1) = 1 - P(X = 0) = 1 - \frac{e^{-0.03} \cdot (0.03)^0}{0!} = 1 - 0.97 = 0.03$$

$$(d) P(X \leq 1) = P(X = 0) + P(X = 1) = \frac{e^{-0.03} \cdot (0.03)^0}{0!} + \frac{e^{-0.03} \cdot (0.03)^1}{1!} = 0.97 + 0.0291 = 0.9991$$

Example 6.1.5 On the average a school remains closed for 2 days a week due to heavy snowfall. What is the probability the school will remain closed for (a) no day (b) 1 or 2 days a week.

Solution: Here $\mu = 2$ and by using the Poisson distribution $P(X = x) = P(\mu, x) = \frac{e^{-\mu} \cdot \mu^x}{x!}$,

we have:

$$(a) P(X = 0) = P(2, 0) = \frac{e^{-2} \cdot (2)^0}{0!} = 0.135.$$

This shows that there is 13.5% chance that school remains closed for no day in a week.

$$(b) P(X = 1) + P(X = 2) = \frac{e^{-2} \cdot (2)^1}{1!} + \frac{e^{-2} \cdot (2)^2}{2!} = 0.54.$$

This means that there is 54 % chance that school remains closed for 1 or 2 days a week.

Theorem: Prove that Poisson distribution is a probability distribution.

Proof: We have to prove that $\sum_x P(\mu, x) = 1$

$$\begin{aligned} \text{Now } \sum_x P(\mu, x) &= \sum_x \frac{e^{-\mu} \mu^x}{x!} = \frac{e^{-\mu} \mu^0}{0!} + \frac{e^{-\mu} \mu^1}{1!} + \frac{e^{-\mu} \mu^2}{2!} + \frac{e^{-\mu} \mu^3}{3!} \dots \\ &= e^{-\mu} \left[1 + \frac{\mu^1}{1!} + \frac{\mu^2}{2!} + \frac{\mu^3}{3!} \dots \right] = e^{-\mu} e^{\mu} = e^0 = 1 \end{aligned}$$

Theorem: Show that mean and variance of Poisson distribution is both equal to np.

Proof: By definition,

$$\begin{aligned} \text{Mean} &= \sum_x x f(x) = \sum_x x \cdot \frac{e^{-\mu} \mu^x}{x!} = 0 \cdot \frac{e^{-\mu} \mu^0}{0!} + 1 \cdot \frac{e^{-\mu} \mu^1}{1!} + 2 \cdot \frac{e^{-\mu} \mu^2}{2!} + 3 \cdot \frac{e^{-\mu} \mu^3}{3!} \dots \\ &= e^{-\mu} \left[\mu + \mu^2 + \frac{\mu^3}{2!} \dots \right] = \mu \cdot e^{-\mu} \left[1 + \mu + \frac{\mu^2}{2!} + \dots \right] = \mu \cdot e^{-\mu} \cdot e^{\mu} = \mu = np \end{aligned}$$

$$\begin{aligned} \text{Variance} &= \sigma^2 = E(X^2) - [E(X)]^2 = \sum_x x^2 \frac{e^{-\mu} \mu^x}{x!} - \mu^2 = \sum_x [x(x-1) + x] \frac{e^{-\mu} \mu^x}{x!} - \mu^2 \\ &= \sum_x [x(x-1)] \frac{e^{-\mu} \mu^x}{x!} + \sum_x x \frac{e^{-\mu} \mu^x}{x!} - \mu^2 \\ &= e^{-\mu} \left[0 + 0 + 2 \cdot \frac{\mu^2}{2!} + 3 \cdot 2 \cdot \frac{\mu^3}{3!} + 4 \cdot 3 \cdot \frac{\mu^4}{4!} + 5 \cdot 4 \cdot \frac{\mu^5}{5!} + \dots \right] + \mu - \mu^2 \\ &= e^{-\mu} \mu^2 \left[1 + \mu + \frac{\mu^2}{2!} + \frac{\mu^3}{3!} + \dots \right] + \mu - \mu^2 = e^{-\mu} e^{\mu} \mu^2 + \mu - \mu^2 = \mu = np \end{aligned}$$

Thus mean and variance of Poisson distribution are both equal to np.

Poisson Process

The number of outcomes occurring during a given time interval or in specified region are often called **Poisson Process**. Hence a Poisson process might generate observations for random variable X representing, for example, the number of telephone calls per hour, the number of days school is closed due to snow fall, etc. Poisson process possesses the following properties:

1. The number of outcomes in one time interval or specified region is independent of the number that occurs.
2. The probability that a single outcome will occur during a very short time interval or in a small region is proportional to the length of the time interval or the size of the region and does not depend on the number of outcomes occurring outside this time interval or region.
3. The probability that more than one outcome will occur in such a short time interval is negligible.

Under these properties, it can be shown that the probability for the number of occurrences of a random event in an interval of stated length t is given by Poisson distribution with the parameter λt . That is the Poisson formula is:

$$P(X = x) = P(\lambda t, x) = \frac{e^{-\lambda t} \cdot (\lambda t)^x}{x!}$$

Where t = number of units of times

x = number of occurrences in t unit of time

λ = average number of occurrences per unit of time

Example 6.1.6 On the average, five telephone calls arrive over a period of one minute to the switchboard of certain organization. Find the probability that over a period of one minute (a) exactly five phone calls will arrive (b) from four to six phone calls will arrive.

Solution: Let X be a random variable of phone calls arriving at the switchboard over a period of one minute. Using Poisson distribution with $\mu = 5$,

$$(a) P(X = 5) = \frac{e^{-5} 5^5}{5!} = 0.1755$$

This means that there is about 18% chance that in a period of four minutes five telephone calls will arrive.

$$(b) P(4 \leq X \leq 6) = P(X = 4) + P(X = 5) + P(X = 6)$$

$$= \frac{e^{-5} 5^4}{4!} + \frac{e^{-5} 5^5}{5!} + \frac{e^{-5} 5^6}{6!} = 0.4972$$

This means that there is about 50% chance that in a period of one minute from four to six telephone calls will arrive.

Example 6.1.7 On the average, four telephone calls arrive over a period of one minute to the switchboard of certain organization. Assuming a Poisson process, find the probability that in a 15-second interval, there will be four to six calls arrive at switchboard.

Solution: The readers may observe the difference between examples 4 and 5. Here one minute = 60 seconds, hence 15 seconds are the 1/4 the of unit time which is one minute. Hence $t = 1/4$. Therefore, average number of calls per 15-second interval will be

$$\lambda t = 4(1/4) = 1. \text{ Thus using the Poisson process, } P(X = x) = P(\lambda t, x) = \frac{e^{-\lambda t} \cdot (\lambda t)^x}{x!}, \text{ we have}$$

$$P(4 \leq X \leq 6) = P(X = 4) + P(X = 5) + P(X = 6)$$

$$= \frac{e^{-1}1^4}{4!} + \frac{e^{-1}1^5}{5!} + \frac{e^{-1}1^6}{6!} = 0.0189$$

Example 6.1.8 A company that prints fabrics has discovered that on the average there is a flaw in the fabric every 20 meter. The company buys the fabric in 60 meter lengths and wants to find out the probability that a 60 meter length will have more than two flaws.

Solution: In 20 meter fabric averagely there is one flaw hence in 60 meter fabric there will averagely be three flaws. Hence $\lambda t = 60(1/20) = 3$. Thus using the Poisson process,

$$P(X = x) = P(\lambda t, x) = \frac{e^{-\lambda t} \cdot (\lambda t)^x}{x!} \text{ we have:}$$

$$P(X > 2) = 1 - P(X \leq 2) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

$$= 1 - \left(\frac{e^{-3}3^0}{0!} + \frac{e^{-3}3^1}{1!} + \frac{e^{-3}3^2}{2!} \right) = 0.5768$$

WORKSHEET 06

1. The probability that a patient recovers from a delicate heart operation is 0.9. What is the probability exactly 5 of the next 7 patients having this operation survive?
2. The probability that a student is not a smoker is $2/3$. Find the probability that out of 5 students selected at random (i) 4 are smokers (ii) at least 4 are smokers?
3. 30% students have passed their year's Degree Examination. A random sample of 7 students is selected. Find the probability that:
 - (i) Exactly 5 students passed the examination
 - (ii) 3 to 5 students passed the examination.
4. If 10% of the television pictures tubes burn out before their guarantee period, what is the probability that out of 6 tubes 3 are burnt before guarantee expires?
5. A salesman plans his work so that he makes 6 business calls in a day. The probability that he will make a sale at randomly selected establishment is 0.3, what is the probability of 3 sales?
6. A survey report showed that in a locality 60% of the families earning more than Rs.15,000 own a car. If a sample of 18 families in this category is taken, find the probability that exactly 12 of them own a car?
7. Assume that, it is known that 10% of all electronic calculators are faulty. If a firm purchases five such calculators, what is the probability that there will be:
 - (i) Exactly 3 fault calculators
 - (ii) Between 1 and 4 faulty calculators?
8. A plant manufactures 8% defective items. A sample of 6 is selected. What is the probability that:
 - (i) At most two are defective
 - (ii) All are defective?
9. The probability that a certain kind of component will survive a given shock test is $3/4$. Find the probability that exactly 2 of the next 4 components tested survive.
10. A Binomial random variable has Mean 1.8 and Variance 1.44. Find n, p, and q?
11. The number of accidents per month in a city follows a Poisson distribution with the average of 5. Find the probability of 4 accidents in a month?
12. A manufacturing company's switch board receives an average of 4 incoming calls per minute. Find the probability of 5 incoming calls in a given minute?

13. (a) In a steel industry, the average number of fatal accidents per month is 0.5. What is the probability that there will be no fatal accident in the next month?
(b) In a steel industry, the average number of fatal accidents per month is 0.5. What is the probability that there will be one fatal accident in the next two months?
(c) The average number of cars arriving at petrol pump is 2 per minute. What is the probability that at any randomly selected four – minutes interval, 5 cars will arrive at the station?
14. Certain types of photographic film are produced and sold in rolls of 100 meter. If the expected number of defects on film rolls is 15 per 100 meter, what is the probability that no more than two defects occur in any 10 meter length?
15. If 5% of the electric bulbs manufactured by a company are defective, use Poisson distribution to find the probability that in a sample of 100 bulbs:
(i) none is defective (i) 5 bulbs will be defective.
16. During a laboratory experiment the average number of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond?
17. At a fair ground stall, there are a number of identical games and the probability that a single player wins a single game is 0.4. How many players need to play the game to ensure that at least one person wins is approximately 0.98?
18. An electronic product contains 40 integrated circuits. The probability of an integrated circuit being defective is 0.01. The product operates only if there are no defective integrated circuits. What is the probability that product operates.
19. A company manufactures relays and by random experiment it was found that the defective relays manufactured are 10%. The company dispatches 10 relays in a lot to the market. What is the probability that of (i) one defective relay (ii) less than 2 defective relays in a given lot? What will be the reliability of such a work?
20. A company manufactures screw which are send in a lot of 10, 000. The manufacturer operates a sampling scheme whereby a random sample of 10 is taken from each lot ready for dispatch and they are released if the number of defective screws in a sample is less than 3. Otherwise, the lot is rejected and reproduced.
i. If 10% of all the screws produced are known to be defective, find the portion of lots that will be rejected?
ii. If manufacturer replaces all his screws producing machines causing the proportion of defective screws drop to only 4% and now releases lots only if 2 out of 10 are defective, will the proportion of lots rejected decreases? If so, how many screws per lot will the manufacturer expects to save?
21. If 10% of television picture tubes burn before their guaranty period has expired, what is the probability that a merchant who has sold 200 such tubes will be forced to replace 20 of them?
22. A manufacturer of a consumer electronics product expects 2% of units to fail during the warranty period. A sample of 500 independent units is traced for warranty performance.
i. What is the probability that none fails during the warranty period?
ii. What is the expected number of failures during the warranty period?
iii. What is the probability that more than two units fail during the warranty period?

23. In a time sharing computer system, the number of teleport enquires averages 0.2 per millisecond and follows Poisson distribution.

- i. Find the probability that no enquires are made during the next millisecond?
- ii. Find the probability that no enquires are made during three millisecond?

24. Contamination is a problem in the manufacturing of optical storage disks. The number of contamination that occurs on the optical disk has Poisson distribution and the average number of particles per centimeter squared of media surface is 0.3. The area of disk under study is 200 cm^2 . Find the probability that 18 particles occur in the area of disk under study.

25. On the average a certain intersection results in 3 traffic accidents per month. What is the probability that in any given month at this intersection:

- (i) Exactly 5 accidents will occur
- (ii) Less than 3 accidents will occur
- (iii) At least 2 accidents will occur

26. A machine producing thin metallic strips is estimated to average one minor flaw per 24 meters. If flaws occur randomly, estimate the probability of

- (i) Two flaws in a 3 meter length
- (ii) No flaw in a 12 meter length

27. The number of cracks in a certain inter-state highway that are significant to require repair is assumed to follow Poisson distribution with a mean of two cracks per mile.

- (i) What is the probability that there is no crack in 5 miles of highway?
- (ii) What is the probability that at least one crack require repair in one half mile of highway?

28. If X has the discrete uniform distribution $f(x) = 1/k$, for $x = 1, 2, \dots, k$, show that

(a) Its mean is $\mu = \frac{k+1}{2}$ (b) Its variance is $\sigma^2 = \frac{k^2-1}{12}$

29. A multiple choice test consist eight questions and three answers to each question (of which one is correct). If a student answers each question by rolling a balanced die and checking the first answer if he gets a 1 or 2, the second answer if he gets a 3 or 4, and the third answer if he gets a 5 or 6, what is the probability that he will get exactly four correct answers?

30. An automobile safety engineer claims that one in 10 automobiles accidents is due to driver fatigue. Using the formula for the binomial distribution and rounding to four decimals, what is the probability that at least 3 of 5 automobile accidents are due to driver fatigue?

31. If 40% of the mice used in an experiment will become very aggressive within one minute after having been administrated an experiment drug, find the probability that exactly six of 15 mice that have been administrated the drug will become very aggressive within one minute, using the formula for the binomial distribution.

32. In a certain city, incompatibility is given as the legal reason in 70% of all divorce cases. Find the probability that five of the next six divorce cases filled in the city will claim incompatibility as the reason, using the formula for the binomial distribution.

33. A social scientist claims that only 50% of all high school seniors capable of doing college work actually go to college. Assuming that this claim is true, use 'Table-I' to find the probabilities that among 18 high school seniors capable of going college work (a) exactly 10 will go to college (b) al least10 will go to college (c) at most eight will go to college.

34. In planning the operation of a new school, one school board member claims that four out of five newly hired teachers will stay with the school for more than a year, while another school board claims that it would be correct to say three out of five. In the past, the two board members have been about equally reliable in their predictions, so in the absence of any other information we would assign their judgments equal weight. If one or the other has to be right, what probabilities would we assign to their claims if it were found that 11 of 12 newly hired teachers stayed with the school for more than a year?

35. A manufacturer claims that at most 5% of the time a given product will sustain fewer than 1,000 hours of operation before requiring service. Twenty products were selected at random from the production line and tested. It was found that three of them required service before 1,000 hours of operation. Comment on manufacturer's claim.

CHAPTER SEVEN

NORMAL DISTRIBUTION

7.1 INTRODUCTION

The normal probability distribution, which is considered as the foundation of modern statistical theory, was discovered by Abraham de Moivre's (1667 – 1754) as the limiting form of binomial distribution with the assumption that n (number of trials) is getting larger for a fixed value of probability p of success. This paper of de Moivre's remained unnoticed until 1924 when it was found in a library by Karl Pearson (1857 – 1936). The normal distribution is also called the **Gaussian distribution (1777 – 1855)**, who derived its equation using probability distribution of errors of measurements. It was Karl Pearson who called it normal distribution.

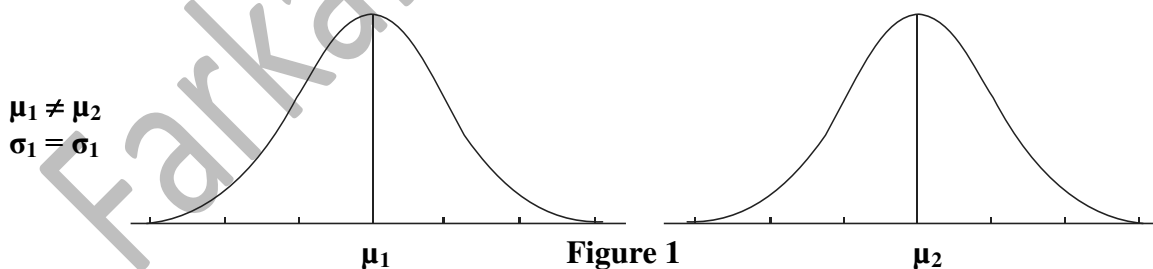
Normal Curve

Normal distribution is one of the most important continuous distributions whose applications are found almost in every walk of life such as metrological experiments; rainfall studies, heights and weights of males of similar age; the time to complete certain job or accuracy of measurements.

The probability distribution function (p.d.f) is called normal curve and is defined by:

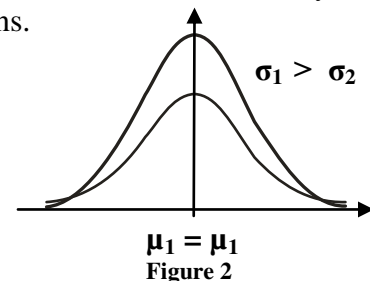
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{(x-\mu)}{\sigma}\right]^2}, \text{ for } -\infty < x < \infty, \text{ and } \sigma > 0$$

Here μ and σ respectively are the mean and standard deviation of normal curve and are known as its parameters. The graph of $f(x)$ defined above is symmetrical bell-shaped curve. The location and shape of the normal curve are determined by the parameters μ and σ . To understand this precisely, let us look at the following two figures.



In figure 1 we see that there are two normal curves with different means but same standard deviations. We observe that the shape of both curves is same where as they are located at different points which, are the values of their means.

In figure 2, we see two different normal curves with same mean but different standard deviations. We observe that the normal curve with larger standard deviation σ is spread out distribution.



Properties of Normal curve

There are several important properties of a normal curve. Some of these are:

1. The total area under the curve is equal to 1 since $f(x)$ is p.d.f.
2. The curve approaches the horizontal axis asymptotically on either side.
3. The highest point on the curve occurs at the mean, μ ; which is also the mode and median.
4. The curve is symmetrical about the vertical line $x = \mu$; that is, the area to the right of mean is the same as the area to the left of mean.
5. The position of the curve depends upon the value of μ .
6. The shape of the curve depends on the value of standard deviation σ . A larger value of σ results in a flatter, wider spread distribution.
7. Probabilities of normal random variables are given by areas under the normal curve.

Theorem: Show that normal curve is probability density function.

Proof: It has to be proved that $\int_{-\infty}^{\infty} f(x) dx = 1$ where $f(x)$ is p.d.f of normal random variable. Now,

$$\int_{-\infty}^{\infty} f(x) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left[\frac{(x-\mu)}{\sigma}\right]^2} dx$$

Substituting $z = (x - \mu)/\sigma$ or $\sigma dz = dx$. The limits will remain same for variable z . Thus,

$$\int_{-\infty}^{\infty} f(x) dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} dz = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-z^2/2} dz = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-z^2/2} dz \quad (1)$$

[Using the property of even function]

Putting $z^2/2 = u$ or $z dz = du$. Thus, $dz = du/z = z^{-1} du = (2u)^{-1/2} du$.

Thus equation (1) becomes

$$\int_{-\infty}^{\infty} f(x) dx = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-u} (2u)^{-1/2} du = \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-u} (u)^{-1/2} du = \sqrt{\frac{1}{\pi}} \Gamma(1/2) = \sqrt{\frac{1}{\pi}} \cdot \sqrt{\pi} = 1$$

Area under Normal Curve

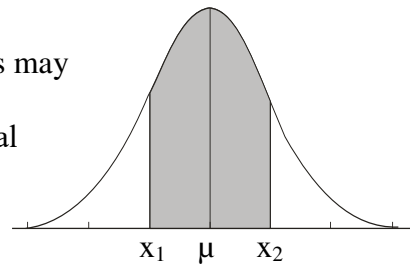
As mentioned above that normal distribution is a continuous type of distribution. The probability when the random variable X assumes the value in the range (x_1, x_2) is associated with evaluating the integral:

$$P(x_1 < X < x_2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{x_1}^{x_2} e^{-\frac{1}{2}\left[\frac{(x-\mu)}{\sigma}\right]^2} dx$$

This is shown in the figure. Now some important points may be noted before further discussions.

- i. If any one of μ and σ is changed the area under normal curve will also change.
- ii. If any one of limits x_1 and x_2 or both change, the area under normal curve will also change.

In any case one has to evaluate above integral which is laborious and time consuming. To save evaluating integral, a method has been devised to do away at least a problem of mean and standard deviation. A new curve then formed is known as normalized or



standardized normal curve. The probability (area under the normal curve) of normal random variable has been tabulated. These tabulated values are shown in “Appendix A”.

Normalized Normal Curve

As mentioned above a method has been devised to do away a problem of mean and standard deviation. If we substitute $Z = (X - \mu) / \sigma$, then it is proved that mean and standard deviation of random variable Z are zero and one respectively. It then becomes easy to handle the values x_1 and x_2 . Before we proceed to compute the probability say $P(x_1 < X < x_2)$ or area under normal curve between two points x_1 and x_2 , we prove that under the substitution $Z = (X - \mu) / \sigma$, the new random variable will have mean equal to zero and standard deviation equal to one.

Theorem: Prove that normalized normal random variable has mean equal to zero and standard deviation equal to one.

Proof: Mean of $Z = E(Z) = \mu_Z = E\left(\frac{X - \mu}{\sigma}\right) = \frac{1}{\sigma} [E(X) - E(\mu)] = \frac{1}{\sigma} (\mu - \mu) = 0$

$$\text{Var}(Z) = \sigma_Z^2 = E(Z^2) - \{E(Z)\}^2 = E\left(\frac{X - \mu}{\sigma}\right)^2 - \{0\}^2 = \frac{1}{\sigma^2} E(X - \mu)^2 = \frac{1}{\sigma^2} \sigma^2 = 1$$

Since, $\sigma_Z^2 = 1$ or $\sigma_Z = 1$.

Thus no matter what may be the value of μ and σ , the normalized normal curve will have mean equal to zero and standard deviation equal to one.

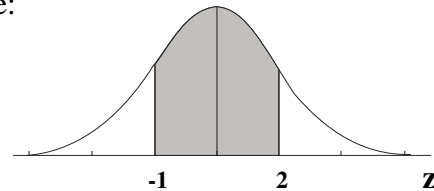
Area under Normalized Normal Curve

We produce below couple of examples that will make the readers how to use the normal table for calculating the required area/probabilities.

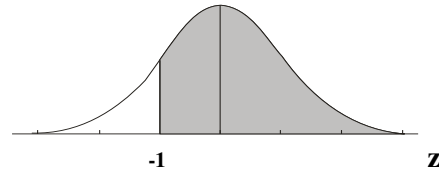
Example 7.1.1 By using area under normal curve table given in appendix A, find (i) $P(-1 < Z < 2)$, (ii) $P(Z > -1)$, (iii) $P(Z > 2)$, (iv) $P(Z < 3)$ and (v) $P(Z < -2)$.

Solution: Refer to appendix A the required values are:

(i) $P(-1 < Z < 2) = P(Z < 2) - P(Z < -1)$
 $= 0.9772 - 0.1587 = 0.8185$

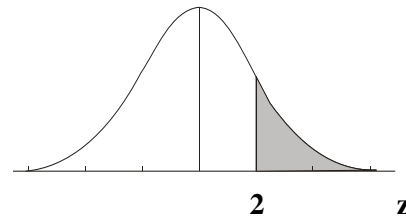
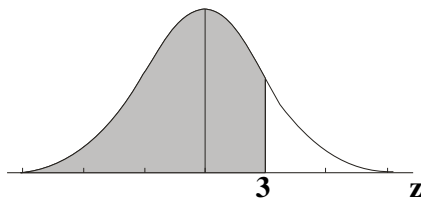


(ii) $P(Z > -1) = 1 - P(Z < -1) = 1 - 0.1587 = 0.8413$

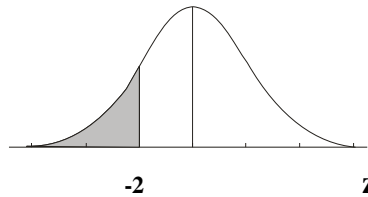


(iii) $P(Z > 2) = 1 - P(Z < 2) = 1 - 0.9772 = 0.0228$

(iv) $P(Z < 3) = 0.9987$



(v) $P(Z < -2) = 0.0228$



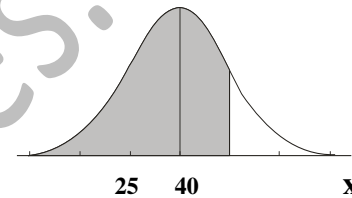
REMARK: Students are advised to ask their tutors to make them understand how to use the table giving the area under normal curve]

Determining Normal Distribution Probabilities

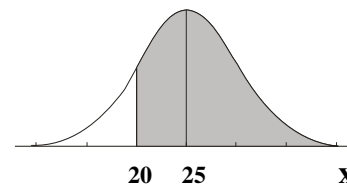
We have learnt how to compute the areas/probabilities under normalized normal curves; that is, the normal curve whose mean is zero and whose standard deviation is one. Now we are ready to find the area under any normal curve with any mean and standard deviation. To do this we use the transformation $Z = (X - \mu)/\sigma$ which will change the mean and standard deviations to zero and one respectively.

Example 7.1.2 If X is a normal random variable with mean 25 and standard deviation 10, find the following probabilities: (a) $P(X < 40)$, (b) $P(X > 20)$, (c) $P(15 < X < 40)$. Also find the point 'k' below which the total area is 45% and another point 'm' above which the total area is 85%.

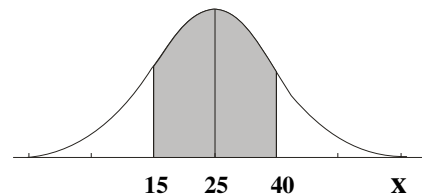
Solution: (a) $P(X < 40)$: Using the transformation $Z = (X - \mu)/\sigma$, we get: $Z = (40 - 25)/10 = 1.5$. Thus $P(X < 40) = P(Z < 1.5) = 0.9332$



(b) $P(X > 20)$: Using the transformation $Z = (X - \mu)/\sigma$, we get: $Z = (20 - 25)/10 = -0.5$. Thus $P(X > 20) = P(Z > -0.5) = 1 - P(Z < -0.5) = 1 - 0.3085 = 0.6915$



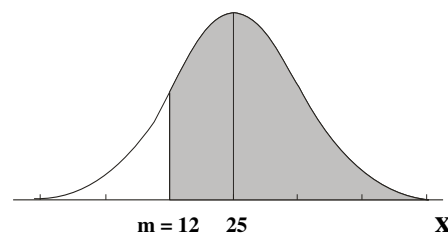
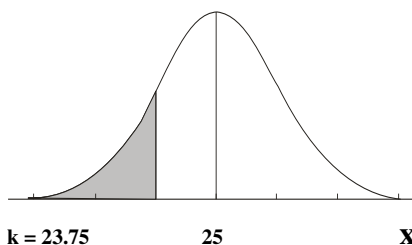
(c) $P(15 < X < 40)$ Using the transformation $Z = (X - \mu)/\sigma$, we get: $Z_1 = (15 - 25)/10 = -1$ and $Z_2 = (40 - 25)/10 = 1.5$. Thus $P(15 < X < 35) = P(-1 < Z < 1.5) = P(Z < 1.5) - P(Z < -1) = 0.9332 - 0.1587 = 0.7745$



Now, as given $P(X < k) = 45\% = 0.45$

Using the transformation $Z = (X - \mu)/\sigma = (k - 25)/10$

$P(X < k) = P[Z < (k - 25)/10] = 0.45$. From the table the point where 0.45 probability occurs is -0.125. Thus $(k - 25)/10 = -0.125$ or $k = 23.75$



Finally, as given $P(X > m) = 85\% = 0.85$. Using the transformation

$$Z = (X - \mu)/\sigma = (m - 25)/10,$$

$$P(X > m) = P[Z > (m - 25)/10] = 0.85 \text{ or } 1 - P[Z < (m - 25)/10] = 0.85.$$

$$\text{Or } P[Z < (m - 25)/10] = 0.15.$$

From the table the point where probability 0.15 occurs is -1.3. Thus,

$$(m - 25)/10 = -1.3 \text{ or } m = 12.$$

Applications of Normal Distribution

In this section we present application problems on normal distribution arising from engineering, science, education, management; etc.

Example 7.1.3 A company manufactures component for lath machine and claims that the lifetime of the component is normally distributed with mean life of 30 months and a standard deviation of four months. Find the probability that a component selected at random will last (a) less than 40 months (b) less than 25 months (c) more than 37 months and (d) between 26 and 32 months.

The company guarantees to replace any component that do not last more than 20 months. Find the probability that a component will have to be replaced under guarantee.

Solution: Let X be a normal random variable that represents the lifetimes of the components produced. As given $\mu = 30$ months and $\sigma = 4$ months. We are asked to find:

$$(a) P(X < 40) = P[Z < (40 - 30)/4] = P(Z < 2.5) = 0.9938.$$

This shows that there is 99% chance that selected component will have lifetime of less than 40 months.

$$(b) P(X < 25) = P[Z < (25 - 30)/4] = P(Z < -1.25) = 0.1056.$$

This means there is about 11% chance that selected component will have lifetime of less than 25 months.

$$(c) P(X > 37) = 1 - P(X < 37) = 1 - P[Z < (37 - 30)/4] = 1 - P(Z < 1.75) = 1 - 0.9599 = 0.0401$$

This shows that there is 4% chance that selected component will have lifetime of greater than 37 months.

$$(d) P(26 < X < 32) = P[Z < \{(26 - 30)/4\} - \{(32 - 30)/4\}] = P(-1 < Z < 0.5) \\ = P(Z < 0.5) - P(Z < -1) = 0.6915 - 0.1585 = 0.5328$$

This indicates that there is 53% chance that selected component will have lifetime between 26 and 32 months.

Now if company gives guarantee that any component will last for more than 0 months it means that replacement of the component will take place if the lifetime of the component is less than 20 months. Thus probability of replacement of component is:

$$P(X < 20) = P[Z < (20 - 30)/4] = P(Z < -2.25) = 0.0122.$$

Example 7.1.4 The management of certain manufacturing company of computer memory chips has observed that to complete designing a new compute memory chip follows a normal distribution with an average time of 50 hours and a standard deviation of 7 hours. What is the probability that technical hand will take (a) more than 60 hours (b) less than 30 hours and (c) between 50 and 60 hours to complete the chip design?

Solution:

$$(a) P(X > 60) = 1 - P(X < 60) = 1 - P[Z < (60 - 50)/7] = 1 - P(Z < 1.43) = 1 - 0.9236 = 0.0764$$

This shows that there is about 8% chances that chip will take more than 60 hours to be designed.

$$(b) P(X < 30) = P[Z < (30 - 50)/7] = P(Z < -2.85) = 0.0022.$$

This shows that there is 0.2% chance that chip will be designed for less than 30 hours.

$$(c) P(50 < X < 60) = P[(50 - 50)/7 < Z < (60 - 50)/7] = P(0 < Z < 1.43) \\ = P(Z < 1.43) - P(Z < 0) = 0.9236 - 0.5 = 0.4236$$

This means that there is 42% chance that chip will be designed between 50 and 60 hours.

Example 7.1.5 Manufacturing of a steel block follows a normal distribution and can be characterized in three categories: It takes less than 50 hours, between 50 and 60 hours and more than 60 hours. If 60%, 35% and 5% of these blocks were manufactured in the above categories, find the mean and standard deviation of this distribution.

Solution: Given that

$$P(X < 50) = 60\% \quad \text{or } P[Z < (50 - \mu)/\sigma] = 0.6, \quad \text{or } (50 - \mu)/\sigma = 0.26 \quad (1)$$

$$\text{Also } P(X > 60) = 5\% \quad \text{or } 1 - P(X < 60) = 0.05, \quad \text{or } -P(X < 60) = -0.95$$

$$\text{Or, } P(X < 60) = 0.95 \quad \text{or } P[Z < (60 - \mu)/\sigma] = 0.95, \quad \text{or } (60 - \mu)/\sigma = 1.65 \quad (2)$$

Solving equation (1) and (2), we get: $\sigma = 6.99$ and $\mu = 48.18$.

Example 7.1.6 If X is a normal distribution with mean equal to 40 and probability that its value lies between 25 and 55 is 0.8662, find the probability of X between 20 and 60.

Solution: Given that X is a normal distribution with $\mu = 40$ and that $P(25 < X < 55) = 0.8662$. Since the mean 40 lies in the mid of (25, 55) hence,

$$P(25 < X < 55) = 2 P(40 < X < 55) = 0.8662$$

$$\text{or } P(40 < X < 55) = 0.4331$$

$$\text{or } P[0 < Z < (55 - 40)/\sigma] = 0.4331$$

$$\text{Thus } P[Z < (55 - 40)/\sigma] - P(Z < 0) = 0.4331$$

$$\text{or } P[Z < (55 - 40)/\sigma] - 0.5 = 0.4331$$

$$\text{or } P[Z < (55 - 40)/\sigma] = 0.9331$$

$$\text{or } (55 - 40)/\sigma = 1.5. \quad \text{Thus, } \sigma = 10.$$

$$P(20 < X < 60) = P[Z < [(60 - 40)/10] < Z < [(20 - 40)/10]]$$

$$= P(2 < Z < -2) = P(Z < 2) - P(Z < -2) = 0.97725 - 0.01831 = 0.95894.$$

Example 7.1.7 The time to solve each MCQ question is found to follow normal distribution with mean of 30 seconds and standard deviation of 10 seconds. Find the 10th and 90th percentiles.

Solution: 10th percentile means the time taken to solve 10 % of MCQs and 90th percentile means time taken to solve 90% of all MCQs. Now

$$P(X < k) = P[Z < (k - 30)/10] = 0.1 \quad \text{or } (k - 30)/10 = -2.22 \quad \text{or } k = 7.8 \text{ sec.}$$

This means that time taken to complete the 10% of all MCQs is 7.8 seconds.

$$\text{Similarly, } P(X < k) = P[Z < (k - 30)/10] = 0.9 \quad \text{or } (k - 30)/10 = 1.28 \quad \text{or } k = 42.8 \text{ s}$$

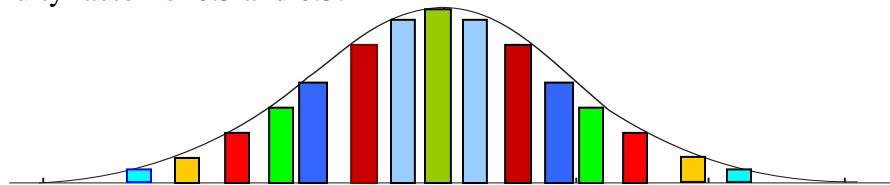
This means that time taken to complete the 19% of all MCQs is 42.8 seconds.

The Normal Approximation to Binomial Distribution

We know that binomial distribution is difficult to calculate for large value of n . If p is very small and n is very large then Poisson distribution gives better results. But if p is near to 0.5 and n is large then it has been observed that binomial distribution is approximated by normal distribution with $\mu = np$ and $\sigma = \sqrt{npq}$. This type of adjustment between discrete and continuous distributions is known as **continuity correction**. In this case,

$$P(X = x) = P(x - 0.5 < X < x + 0.5)$$

Thus continuity factor is -0.5 and 0.5.



Example 7.1.8 If X is a binomial random variable with $n = 20$ and $p = 0.47$, find (i) $P(X < 9)$ (ii) $P(X > 12)$.

Solution: Here, $\mu = np = 20(0.47) = 9.4$ and $\sigma = \sqrt{npq} = \sqrt{20(0.47)(1-0.47)} = 2.232$.

(i) Thus by continuity factor we are required to find $P(X < 8 + 0.5) = P(X < 8.5)$

Now, $P(X < 8.5) = P[Z < (8.5 - 9.4)/2.232] = P(Z < -0.403) = 0.3446$.

(ii) By continuity factor, we require to find $P(X > 12 - 0.5) = P(X > 11.5)$

Now, $P(X > 11.5) = 1 - P(X < 11.5) = 1 - P[Z < (11.5 - 9.4)/2.232]$
 $= 1 - P(Z < 0.941) = 1 - 0.8264 = 0.1736$.

WORKSHEET 07

- Given a normal distribution $\mu = 12$ and $\sigma = 2$ find
 - The area below 13.5
 - The area above 10
 - The area between 6 and 18.
- The burning time of an experimental rocket is a random variable having the normal distribution with $\mu = 4.76$ seconds and $\sigma = 0.04$ seconds. What is the probability that this kind of rocket will burn:
 - less than 4.66 seconds
 - more than 4.80 seconds
 - anywhere from 4.70 to 4.82 seconds.
- A certain type of storage battery lasts on average 3.0 years with a standard deviation of 0.5 year. Assuming that the battery lives are normally distributed, find the probability that a given battery will last less than 2.3 years?
- An electrical firm manufactures light bulbs that have a life, before burnt out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours?
- A certain machine makes electrical resistors having a mean resistance of 40 ohms and a standard deviation of 2 ohms. Assuming that the resistance follows a normal distribution and can be measured to any degree of accuracy, what percentage of resistors will have a resistance exceeding 43 ohms?
- A research scientist reports that mice will live an average of 40 months when their diets are sharply restricted and then enriched with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months, find the probability that a given mouse will live:
 - more than 32 months
 - less than 28 months
 - between 37 and 49 months.
- A petrol station sells, on the average, 14500 liters of petrol, per day with a standard deviation of 2500 liters. If a manager stocks 20,000 liters on a particular day, what is probability that:
 - At most 20,000 liters, will be sold
 - More than 10,000 liters, will be sold.
- Given a normal distribution with $\mu = 40$ and $\sigma = 6$, find the value x that has (i) 38% of the area below it and (ii) 5% of the area above it.
- On an examination the average grade was 74 and the standard deviation was 7. If 12% of the class are A's, and the grades are curved to follow a normal distribution, what is the lowest possible A and the highest possible B?

10. The average life of a certain type of small motor is 10 years, with a standard deviation of 2 years. The manufacturer replaces free all motors that fail while under guarantee. If he is willing to replace only 3% of the motors that fail, how long a guarantee should he offer? Assume that the lives of the motors follow a normal distribution.
11. Using normal approximation to binomial distribution find (i) $P(X < 30)$, (ii) $P(X > 35)$ and (iii) $P(22 < X < 32)$ if $p = 0.6$ and $n = 40$.
12. The time required to assemble a particular machine is normally distributed with a mean of 80 minutes & Variance of 49 min.
- What is the probability of someone assembling a machine in one hour or less?
 - What is the probability of someone to assemble the machine in ore than 50 minutes but less than 75 minutes?
13. The spot speeds at a particular location are normally distributed with a mean of 51.7 km/h and standard deviation of 8.3 km/h. What is the probability that:
- The speed exceeds 65 km/h?
 - The speed lies between 40 and 70 km/h?
 - What is the 85th percentile speed?
14. The Line width of a tool used for semiconductor manufacturing is assumed to be normally distributed with a mean of 0.5 micrometer and standard deviation of 0.05 micrometer.
- What is the probability that the line width is greater than 0.62 micrometer?
 - The Line width of 90% tool is below what value?
15. An electric wires produced by some company for use in certain system have a normally distribution with a mean of 0.13 ohms and standard deviation of 0.05 ohms.
- What is the probability that randomly selected wires will have a resistance between 0.12 and 0.14?
 - Determine 96% of confidence line for resistance of wires?
16. Assume that the current measurement in a strip of a wire follows a normal distribution with a mean of 10 m-A and a standard deviation of 2 m-A. What is the probability that a measurement exceeds 13 m-A?
17. If 10% of TV picture tubes burn out before their guarantee has expired,
- What is the probability that a merchant who has sold 100 such tubes will be forced to replace 20 of them?
 - What is the probability that out of 100 TVs, merchant will replace at least 5 and more than 15 tubes?
18. The Past experience shows that the number of bungalows constructed by a construction company is normally distributed with a mean of 400 Bungalows and standard deviation of 44. A company is given a new project to construct houses. What is the probability that the company will construct
- Between 350 & 375 bungalows
 - More than 300 bungalows
 - Less than 350 bungalows at the end if required period.
19. If Z is a random variable having the standard normal distribution, find the probabilities that it will take on a value (a) greater than 1.14 (b) greater than -0.36 (c) between -0.46 and -0.09 (d) between -0.58 and 1.12.
20. If Z is a random variable having the standard normal distribution, find:
- (a) $P(Z < 1.33)$ (b) $P(Z \leq -0.79)$ (c) $P(0.55 < Z < 1.22)$ (d) $P(-1.90 \leq Z \leq 0.44)$

21. Find z if the standard normal curve area (a) Between 0 and z is 0.4726 (b) to the left of z is 0.9868 (c) to the right of z is 0.1314 (d) between $-z$ and z is 0.8502.
22. Suppose that during periods of transcendental meditation the reduction of a person's oxygen consumption is a random variable having a normal distribution with $\mu = 37.6$ cc per minute and $\sigma = 6.6$ cc per minute. Find the probabilities that during a period of transcendental meditation a person's oxygen consumption will be reduced by (a) at least 44.5 cc per minute (b) at most 35.0 cc per minute (c) anywhere from 30 to 40 per minute.
23. In a photographic process, the developing time of prints may be looked upon as a random variable having the normal distribution with $\mu = 15.4$ sec and $\sigma = 0.48$ sec. Find the probabilities that the time it takes to develop one of the prints will be (a) at least 16 sec (b) at most 14.2 sec (c) anywhere from 15 to 15.8 sec.
24. A random variable has a normal distribution with $\sigma = 10$. If the probability that the random variable will take on a value less than 82.5 is 0.8212, what is the probability that it will take on a value greater than 58.3?
25. Use the normal approximation to the binomial distribution to determine (to four decimal places) the probability of getting seven heads and seven tails in 14 flips of a balanced coin.
26. If 23% of all patients with high blood pressure have bad side effects from a certain kind of medicine, use the normal approximation to find the probability that among 120 patients with high blood pressure treated this medicine more than 32 will have bad side effects?
27. If the probability is 0.20 that a certain bank will refuse a loan application, use the normal approximation to determine (to three decimals) the probability that the bank will refuse at most 40 of 225 loan applications.
28. To illustrate the law of large numbers, use the normal approximation to the binomial distribution to determine the probability that the proportion of heads will be anywhere from 0.49 to 0.51 when a balanced coin is flipped (a) 100 times (b) 1, 000 times (c) 10,000 times.

CHAPTER EIGHT

SAMPLING DISTRIBUTION AND INTERVAL ESTIMATION

8.1 INTRODUCTION

In chapter one it discussed that there are two ways to collect the data, through experiments and by surveying. An experiment is preferred when the population is under control and survey is used when it is not possible to control the population.

It is often very difficult to make a survey as the population contains large number of items and it is not only time consuming but expensive as well. Instead, a smaller group or a **sample** is used to decide the population characteristics, such as, population mean and population standard deviation. For example, it is not possible to test every bulb produced in a factory for its life in hours. Instead we take couple of bulbs and test them for their lives. It is important to note that the sample must be truly representative of the population otherwise the conclusion made on the basis of sample may be misleading. For example, if a survey is made to know which lipstick color is more liked by ladies and the question is asked to gents, then certainly misleading results will occur.

Question now arises how to collect the sample. There are many methods of collecting the samples. All these methods aim to ensure that sample is the representative of the population. Of these entire methods one which is most simple and universally recognized is to use *Simple Random Sample (SRS)* method. In this method items are selected in such a way that each one is equally likely. SRS method makes use of table of *Random Numbers* given in Appendix 'B'. The method is illustrated by the following example.

Example 8.1.1: A computer memory chips manufacturer employees 1000 people. It is desired by the management to make a survey to see how many workers have remained absent from the job due to sickness throughout a year. A sample of 10 workers is chosen. Estimate the average number of days of absenteeism due to sickness.

Solution: The first step is to allocate ID Numbers to each employee and let these be from 0001 to 1000. Now pick an arbitrary point in the random number table (APPENDIX-II), say for instance a number in the first row and first column. Start with this number, search in this column ten numbers which are less than or equal to 1000 taking the right four digits of these numbers. Continue the process till 10 numbers are chosen. It may be noted that if any a number chose is repeated delete it from the list and choose another number that is less than or equal to 1000. The numbers are, 0629, 0572, 0542, 0504, 0368, 0325, 0693, 0483, 0274 and 0935.

The management now finds from the leave record the number of days the employee holding the above ID numbers who have remained absent due to sickness. The result for example is as follows:

ID#	0629	0572	0542	0504	0368	0325	0693	0483	0274	0935	Total
Days	2	2	4	3	0	1	4	5	1	3	25

Now compute the mean and standard deviation. They are,

Mean = $\bar{x} = \Sigma x_i / n = 25/10 = 2.5$ days.

Variance = $s^2 = \Sigma (x_i - \bar{x})^2 / (n - 1) = 2.5$. Thus, standard deviation = $s = 1.5$.

This shows that mean number of days the employees remain absent from the factory is 2.5 days with standard deviation equal to 1.5 days.

Point Estimation

A sample is used to estimate some of the features of population known as *population parameters* such as population mean and population standard deviation. A *point estimate* is a number which represents an estimate of the population parameters and is based on a sample. It may be noted that sample mean and population mean are denoted by $\bar{x}$ and μ respectively. Similarly, sample standard deviation and population standard deviation are denoted by $\bar{s}$ and σ respectively. Thus, in the above example we may estimate that $\bar{x} = 2.5 \sim \mu$ and that is, we make an inference that mean number of days the workers remained absent due to sickness in a year is 2.5 days. Similarly, $\bar{s} = 1.5 \sim \sigma$.

Sampling Distribution

We know that when a sample is taken from a population it can be used to estimate the population mean and population standard deviation. If many samples of the same size are taken from a population and a point estimate of the population mean is calculated for all samples, then each estimate will be different from each other.

The main draw back of using the random numbers is that we only use one sample and point estimation so obtained is not very much informative.

To overcome this problem, if instead we take all possible samples, a frequency distribution is obtained which is known as *sampling distribution of sample means*. This sampling distribution of sampling means is usually denoted by $\bar{X}$ or $\mu_{\bar{x}}$. To illustrate the idea more clearly, let us consider a population consisting of four elements says 1, 2, 3 and 4 with a sample size of 2. It may be noted that sampling distribution is done in two ways.

(a) Sampling with replacement

(b) Sampling without replacement

Sampling Distribution with Replacement

Let the population be, 1, 2, 3, 4 then population size $N = 4$. Let sample size $n = 2$.

Therefore $\mu = (1 + 2 + 3 + 4)/4 = 2.5$ and $\sigma = 1.118$

Sample	Mean ($\bar{x}$)
(1, 1)	1
(1, 2)	1.5
(1, 3)	2
(1, 4)	2.5
(2, 1)	1.5
(2, 2)	2
(2, 3)	2.5
(2, 4)	3
(3, 1)	2
(3, 2)	2.5
(3, 3)	3
(3, 4)	3.5
(4, 1)	2.5
(4, 2)	3
(4, 3)	3.5
(4, 4)	4

Sampling Distribution with Replacement			
Mean ($\bar{x}$)	f	$\bar{x} f_i$	$(\bar{x} - \bar{X})^2 f$
1	1	1	2.25
1.5	2	3	2
2	3	6	0.75
2.5	4	10	0
3	3	9	2.25
3.5	2	7	2
4	1	4	2.25
Sum	16	40	10

$$\mu_{\bar{x}} = \bar{X} = \frac{\Sigma \bar{x} f}{\Sigma f} = \frac{40}{16} = 2.5,$$

$$\sigma_{\bar{x}} = \sqrt{\frac{\Sigma (\bar{x} - \bar{X})^2 f}{(\Sigma f)}} = \sqrt{\frac{10}{16}} = 0.79$$

Sampling Distribution without Replacement

Using the above information, we have

Sample		Mean $\bar{x}$
(1, 2)		1.5
(1, 3)		2
(1, 4)		2.5
(2, 3)		2.5
(2, 4)		3
(3, 4)		3.5

Sampling Distribution without Replacement			
Mean ($\bar{x}$)	f_i	$\bar{x} f_i$	$((\bar{x} - \bar{X})^2 f_i)$
1.5	1	1.5	1
2	1	2	0.25
2.5	2	5	0
3	1	3	0.25
3.5	1	3.5	1
Sum	6	15	2.5

$$\mu_{\bar{x}} = \bar{X} = \frac{\sum \bar{x} f}{\sum f} = \frac{15}{6} = 2.5,$$

$$\sigma_{\bar{x}} = \sqrt{\frac{\sum (\bar{x} - \bar{X})^2 f}{(\sum f)}} = \sqrt{\frac{2.5}{6}} = 0.6457$$

We may observe that mean of population is same as that of sampling distribution which may be done with or without replacement. But the standard deviation of the population and that of sampling distribution are different. It has been proved that the relation between standard deviation of population and sampling distribution is given by:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \quad \text{if the sampling is done with replacement and}$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \quad \text{if the sampling is done without replacement or sample size is less than 30.}$$

The factor $\sqrt{\frac{N-n}{N-1}}$ is known as *finite population correction* (fpc) factor.

Let us see this:

Now, for sampling distribution with replacement

$$\frac{\sigma}{\sqrt{n}} = \frac{1.118}{\sqrt{2}} = 0.79 = \sigma_{\bar{x}}.$$

And for sampling distribution without replacement

$$\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} = \frac{1.118}{\sqrt{2}} \times \sqrt{\frac{2}{3}} = 0.79 = 0.645 = \sigma_{\bar{x}}.$$

These results agree with the results obtained in the above tables.

It may be noted that $\sigma_{\bar{x}}$ is known as *standard deviation of sampling distribution of the sample means* and is usually called 'Standard Error of Sample Means'.

Further to note that if standard deviation of population is not known then 'Estimated Standard Error of Sample Means' may be used in place of it which is defined as:

$$s_{\bar{x}} = s / \sqrt{n}$$

Example 8.1.2 A population is known to have a standard deviation of 25. A sample of 10 items is taken from this population. Find the standard error of the sample means.

Solution: Here, $\sigma_{\bar{x}} = \sigma / \sqrt{n} = 25 / \sqrt{10} = 7.91$

Example 8.1.3 In the example for the leave record of employees working with industry manufacturing the computer chips, find the 'Estimated Standard Error of sample means.

Solution: Here, $s_{\bar{x}} = s / \sqrt{n} = 1.5 / \sqrt{10} = 0.474$

Sampling from a Normal and Non-Normal Population

It has been seen that if population distribution is normal then the distribution of the sample means is also normal with mean equal to $\mu_{\bar{x}}$ and variance equal to $\sigma_{\bar{x}}^2 = \sigma^2 / n$.

If the population is not normally distributed, the distribution of sampling means approaches a normal distribution with mean equal to μ and variance of $\sigma_{\bar{x}}^2 = \sigma^2 / n$, as above, provided the sample size n becomes large. This is known as 'Central Limit Theorem'.

Applications of Central Limit Theorem

Example 8.1.4 From the past experience it is known that average number of cracks on a national highway is 4 per kilometer with standard deviation of 15. What is the probability that a random sample of 100 kilometers of highway will yield a mean number of cracks between (a) between 2 to 5 cracks (b) less than 2 cracks and (c) more than 3 cracks per kilometer (d) If mean is not known then find the probability that mean value differs from the true mean by more than 3 cracks.

Solution: The sample of 100 is well above 30 and can be regarded as large. Thus the sampling distribution of the sample mean can be close to normal. Thus central limit theorem can be invoked. Now the required probabilities are found as:

Using the transformation: $\bar{Z} = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$

$$\text{a. } \bar{z}_1 = \frac{\bar{x}_1 - \mu}{15 / \sqrt{100}} = \frac{2 - 4}{15 / \sqrt{100}} = -2 / 1.5 = -1.34, \quad \bar{z}_2 = \frac{\bar{x}_2 - \mu}{15 / \sqrt{100}} = \frac{5 - 4}{15 / \sqrt{100}} = 1 / 1.5 = 0.67$$

$$\text{Thus } P(2 < \bar{X} < 5) = P(-1.34 < \bar{Z} < 0.67) = 0.74857 - 0.09012 = 0.65845$$

This shows that there is 66% chance that average number of cracks per kilometer on national highway is between 2 to 5.

$$\text{b. } \bar{z} = \frac{\bar{x} - \mu}{15 / \sqrt{100}} = \frac{2 - 4}{15 / \sqrt{100}} = -2 / 1.5 = -1.34$$

$$\text{Thus, } P(\bar{X} < 2) = P(\bar{Z} < -1.34) = 0.09012$$

This shows that there is 9% chance that average number of cracks per kilometer on national highway is less than 2.

$$\text{c. } \bar{z} = \frac{\bar{x} - \mu}{15 / \sqrt{100}} = \frac{3 - 4}{15 / \sqrt{100}} = -0.67$$

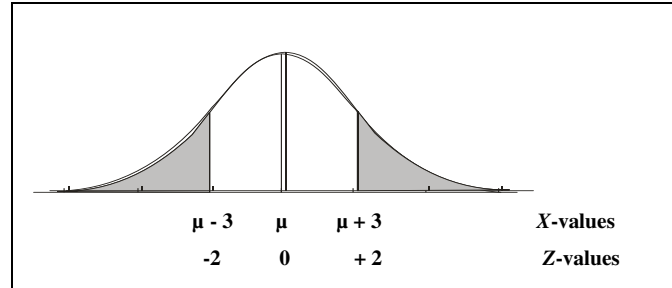
$$\text{Thus } P(\bar{X} > 3) = 1 - P(\bar{X} < 3) = 1 - P(\bar{Z} < -0.67) = 1 - 0.25143 = 0.74857$$

This shows that there is 75% chance that average number of cracks per kilometer on national highway is greater than 3.

$$\text{d. } \bar{z}_1 = \frac{(\mu - 3) - \mu}{15 / \sqrt{100}} = \frac{-3}{1.5} = -2 \quad \text{and} \quad \bar{z}_2 = \frac{(\mu + 3) - \mu}{15 / \sqrt{100}} = \frac{3}{1.5} = 2$$

$$\begin{aligned}
 P(\bar{X} < \mu - 3) + P(\bar{X} > \mu + 3) &= P(\bar{Z} < -2) + P(\bar{Z} > 2) \\
 &= 2P(\bar{Z} < -2) = 2(0.01831) = 0.03662 \approx 0.04
 \end{aligned}$$

Thus if average number of cracks is not known even we can predict that there is 4% chance that the number of cracks per kilometer will differ from the true population mean by more than three cracks. This is depicted in the following figure.



8.2 INTERVAL ESTIMATION OF POPULATION MEAN

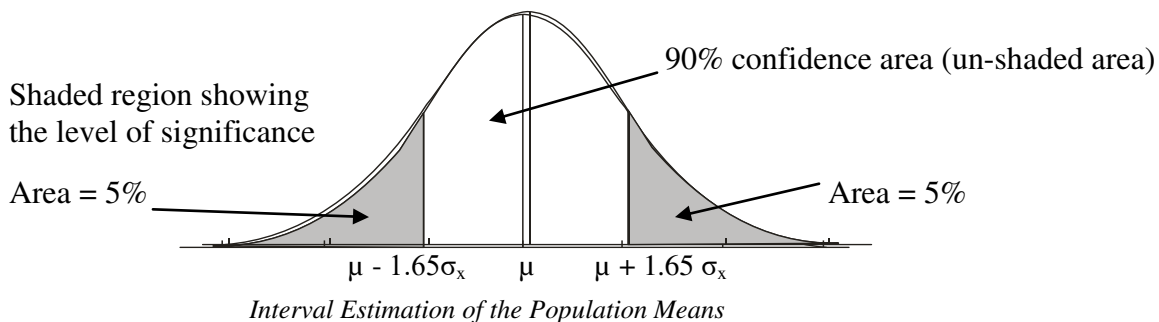
With point estimation, the sample mean is considered as a good estimator of the unknown population mean. But we know that it is unlikely that it will be exactly equal to unknown population parameters. Therefore, *interval estimation* is the best way to estimate the population parameters. It uses the principle to define an interval within which, we can state with certain level of confidence where the true mean lies. The probability that a specified interval contains the unknown population parameter is known as *confidence* and the interval itself is called *confidence interval* or *interval estimate*. It is defined as

$$\mu = \bar{X} \pm \bar{Z}_{\alpha/2} \sigma / \sqrt{n}$$

It is common to choose a 95 per cent confidence level, that is to say, an interval such that there is 95% chance that it will contain population mean. Other confidence intervals used are 90% and 99%. From the normal distribution table, one can see that points which contain 90%, 95% and 99% of the area under normal curve are as follows.

α	$1 - \alpha$	Value of $Z_{\alpha/2}$	Area in the interval $\mu \pm Z_{\alpha/2}$
10%	90%	± 1.65	0.90
5%	95%	± 1.96	0.95
1%	99%	± 2.58	0.99

Here α is known as level of significance. This means that if we are 90% confident that population mean lies in the interval $\mu = \bar{X} \pm \bar{Z}_{\alpha/2} \sigma / \sqrt{n}$ then there is 10% chance that it may not be true. This is called *level of significance*. This is shown in the following figure.



Now 90% confidence interval for the population mean is calculated as

$$\mu = \bar{X} + Z_{\alpha/2} \sigma_{\bar{X}} = \bar{X} \pm 1.65 \sigma / \sqrt{n}$$

Example 8.2.1 It is known that length of lives of CD follow a normal distribution with standard deviation of 11 weeks. Construct 95% and 99% confidence interval for the population mean given that a sample of 100 CD gives a mean lives of 8 weeks.

Solution: Using the formula, $\mu = \bar{X} + Z_{\alpha/2} \sigma_{\bar{X}} = \bar{X} \pm 1.65 \sigma / \sqrt{n}$

(a) 95% confidence interval gives, $\mu = \bar{X} + Z_{\alpha/2} \sigma_{\bar{X}} = 8 \pm 1.96(11) / \sqrt{100} = 8 \pm 2.156$

or $\mu \in (5.844, 10.156)$

This means that with 90% confidence one may claim that length of life of CD selected at random from the population will have a man life between 6 and 10 weeks.

(b) 99% confidence interval gives, $\mu = \bar{X} + Z_{\alpha/2} \sigma_{\bar{X}} = 8 \pm 2.58(11) / \sqrt{100} = 8 \pm 2.838$

or $\mu \in (5.162, 10.838)$

This means that with 99% confidence one may claim that length of life of CD selected at random from the population will have a man life between 5 and 11 weeks.

Measurement of Standard Error and Probability (σ is unknown)

t - Distribution

What has been discussed above is based on the assumption that standard deviation σ is known. However, in practice standard deviation is not known and as a result it has to be estimated as well. This is because the information comes from the sample. We know that a sample mean is balanced estimator of the population mean μ , but the sample standard deviation, s , is reasonable estimator of the population standard deviation, σ , only when it is calculated by using the formula

$$s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{(n-1)}} \equiv \text{Balanced estimator of } \sigma$$

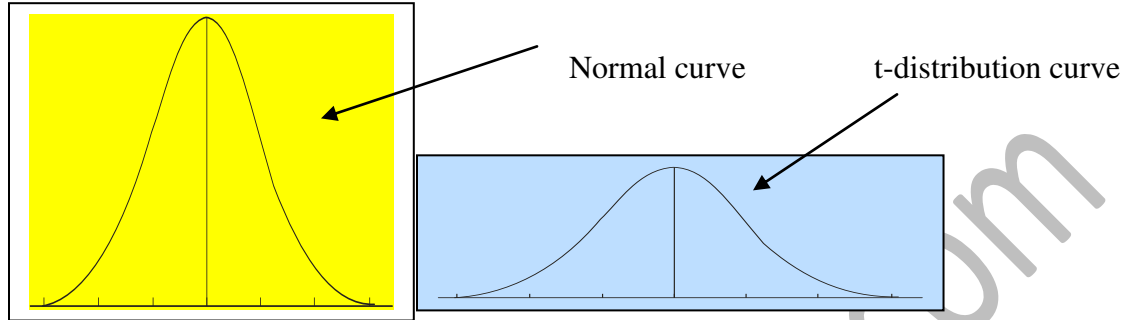
Having shown that σ is unknown it may be estimated by using an accustomed value of s . The question raised here is whether or not the normal distribution can continue to be used for measuring the probability as before. Unluckily, the answer is no. A different distribution which is identified as *t-distribution* has to be used. It has been observed that for large samples of sizes around 30 or more, the *t-distribution* is very much close to normal distribution and for practical purpose the sampling distribution may be regarded as normal. Thus the procedure for making interval estimation for population mean μ is same as before subject to condition that estimate of $\sigma_{\bar{X}}$ is made in the same way described above, that is; $\sigma_{\bar{X}} \cong s_{\bar{X}} = s / \sqrt{n}$ where s is defined as above. Thus *t-distribution* is mainly important for small samples ($n < 30$). It may be noted that if n is less than 30 the denominator of $\frac{\bar{X} - \mu}{s_{\bar{X}}} = \frac{\bar{X} - \mu}{s / \sqrt{n}}$ does not remain constant, but will vary according to each

sample standard deviation s . In fact it results in a distribution which is more spread out than the normal distribution and is known as *t-distribution* as mentioned above. Here in place of Z-statistic, we calculate t-statistic which is defined as

$$t = \frac{\bar{X} - \mu}{s_{\bar{X}}} = \frac{\bar{X} - \mu}{s / \sqrt{n}}$$

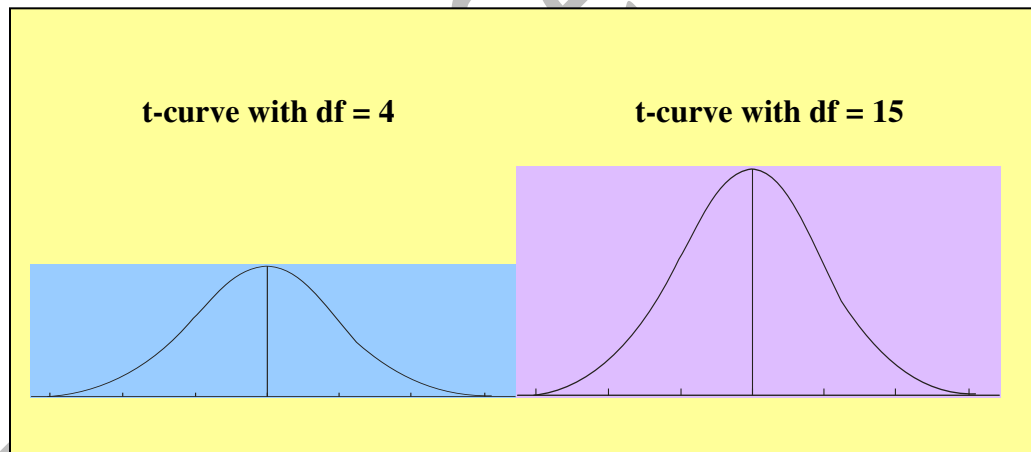
with $(n - 1)$ degrees of freedom.

The difference between both statistics is that in Z-statistic we know the standard deviation where as in t-statistic it has to be estimated. The following figure shows the difference between both statistics. For the Z-statistic the normal distribution provides a fixed model and for t-statistic there is a family of distributions that vary according to sample size also known as *degrees of freedom* (df).



Characteristics of t-Distribution

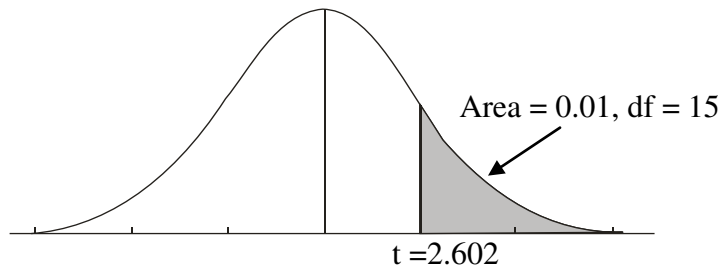
A t-distribution resembles the Z-curve as both are symmetrical about the mean $\mu_z = \mu_t = 0$. But *t-distributions* are flatter than the normal curve. However, as the sample size n increases, so too the number of degrees of freedom and more closely the t-curve will be to normal curve. If n tends to infinity, the t-curve becomes identical to normal curve.



Procedure to Use t-Distribution Probability Table

Like the table showing the areas under normal curves, areas under the *t*-distribution curves allow to find the probability statements are available in 'Appendix C'. However, as the *t*-distribution is not a single distribution but a family of distributions hence the table is laid out differently from the table of normal curve. In this table each row indicates a separate *t*-distribution according to the number of degrees of freedom as well as the area in the upper tail. For example, for 15 degrees of freedom, the value of the *t* which leaves an area of 0.01 in the tail of the distribution, that is; $t_{15, 0.01} = 2.602$.

Remark: Students are advised to learn the use of t-table from their tutors).



Interval Estimation Using t- Distribution

Interval estimation using the t-distribution follows the same principles as those using the normal distribution in terms of Z distribution. The Z confidence interval for the population mean μ is defined as $\mu = \bar{X} + Z_{\alpha/2} \sigma_{\bar{X}}$ where the level of confidence is $(1 - \alpha)\%$. Similarly, the t-confidence interval for the population mean μ with $(1 - \alpha)\%$ confidence interval is defined as $\mu = \bar{X} + t_{v, \alpha/2} s_{\bar{X}}$ where $v = (n - 1)$ is number of degrees of freedom.

Example 8.2.2 The amounts spent (in dollars) by customers in a shop on a particular day of a week are normally distributed. A random sample of 16 customers is given as under: 1900, 2100, 3500, 2900, 1300, 3500, 700, 1800, 2100, 1400, 2900, 2000, 1200, 2400, 3200, 2300. Estimate a 99% confidence interval for the population mean.

Solution: The sample mean is $\bar{X} = \frac{\sum x}{n} = 2200$. The sample standard deviation and standard error respectively are:

$$s = \sqrt{\frac{\sum (x - \bar{X})^2}{n-1}} = 833 \quad \text{or} \quad s_{\bar{X}} = s / \sqrt{n} = 833 / 4 = 208.25.$$

Number of degrees of freedom $v = (n - 1) = (16 - 1) = 15$.

Level of confidence = 99% giving level of significance $\alpha = 0.01$ or $\alpha/2 = 0.005$.

Thus $t_{v, \alpha/2} = t_{15, 0.005} = 2.947$, therefore, 99% confidence interval is:

$$\mu = \bar{X} \pm t_{v, \alpha/2} s_{\bar{X}} = 2200 \pm 2.947(208.25) = 2200 \pm 613.7 \quad \text{or} \quad \mu \in (1586.3, 2813.7)$$

Thus with 99% confidence one can claim that minimum / maximum amount spent by all customers on a particular day of the week is between \$1586.3 and \$2813.7.

Estimating the Difference between Two Population Means

If we have two populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 respectively, then a point estimator of difference between μ_1 and μ_2 is given by the statistics $\bar{X}_1$ and $\bar{X}_2$. Therefore, to obtain a point estimator of $\mu_1 - \mu_2$, we shall select two independent random variables, one from each population, of size n_1 and n_2 and compute the difference $\bar{X}_1 - \bar{X}_2$ of the sample means. The confidence interval for difference between two means when σ_1^2 and σ_2^2 are known is defined as:

$$(\bar{X}_1 - \bar{X}_2) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{X}_1 - \bar{X}_2) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Here, $z_{\alpha/2}$ is the z-value leaving an area of $\alpha/2$ to the right. The degree of confidence is exact if samples are selected from normal population. If the population is not normal then

central limit theorem is invoked that gives a good approximation for reasonable size samples.

Example 8.2.3 The following data shows a comparison between the mileage per gallon for two engines A and B.

	Experiment conducted	Mean	Standard Deviation
Engine A	50	36 miles/gallon	6
Engine B	75	42 miles/gallon	8

Find a 95% confidence interval for the difference between their mean mileages.

Solution: Using the formula as stated above and notice that for 95% confidence interval $z_{\alpha/2} = 1.96$. Thus 95% confidence interval for difference between population means is

$$(\bar{x}_1 - \bar{x}_2) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$(42 - 36) - 1.96 \sqrt{\frac{64}{75} + \frac{36}{50}} < \mu_1 - \mu_2 < (42 - 36) + 1.96 \sqrt{\frac{64}{75} + \frac{36}{50}}$$

or $3.43 < \mu_1 - \mu_2 < 8.57$

The above formula for estimating the difference between two means is applicable when population variances are known. Two other cases arise in practice.

Case I: When two variances are equal but unknown.

Case II: When two variances are unknown and also are not equal

In case I the interval estimation for difference between population mean is given by

$$(\bar{x}_1 - \bar{x}_2) - t_{v, \alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{v, \alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where $v = n_1 + n_2 - 2$ is degrees of freedom, and

$$S_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

In case II the interval estimation for difference between means is given by:

$$(\bar{x}_1 - \bar{x}_2) - t_{v, \alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{v, \alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where $v = \frac{(s_1^2 / n_1 + s_2^2 / n_2)^2}{\left[(s_1^2 / n_1) / (n_1 - 1) \right] \left[(s_2^2 / n_2) / (n_2 - 1) \right]}$

Example 8.2.4 The following data shows a comparison between the lives of HD's manufactured in two different companies A and B. It is given that $\sigma_1^2 = \sigma_2^2$ but are unknown.

	Sample size	Mean	Standard Deviation
Company A	12	520 weeks	65 weeks
Company B	18	555 weeks	70 weeks

Find a 95% confidence interval for the difference between their mean lives assuming that the population is approximately normally distributed.

Solution: Since $\sigma_1^2 = \sigma_2^2$ are equal but unknown hence we use the following formula to compute the difference between population means.

$$(\bar{x}_1 - \bar{x}_2) - t_{v, \alpha/2} S_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{v, \alpha/2} S_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Where $v = n_1 + n_2 - 2$ is degrees of freedom, and

$$S_P = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Before we compute the confidence interval we must find S_P .

$$S_P = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(18 - 1)(70)^2 + (12 - 1)(65)^2}{18 + 12 - 2}} = 86.08.$$

Degrees of freedom: $v = n_1 + n_2 - 2 = 28$.

$\alpha/2 = .05/2 = 0.025$ so that $t_{v, \alpha/2} = t_{28, 0.025} = 2.048$. Thus

$$\begin{aligned} & (\bar{x}_1 - \bar{x}_2) - t_{v, \alpha/2} S_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{v, \alpha/2} S_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \\ \Rightarrow & (555 - 520) - 2.048(86.08) \sqrt{\frac{1}{18} + \frac{1}{16}} < \mu_1 - \mu_2 < (555 - 520) + 2.048(86.08) \sqrt{\frac{1}{18} + \frac{1}{16}} \\ \Rightarrow & -25.57 < \mu_1 - \mu_2 < 95.57 \end{aligned}$$

This shows that difference between the lives of HDs manufactured by two different companies is -25.57 to 95.57 weeks. We make this claim with 95% confidence.

Example 8.2.5 Steel bars were produced by two different companies and their breaking strengths were recorded (in 10000) of psi and shown in the following table.

Company A	Company B
0.15, 0.14, 0.11, 0.16, 0.12, 0.13, 0.14, 0.10, 0.14, 0.11	0.16, 0.11, 0.14, 0.13, 0.12, 0.14, 0.13, 0.15, 0.14, 0.10, 0.13, 0.12, 0.11, 0.10, 0.11

If it is known that $\sigma_1^2 = \sigma_2^2$ but unknown, find the confidence interval for the breaking strength of steel bars of two companies assuming that the population is approximately normally distributed.

Solution: Note: The difference between this example and the previous one is that in previous example the means and standard deviations are given whereas we have to compute these for both data. Students are advised to find these. Here we place these values.

Company A
 $n_1 = 10$
 Mean = 0.13
 $s_1 = 0.01$

Company B
 $n_2 = 15$
 Mean = 0.126
 $s_2 = 0.01$

$v = 10 + 15 - 2 = 23$. $\alpha/2 = .05/2 = 0.025$ or $t_{v, \alpha/2} = t_{23, 0.025} = 2.069$. $S_P = 0.01$.

Using the formula:

$$(\bar{x}_1 - \bar{x}_2) - t_{v, \alpha/2} S_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{v, \alpha/2} S_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$\text{or } (0.13 - 0.126) - 2.069(0.01)\sqrt{\frac{1}{10} + \frac{1}{15}} < \mu_1 - \mu_2 < (0.13 - 0.126) + 2.069(0.01)\sqrt{\frac{1}{10} + \frac{1}{15}}$$

or

$$-0.016 < \mu_1 - \mu_2 < 0.024$$

or multiplying by 10000: $-160 < \mu_1 - \mu_2 < 240$

This means that difference between the average breaking strength of steel bars produced in two different companies is between -160 and 240 psi and we say this with 95% confidence.

Example 8.2.6 A study was made at two different lakes to see the level of water pollution. Fifteen samples at “Lake-A” were collected and 12 at “Lake-B”. The average pollution calculated at “Lake-A” was 3.84 mg/l with standard deviation of 3.07 mg/l. The average pollution calculated at “Lake-B” was 1.49 mg/l with standard deviation of 0.08 mg/l. Find 95% confidence interval for the difference in the true averages of pollution at both lakes given that observations come from normal populations with different unknown variations.

Solution: Data;

Lake-A

$$n_1 = 15$$

$$\text{Mean} = 3.84$$

$$s_1 = 3.07$$

Lake-B

$$n_2 = 12$$

$$\text{Mean} = 1.49$$

$$s_2 = 0.08$$

Since variances are different and unknown and the observations follow a normal distribution, we use the formula,

$$(\bar{x}_1 - \bar{x}_2) - t_{v, \alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{v, \alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$\text{where } v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\left[(s_1^2/n_1)/(n_1-1) \right] \left[(s_2^2/n_2)/(n_2-1) \right]}$$

After calculations are made, we get

Degrees of freedom, $v = 16$. $\alpha/2 = .05/2 = 0.025$ or $t_{v, \alpha/2} = t_{16, 0.025} = 2.120$ and

$$2.35 - 2.12 \sqrt{\frac{3.07^2}{15} + \frac{0.08^2}{12}} < \mu_1 - \mu_2 < 2.35 + 2.12 \sqrt{\frac{3.07^2}{15} + \frac{0.08^2}{12}}$$

This simplifies to $0.60 < \mu_1 - \mu_2 < 4.10$.

Hence we are 95% confident that difference between two population means (mean pollution in the two lakes) lies within the limits 0.60 mg/l to 4.10 mg/l.

8.3 SAMPLING DISTRIBUTION OF SAMPLE VARIANCE

Chi Squared Distribution

In the previous section we discussed the sampling distribution of means $\bar{X}$. The central limit theorem allowed us to make the substitution $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ which turns the mean and

standard deviation of normal distribution equal to zero and unity respectively provided n gets larger. In fact, sampling distribution of important statistics, allow us to learn information from the problems under consideration. For example, if an engineer is

interested in the population breaking strength of steel rods, the sampling distribution of $\bar{X}$ will be studied after the required data is being collected. On the other hand, if the variability in breaking strength is to be studied, the sampling distribution of s^2 will be used in learning its counterpart σ^2 , the population variance.

For a random sample of size n from a normal population with mean μ and variance σ^2 , if sample variance s^2 is computed, we obtain a value of the statistic χ^2 , known as Chi-squared distribution and is defined as:

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{\sum (X_i - \bar{X})^2}{\sigma^2}$$

with $v = n - 1$ degrees of freedom. The values of random variables χ^2 are computed from each sample by using above formula. The probability that a random sample produces a χ^2 value greater than some specified value is equal to the area under the curve to the right of this value. The shape of χ^2 distribution is shown in the following figure.

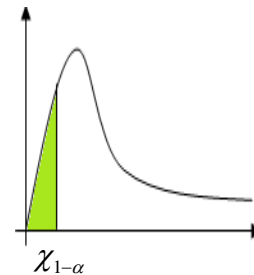
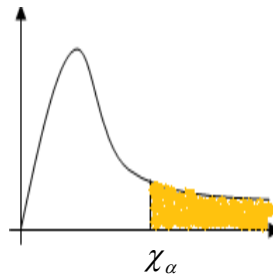
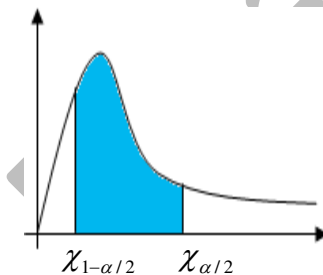
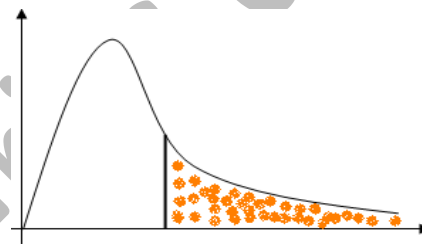
Appendix D gives values of χ^2 for various values of v and α . The areas α are the column headings whereas v shows the degrees of freedom and is given in the leftmost column.

For example, $\chi^2_{7, 0.05}$ indicates the value of χ^2

with 7 degrees of freedom and 0.05 of the area to its right. This value can be seen from the

table which is 14.067. The χ^2 -value with 7 degrees of freedom and 0.95 areas to the right is 2.167. Generally speaking, chi-squared distribution provides interval estimation for population variance with certain degree of significance and degrees of freedom.

This is depicted in the following figures.



Example 8.3.1 A manufacturer of Jack Pot guarantees that his manufactured machines will averagely last for 3 years with standard deviation of 1 year. Five of the Jack Pots were selected at random from a population and were tested. Their lives in years were as follows: 4.2, 1.9, 3.0, 2.4, and 3.5. Is the manufacturer true in his claim?

Solution: The mean life of five Jack Pots is

$$\bar{X} = \frac{4.2 + 1.9 + 3.0 + 2.4 + 3.5}{5} = \frac{15}{5} = 3$$

Using the formula for sample variance, we get

$$s^2 = \frac{\sum (x_i - \bar{X})^2}{n-1} = 0.815$$

Another formula for computing s^2 is: $s^2 = \frac{n \sum x_i^2 - (\sum x_i)^2}{n(n-1)}$. The excellence of this formula is that without computing the mean we can directly find the sample variance. Now using the formula for the computation of χ^2 , we get

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(5-1)(0.815)}{1^2} = 3.26.$$

It may be noted that degrees of freedom $v = n - 1 = 5 - 1 = 4$. Thus above value of χ^2 with four degrees of freedom lies in the range (0.484, 11.134) with 97.5% accuracy. This can be seen from the table given in Appendix D. Hence the manufacturer should not suspect his claim that his Jack Pot machine will have variance life of 1 squared year.

Estimating the Variance

In the above section we discussed how χ^2 distribution is used to test the population variance with certain degrees of freedom. In this section, we shall study how a confidence interval is used to see the range where the population variance lies. We state the theorem that states the confidence interval for the population variance.

Theorem: If s^2 is the variance of a random variable of size n from a normal population then a $(1 - \alpha)$ 100% confidence interval for population variance σ^2 is given by

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}$$

where $\chi^2_{\alpha/2}$ and $\chi^2_{1-\alpha/2}$ are χ^2 values with $v = n - 1$ degrees of freedom, leaving area $\alpha/2$ and $(1 - \alpha/2)$, respectively to the right of χ^2 - curve/distribution.

Example 8.3.2 The health department wishes to see the variance in height squared from a population consisting of pupils of class VIII of certain government school. A sample of 10 pupils was selected at random and their heights in inches were recorded as under, 46.0, 45.2, 46.4, 46.1, 46.9, 45.47, 45.8, 45.9, 46.1 and 47.0. Find a 99% confidence interval for population variance of all students of class VIII.

Solution: Here $n = 10$ so that $v = n - 1 = 9$. Since level of confidence is 99% hence level of significance $\alpha = 0.01$ and therefore, $\alpha/2 = 0.005$ which implies that $(1 - \alpha/2) = 0.995$.

Thus $\chi^2_{v, \alpha/2} = \chi^2_{9, 0.005} = 23.589$ and $\chi^2_{v, 1-\alpha/2} = \chi^2_{9, 0.995} = 1.735$

Now using the formula for computing the sample variance, we get

$$s^2 = \frac{n \sum x_i^2 - (\sum x_i)^2}{n(n-1)} = \frac{10(21273.12) - (461.2)^2}{10(9)} = 0.286$$

Now estimation of population variance is given by

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}} \quad \text{or} \quad \frac{9(0.286)}{23.589} < \sigma^2 < \frac{9(0.286)}{1.735} \quad \text{or} \quad 0.109 < \sigma^2 < 1.48$$

This shows that variance squared heights of pupils of class VIII lies between 0.109 and 1.48 and this claim is made with 99.5% confidence.

8.4 F-DISTRIBUTION

We know that Z and t (statistics) distributions are used to estimate one of the population parameters namely the population mean and σ^2 along with their interval estimations. The chi-square (χ^2) distribution is used to find the interval for population variance. It is often the case that sometimes it is required to compare the variability of two or more random variables. F-distribution is the right choice to compare the variances of two or more independent samples taken from different populations. For example, a civil engineer may be interested to compare the sample variances of concrete supplied by two different companies.

The F-statistic is defined as the ratio of two independent chi-squared distributions each divided by its number of degrees of freedom. Thus

$$F = \frac{\chi^2_X / v_1}{\chi^2_Y / v_2}$$

Here X and Y are independent random variables with chi-squared distributions χ^2_X , and χ^2_Y respectively.

We shall now define the p.d.f of F-distribution and draw its graph for the sake of completeness.

We shall not further discuss the p.d.f of F-distribution instead we shall use the tabular values as we have been doing in the previous sections.

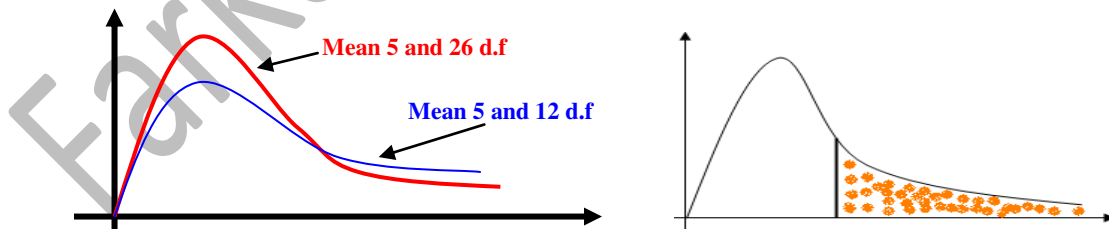
The p.d.f of F-distribution is defined as

$$f(F) = \begin{cases} \frac{\Gamma[(v_1 + v_2)/2] (v_1/v_2)^{v_1/2} F^{v_1/2 - 1}}{\Gamma[(v_1)/2] \cdot \Gamma[(v_2)/2] (1 + v_1 F / v_2)^{(v_1 + v_2)/2}}, & 0 < F < \infty \\ 0, & -\infty < F < 0 \end{cases}$$

It may be noted that the curve of F-distribution has two characteristics:

- It depends on v_1 and v_2 .
- It depends on the order in which v_1 and v_2 are taken.

After these values are taken at hand, we can identify the curve. The following figure shows the effect of degrees of freedom on the shape of two F-curves.



Let F_α be the F value above which we find an area equal to α . This is shown by shaded region in the above figure.

The table given in Appendix E gives the values for $\alpha = 0.05$ (95% confidence) and $\alpha = 0.01$ (99% confidence) for various combinations of degrees of freedom v_1 and v_2 . Hence, the F value with 5 and 26 degrees of freedom, leaving an area of 0.05 to the right is $F_{0.05}(5, 26) = 2.59$.

The following theorem will allow computing $F_{0.95}$ and $F_{0.99}$. The proof is beyond the scope of this book.

Theorem: The F distribution with degrees of freedom v_1 and v_2 satisfy the condition

$$F_{\alpha}(v_1, v_2) = \frac{1}{F_{1-\alpha}(v_2, v_1)}$$

Using this theorem, we see that $F_{0.95}(26, 5) = 1 / F_{0.05}(5, 26) = 1 / 2.59 = 0.39$.

Estimating the Ratio of Two Variances

As stated above that in some cases it is important to find the confidence interval for required level of confidence the ratio between the variances of two independent populations. If s_1^2 and s_2^2 are the variances of two independent samples of size n_1 and n_2 taken from the normal populations then $(1 - \alpha)$ 100% confidence interval for σ_1^2 / σ_2^2 is

given by:
$$\frac{s_1^2}{s_2^2} \cdot \frac{1}{F_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} \cdot F_{\alpha/2}(v_2, v_1)$$

Here $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ are the degrees of freedom for the two samples.

Example 8.4.1 A study was made at two different lakes to see the level of water pollution. Fifteen samples at “Lake-A” were collected and 12 at “Lake-B”. The average pollution calculated at “Lake-A” was 3.84 mg/l with standard deviation of 3.07 mg/l. The average pollution calculated at “Lake-B” was 1.49 mg/l with standard deviation of 2.88 mg/l, find a 98% confidence interval for the ratio of population variances of pollution σ_1^2 / σ_2^2 at both lakes given that observations come from normal populations with different unknown variations.

Solution: Data

Lake-A	Lake-B
$n_1 = 15$	$n_2 = 12$
$v_1 = 14$	$v_2 = 11$
Mean = 3.84	Mean = 1.49
$s_1 = 3.07$	$s_2 = 2.88$

taking $\alpha = 0.2$ then $\alpha / 2 = 0.01$. Using the formula:

$$\frac{s_1^2}{s_2^2} \cdot \frac{1}{F_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} \cdot F_{\alpha/2}(v_2, v_1)$$

or
$$\frac{3.07^2}{2.88^2} \cdot \frac{1}{F_{0.01}(14, 11)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{3.07^2}{2.88^2} \cdot F_{0.01}(11, 14)$$

Or
$$(1.1363) \cdot \frac{1}{4.3} < \frac{\sigma_1^2}{\sigma_2^2} < (1.1363)(3.87). \text{ Thus, } 0.265 < \frac{\sigma_1^2}{\sigma_2^2} < 4.4$$

WORKSHEET 08

1. An electronic company manufactures resistors that have a mean resistance 100 Ω and standard deviation of 10 Ω . The distribution of resistance is normal. Find the probability that a random sample of 20 Ω will have an average resistance less than 95 Ω .

2. The capacitance of certain types of capacitors are given by 95 , 50 , 51 , 53 , 49 , 52 , 53 , 51 , 50 , 49. Find the probability that a random sample of 32 capacitors selected at random with replacement will yield a sample mean

- (i) Greater than 50 but less than 52 Farads.
- (ii) Exceeds 52 Farads.

3. The spot speed at a particular location on a highway is known to be normally distributed with a mean of 80 km/h and S.D of 5 km/h. What is the probability that if a sample of 100 vehicles are tested that the mean speed observed will exceed 75 km/h?
4. The diameters of shaft made by a certain manufacturing process are known to be normally distributed with a mean of 2.5 cm and SD of 0.009 cm. Find the probability that a random sample of 49 such shaft diameters will yield a sample mean greater than 2.505 cm.
5. A random variable is drawn from a population with mean equal to 60 and a SD of 2.0. Find the standard error of the sample mean if the sample size is (i) 16 (ii) 36 and (iii) 81. What sample size would give a standard error equal to 0.2?
6. The mean sampling distribution of means of 36 items is 60 and its standard deviation is 12. Assume the sampling distribution is normal,
 - (i) What percentage of sample means will be between 55 and 65?
 - (ii) What percentage of sample means will be between 57.5 and 62.5?
7. A type of concrete block has been found to have a mean life of 2000 months and a S.D of 250 months. What is the probability that a sample of 81 concrete blocks will have an average life of less than 1920 months?
8. A series of 369 observations of spot speed of vehicles was taken from a normal population with unknown mean and variance. The mean and standard deviation of the sample have been worked out as 39.7 Km/h and 6.8 Km/h respectively. Give a 95% confidence interval for the population mean speed.
9. The mean spot speed of 20 vehicles passing a given location was 68.3 km/h and standard deviation 8.6 km/h. Give a 95% confidence interval for the mean speed of vehicles at that location.
10. The resistance of 10 resistors is given as 50, 51, 52, 53, 49, 52, 53, 51, 50, and 49. Find the 99% confidence interval for the mean resistance of such resistors.
11. A standardized mathematics test was given to 16 girls and 11 boys. The girls made average grade of 76 with a standard deviation of 6, while boys made an average grade of 82 with a standard deviation of 8. Find a 98% confidence interval for σ_1^2 / σ_2^2 , where σ_1^2 is the variance of all boys and σ_2^2 is the variance of all girls who might have taken this test.
12. What is point estimate? Why is it used as point estimate of the population mean μ and point estimate of the population standard deviation σ ?
13. A population has a standard deviation of 126.94. A sample of 25 items is taken from this population. Calculate the standard error of the sample means.
14. A sample of 40 items is taken from a population of 100000 items. The sample has mean 5371.03 and standard deviation 374.64. Calculate a point estimate of the standard error of the sample means.
15. Find the following values: (a) $t_{1, 0.05}$ (b) $t_{2, 0.05}$ (c) $t_{14, 0.01}$ (d) $t_{22, 0.1}$ (e) $t_{35, 0.025}$
16. Find the value x of the t-distribution with 19 degrees of freedom such that $P(t > x) = 0.1$.
17. Find the value x of the t-distribution with 8 degrees of freedom such that $P(t < x) = 0.005$

18. State the standard error of the sample mean when (a) the population standard deviation is known (b) the population standard deviation is not known. A population is known to have a standard deviation of 25. A sample of 10 items is taken from this population. The sample mean is 92.6 find (i) point estimate of the mean of the sample means (ii) standard error of the sample means (iii) what distribution is used for the sampling distribution of the sample means?

19. The amount of soft drink (in ml) poured into bottles by filling machine is approximately normally distributed. Eight bottles were taken at random and the amount of drink was found as: 480, 478, 483, 477, 479, 481, and 485. Construct a 99% confidence interval for the population mean and interpret your results.

20. A survey was conducted to see how many people watch Geo-Capital-Talk program. One hundred people were interviewed and the sample mean was 75 with sample standard deviation 10 people. Find the 95% confidence interval for the population mean and interpret your results.

21. A random sample of regular gas prices at 36 stations in Karachi has a mean of Rs. 46.7 and the sample standard deviation of Rs. 0.5. Construct confidence interval for the mean gas price at all Karachi stations with confidence coefficient (a) 90%; (b) 99%. Which is the more precise interval?

22. Chairman of a company selling the cars of the opinion that profit on the sale of cars this year is less than the last year although the number of cars sold this year is more than the last year. The Manager takes a sample of 35 sales for both years and finds that average profit on the sale last year was Rs. 32 millions with sample standard deviation of Rs. 1 million and profit on the sale of cars this year is Rs. 30 millions with sample standard deviation of Rs. 0.75 millions. Find the confidence interval of difference between two population means at 98% confidence level.

23. Costly road testing of random samples of cars was carried out to determine if mean mileage (mile per gallon) is greater for road Model-I cars than for Model-II cars. The sample data is as under:

Model-I: $n_1 = 10$	sample mean = 25.9	sample S.D = 1.5
Model-II: $n_2 = 12$	sample mean = 24.5	sample S.D = 1.2

At 2% level, find the confidence interval for the difference in population means.

24. The capacitance of brand I and II split ACs are rated the same. The capacities of random samples of six units of each brand were determined. The sample data in thousands of BTU (British Thermal Units) were:

Brand-I:	6.1	6.5	5.7	6.3	6.5	5.9
Brand-II:	4.9	5.2	5.4	5.2	5.1	5.3

At 5% level determines the confidence interval whether mean BTU consumption for the two brands differ.

25. A company manufactures a new type of signal lamp A sample of 15 lamps is tested and the Sample mean was founded to be 2500 hours with a standard deviation of 250 hours. Determine a 90% Confidence interval for the mean life of the Lamps assuming an approximate normal distribution.

26. Nine people followed a special diet plan for three months. At that time their individual weight loses were 1.8, 2.9, 2.1, 3.5, 1.5, 3.5, 2.5, 2.5, 0.6 and 2.3 pounds.

Construct a 95% confidence interval for average less in weights assuming the population is normal

27. A firm that manufactures electronic Calculators with automatic battery Cells have a length of life that is approximately normally distributed with a S.D of 80 days. If a random sample of 25 such Calculators had an average Life of 860 days. Find a 99% confidence interval for the population mean of all calculators manufactured by this firm.

28. In a management study of workers wages a simple random sample of 196 employees was taken from a production line work force. The Mean of the Sample was to be Rs. 10,000 annum. With S.D of Rs. 760

- a) Determine a 95% confidence interval for workforce
- b) Determine a 99% confidence interval for workforce

29. A car owner wants to know the mean weekly mileage he puts on his car. He records his mileage for 52 weeks and finds a mean of 176 miles per week with S.D of 69 construct a 95% confidence interval for mean weekly mileage.

30. Seventy plants of certain species were divided into two group received normally lighting while other receiving special lighting. At the end of specified growing period, the plant weight was measured in grams. The results are summarized below. Find a 90% confidence interval the mean difference in plant weight for the two groups.

Group	n	Mean	S.D
Normal Weight	31	6.7	1.10
Special Lighting	40	3.8	0.70

CHAPTER NINE

TESTS OF STATISTICAL HYPOTHESES

9.1 INTRODUCTION

In the previous chapter we used the information obtained in a simple random sample to conduct a confidence interval estimate of the unknown population parameters. In this chapter we shall start with an assumed value of a population parameter, then we shall use sample evidence to decide whether the assumed value is unreasonable and should be rejected or accepted.

The assumptions we make about the values of population parameters are called *hypotheses*. Sample evidence is used to test the hypotheses; hence the statistical inferences made here are referred to as hypothesis testing. Some examples of the kinds of decisions to which the principles of hypothesis testing may be applied are as under.

- ❖ Has absence from work in an industry due to sickness increased significantly over its level during this year?
- ❖ Is a manufacturing running under control performs its operation within pre-specified limits or has the machine drifted out of control and thus requires repair?
- ❖ Does a new fertilizer increase the yield of a particular crop significantly?

In each of these examples we start with some predetermined standard or knowledge of the previous value of population parameters and wish to decide whether the new information drawn from a sample, is or is not consistent with having been drawn from the population.

Basic Principles of Hypothesis Testing

The basic idea behind hypothesis testing is that we begin with an assumption or hypothesis about the value of population parameters, such as the population mean and population variance. Then sample information may be used to test the hypothesis. The test is based on comparing the sample statistics with the corresponding hypothesized parameters values. If there is no difference between the two values, the hypothesis may be accepted. However, it is more likely to find some difference either as a result of chance, the hypothesis itself being true, or because it is in fact not true. Thus the process concerned in hypothesis testing is to reject the hypothesis only when the difference is so large that probability of its occurring by chance is very small.

To do this, the best way is to make interval estimation for the population parameters where upper and lower confidence limits are determined with certain level of confidence or probability. We are not still sure that the true population parameters will lie in this interval. Likewise in hypothesis testing a hypothesis may be accepted or rejected with some certainty or confidence.

For a while we shall concentrate only on one population parameter, that is, population mean. Later on we shall make similar assumptions regarding the second population parameter namely population variance.

In hypothesis testing if the hypothesized mean, usually denoted by μ_0 , is regarded as the true mean lying at the centre of a sampling distribution of $\bar{X}$ shown in figure 1. In the previous chapter we observed that for example, 5% probability of a sample means falling outside the limits defined by $\mu_0 \pm 1.96 \sigma_{\bar{X}}$ with the probability level of 5% denoted by α . If 5% probability of error is acceptable, then sample means coming within this range as acceptable evidence that the sample has been drawn from a population with a mean that is no different from the hypothesized value μ_0 . Thus the region within the limits becomes an *acceptance region* for the hypothesis and the regions outside the limits are then considered as *regions of rejection* for the hypothesis. The probability itself is known as the *significance level* of the test. Thus 95% level of confidence interval corresponds to 5% level of significance. Similarly 90% level of confidence interval corresponds to 10% level of significance, etc.

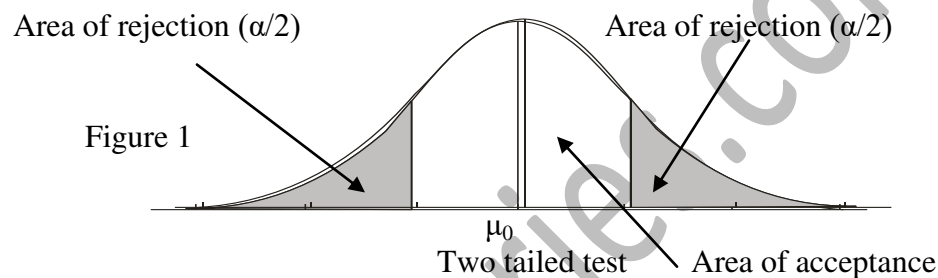
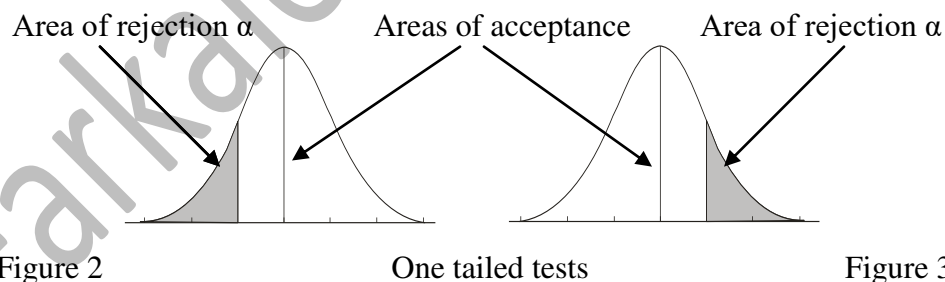


Figure 1 shows that rejection region consists of two 'tails'. As a result, the 5% rejection region is divided into two equal parts with 2½ % of area in each half. On the other hand, *one-tailed* tests deal with the situation where we are interested in testing the hypotheses that the population mean is particularly *greater than* or *less than* some hypothesized value. Figures 2 and 3 depict these facts. In such cases, there is only one rejection region where the whole area equal to α lies at one end (tail).



Difference between Interval Estimation and Hypothesis Testing

Although interval estimation and hypothesis testing look very similar, the fundamental difference between them is of objective. With interval estimation we start with no assumption about the values of population parameters before sample data is collected. The parameter value is unknown and the statistical objective is to estimate it with a given level of confidence. On the other hand, in hypothesis testing, we start with an assumption about the value of the population parameter and then collect sample data to test the hypothesis. Thus, in interval estimation, the sample mean $\bar{X}$ is placed at the centre of the interval, whereas in hypothesis testing, the hypothesized mean μ_0 is placed at the centre.

Conducting Hypothesis Tests: Null and Alternative Hypotheses

In this section we set out the procedures for conducting hypothesis tests for two tailed and one tailed tests. In hypothesis testing, we always talk about two types of tests namely, “Null Hypothesis” and “Alternative Hypothesis” usually denoted by H_0 and H_1 or H_a . Throughout this book, we shall use the notions H_0 and H_1 for null and alternative hypotheses. The null hypothesis H_0 , specifies the value of the parameters to be tested, for example, the population mean μ which is unknown is equal to hypothesized value μ_0 .

Thus:

For two tailed test, the null and alternative hypotheses are expressed as:

Null hypothesis $H_0: \mu = \mu_0$ (while testing the population mean)

Alternative hypothesis $H_1: \mu \neq \mu_0$,

This means that when H_0 is rejected then H_1 is accepted.

In case of one tailed tests, we wish to test the null hypothesis against the alternative hypothesis that the population parameter is not simply different from hypothesized parameter but specifically *greater* or *less* than the hypothesized parameter say population mean.

For one tailed test, the null and alternative hypotheses are expressed as:

Either (i) Null hypothesis $H_0: \mu = \mu_0$ Alternative hypothesis $H_1: \mu > \mu_0$

or (ii) Null hypothesis $H_0: \mu = \mu_0$ Alternative hypothesis $H_1: \mu < \mu_0$

Once hypothesis is formulated, that is, one tailed or two tailed test is required, the next step is to decide which level of significance (α) is chosen to conduct the test and that sample is chosen from normal population in which case Z-distribution is applied or it is approximately normal in which case t-distribution is applied. It is most common to work with 5% or 1% level of significance, however, other levels may also be considered. Once level of significance is decided the next step is to see the areas of rejection and acceptance. The point of division between the two regions marks the critical values of the statistic applied. For two valued test there are two critical values and for one tailed test certainly, there is one critical value. These are already shown figures 1, 2 and 3 above. It may be noted that rejection regions are themselves called *critical values*.

The Test Statistic

There are two ways of stating the test statistic. One way is to calculate a critical value of the statistic itself $\bar{X}$. The other way is to work in terms of the standardized or normalized normal variable Z or t value which ever is appropriate. In practice, statisticians generally use Z values because any level of significance is quickly recognized. We shall use both methods in our first example to show how this is done. However, in most problems we shall directly use Z or t values. The critical values in each case for testing a hypothesized (say) mean μ_0 are as follows.

Type of test	Hypothesis		Critical values using	
	H_0	H_1	$\bar{X}$	Z
Two tailed	$\mu = \mu_0$	$\mu \neq \mu_0$	$\mu_0 \pm Z_{\alpha/2} \sigma_{\bar{X}}$	$\pm Z_{\alpha/2}$
One tailed	$\mu = \mu_0$	$\mu > \mu_0$	$\mu_0 + Z_{\alpha} \sigma_{\bar{X}}$	$+ Z_{\alpha}$
One tailed	$\mu = \mu_0$	$\mu < \mu_0$	$\mu_0 - Z_{\alpha} \sigma_{\bar{X}}$	$- Z_{\alpha}$

The following table shows two tailed and one tailed Z-statistic critical values with different levels of significance. These can be verified from the normal distribution table (Appendix A). The following table shows the critical values of Z-statistic

(α)	One Tail	Two Tails
10% = 0.10	± 1.28	± 1.65
5% = 0.05	± 1.645	± 1.96
1% = 0.01	± 2.33	± 2.58

Decision Rule in Hypothesis Testing

Once the critical values have been determined by using either Z or $\bar{X}$ values, we are in a position to decide whether to accept or reject the null hypothesis. It may be noted that our focus is on the acceptance or rejection of the null hypothesis. Thus rejection of null hypothesis means the acceptance of alternative hypothesis. The decision rule is as follows:

Decision rule for two tailed test

- Using $\bar{X}$ as the test statistic:
Accept H_0 if the actual value of the sample mean $\bar{X}$ falls between the two critical values of $\bar{X}$.
Reject H_0 otherwise.
- ❖ Alternatively, using Z as the test statistic:
Accept H_0 if the actual value of Z corresponding to $\bar{X}$, given by $(\bar{X} - \mu_0) / \sigma_{\bar{X}}$ falls between two critical values of Z.
Reject H_0 otherwise.

Decision rule for one tailed test

- Using $\bar{X}$ as the test statistic:
Accept H_0 if the actual value of the sample mean $\bar{X}$ falls above or below the critical value of $\bar{X}$ depending on whether the lower or upper tail is the appropriate rejection region.
Reject H_0 otherwise.
- ❖ Alternatively, using Z as the test statistic:
Accept H_0 if the actual value of Z corresponding to $\bar{X}$ falls above or below the critical value of Z depending on whether the lower or upper tail is appropriate.
Reject H_0 ; otherwise

Example 9.1.1 A manager in a firm wishes to test whether the average number of days of absence per year due to sickness has changed this year compared with last year which was 8 days per year with a standard deviation of 14 days. A random sample of 64 workers is taken giving a mean of 11 days of absence this year. Test the hypothesis with 5% significant level that number of days of absence has not changed this year for the whole population consisting of all workers of the firm (Two tailed test).

Solution: Here we use normal distribution because value of σ is known.

Given $\mu = 8$, $\sigma = 14$; $n = 64$.

Formulation of hypothesis:

$$H_0: \mu_0 = 8; \quad H_1: \mu_0 \neq 8$$

Level of significance:

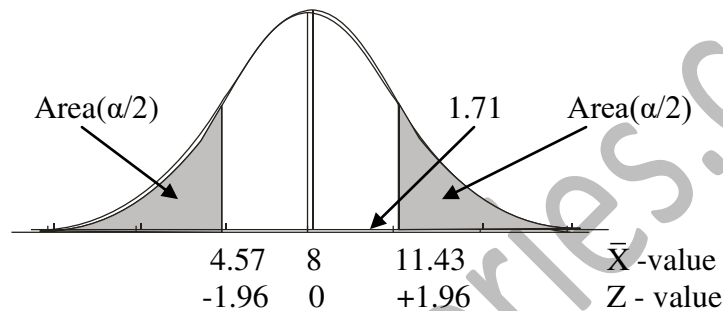
$$\alpha = 0.05, \text{ hence } Z_{\alpha/2} = \pm 1.96$$

Standard error of the mean:

$$\sigma_{\bar{X}} = \sigma / \sqrt{n} = 14 / \sqrt{64} = 14 / 8 = 1.75$$

Critical value using $\bar{X}$ statistic: $\mu_0 \pm Z_{\alpha/2} \sigma_{\bar{X}} = 8 \pm 1.96 (1.75) = (4.57, 11.43)$
 Critical value using Z statistic: $\pm Z_{\alpha/2} = (-1.96, 1.96)$
 Test: (a) Using critical $\bar{X}$ values:
 Here $\bar{X} = 11$ days and we see that $4.57 < 11 < 11.43$
 (b) Using critical Z values:
 $Z = (\bar{X} - \mu_0) / \sigma_{\bar{X}} = (11 - 8) / 1.75 = 1.71$ and we see
 that $-1.96 < 1.71 < +1.96$

Conclusion: The manager must accept the null hypothesis and conclude that according to the evidence, this year averagely the absenteeism per worker significantly is not different from past years. The manager must be 95% sure about this conclusion.



Example 9.1.2 (One tailed test): Using the information in example 1, if the manager is interested in ascertaining whether the absenteeism this year is significantly higher than it has been in the past, that is; whether 11 days is significantly greater than historical value of 8 days?

Solution: The steps are as follows.

Given $\mu = 8$, $\sigma = 14$; $n = 64$

Formulation of hypothesis:

$$H_0: \mu_0 = 8; \quad H_1: \mu_0 > 8$$

Level of significance:

$$\alpha = 0.05, \text{ hence } Z_{\alpha} = +1.645 \text{ (see the normal table)}$$

Standard error of the mean:

$$\sigma_{\bar{X}} = \sigma / \sqrt{n} = 14 / \sqrt{64} = 14 / 8 = 1.75$$

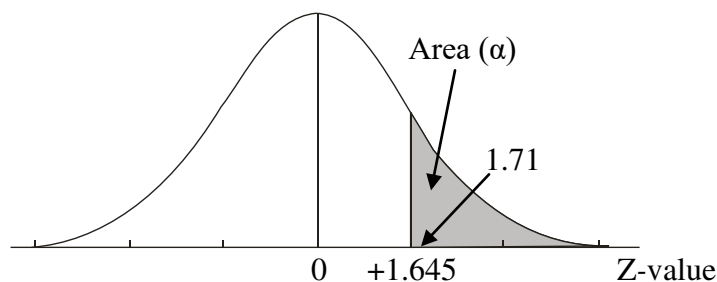
Critical value using Z statistic:

$$Z = Z_{\alpha} = 1.96$$

Test:

Using critical Z value:

$$Z = (\bar{X} - \mu_0) / \sigma_{\bar{X}} = (11 - 8) / 1.75 = 1.71 \text{ and we see that } 1.71 > 1.645.$$



Conclusion: The manager must reject null hypothesis and conclude that according to the evidence, this year averagely the absenteeism per worker is significantly greater than 8 days. The manager must be 95% sure about this conclusion/decision.

t-Statistic in Hypothesis Testing

Z-statistic is used when the population is normal and σ is known. In cases where the t distribution rather than Z distribution applies then, we must expect that t is a better substitute for Z. The circumstances in which t or Z statistic applies depend on sample size and whether or not σ is known. This has already been explained in the previous chapter. We have also explained there that for practical purposes the use of t is generally limited to small samples only. But recall that if the population from which the sample is drawn is itself not normal then sample size must be taken at least 30 so that ‘Central Limit Theorem’ is invoked.

Example 9.1.3 The following sample data is collected from normal population showing the number of deaths due to cancer disease per year.

2319	6237	3422	5601	7390	8125	5133
8218	6634	7340	1278	3195	7321	8830

Perform the following test of hypothesis with 5% level of significance:

$$H_0: \mu_0 = 8000; H_1: \mu_0 \neq 8000$$

Solution: This is a problem of two tailed test. As $n = 14 < 30$ hence we use t-distribution.

Data summary

$$\sum x_i = 81043, \sum x_i^2 = 543,162,959, \bar{x} = 5788.79, s = 2386.22, s_{\bar{x}} = s / \sqrt{n} = 637.75$$

$$\text{Now, } t = (\bar{x} - \mu_0) / s_{\bar{x}} = (5788.79 - 8000) / 637.75 = -3.4672$$

Level of significance $\alpha = 5\%$ or $\alpha/2 = 0.025$.

Degrees of freedom = $v = n - 1 = 13$, hence t-value is: $t_{13, 0.025} = \pm 2.16$.

Now the t-value -3.4672 does not lie in this interval, hence we conclude that null hypothesis is not true which means that average number of deaths due to cancer disease are not 8000 per year and we say this with 95% confidence.

Errors in Hypothesis Testing (Type-I and Type-II Errors)

It may be noted that hypothesis test does not prove the truth or falsity of a hypothesis or assumption made during testing process. Proofs always involve the certainty of an event but, as we are testing the hypothesis on the basis of sample information, we are in fact dealing in probabilities but not certainties. Hence, the terms *acceptance* and *rejection* should properly be understood and interpreted. The acceptance of the null hypothesis indicates only that the sample evidence is simply insufficient to reject it at the desired level of significance, not that it is essentially true. Thus when we deal with hypothesis test where two types of errors may be committed. These are *Type-I* error and *Type-II* error.

When a null hypothesis is rejected but in fact it was true, the error of this type is called ‘Type-I’ error.

When a null hypothesis is accepted but in fact it was false, the error of this type is called ‘Type-II’ error.

Type-I error can be controlled by specifying α , that is; the significance level specifies the risk we are taking to reject a true null hypothesis. But there is no way to control Type-II error. The only way to reduce both types of errors is to increase the sample size, though,

it may not be practicable. Although techniques are available to control these types of errors but it is beyond the scope of this book.

These types of errors are very much important in decision making. For example, consider a company manufacturing cancer drug. If the company has a null hypothesis that the drug is no better than another one, then Type-II error arises if it is found later on that drug was in fact better than the others available in the market. This means that drug will not be applied and people might suffer un-necessarily.

On the other hand if null hypothesis is accepted that drug produced is better than other but latter on it is found that it was not the case. In this case Type-I error is occurred and if the drug is applied, it could even be a dangerous.

Tests of Difference between Two Population Means (Large Samples)

Suppose we start with two normal populations where Population – I has a mean of μ_1 and a standard deviation σ_1 and Population – II has a mean of μ_2 with standard deviation σ_2 . Let n_1 and n_2 be the sizes of two independent samples from these populations respectively. Let $\bar{x}_1 - \bar{x}_2$ be the difference between the sample means. It has been proved that distribution of the difference between is normal and it will have a mean

$$\mu_{\bar{x}_1 - \bar{x}_2} = \mu_1 - \mu_2 \text{ and standard error } \sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

Usually we do not know the values of σ_1 and σ_2 ; so we estimate them by the sample standard deviations s_1 and s_2 . If both sample sizes are greater than 30, we may use the sample standard error of the difference between means,

$$s_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

We can now test the hypothesis about the difference between two population means where this difference may be equal to, less than or greater than zero. Hence, the hypotheses are:

- | | | |
|-------------------------------|-----------------------------|-------------------|
| i. $H_0: \mu_1 - \mu_2 = 0$ | $H_1: \mu_1 - \mu_2 \neq 0$ | (Two tailed test) |
| ii. $H_0: \mu_1 - \mu_2 > 0$ | $H_1: \mu_1 - \mu_2 < 0$ | (One tailed test) |
| iii. $H_0: \mu_1 - \mu_2 < 0$ | $H_1: \mu_1 - \mu_2 > 0$ | (One tailed test) |

The value of Z-statistic is: $z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_{\bar{x}_1 - \bar{x}_2}} = \frac{(\bar{x}_1 - \bar{x}_2)}{s_{\bar{x}_1 - \bar{x}_2}}$, since $(\mu_1 - \mu_2) = 0$

Example 9.1.4 A computer manufacturing company wishes to buy computer motherboards from two companies but is unable to take a decision due to the reason that the mother boards may be faulty. He asks the Statistical department for help in taking the decision that from which company the motherboards are to be bought. Statistical department took a sample 30 motherboards from Company-A, and found that the average number of defective motherboards is equal to 3 with standard deviation of 2. They took a sample of 40 motherboards from Company-B and found that the average number of defective motherboards is 2 with standard deviation of 3. Test the hypothesis at 5% level of significance that there is no difference between the population means.

Solution: (i) $H_0: \mu_1 - \mu_2 = 0$ $H_1: \mu_1 - \mu_2 \neq 0$ (Two tailed test)

(ii) $\alpha = 0.05$, hence critical values are $(-1.96, +1.96)$

(iii) Standard error: $s_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{2^2}{30} + \frac{3^2}{40}} = 0.6$

(iv) Test statistic: $z = \frac{(\bar{x}_1 - \bar{x}_2)}{s_{\bar{x}_1 - \bar{x}_2}} = (3 - 2) / 0.6 = 1.67$

Decision: Since 1.67 lies within the critical values hence we may accept the null hypothesis. This means that motherboards may be purchased from any company then mean number of defective motherboards will be same. We should be 95% confident about this decision.

Example 9.1.5 A large food retailing company is considering installing new electronic checkout systems in its supermarket throughout the country. Two different systems are available and company decides to conduct tests to ascertain whether or not speeding up of customers throughout by System-A is greater than System-B. The two systems are installed and the following sampling information for each system is recorded.

System-A

$$n_1 = 120$$

$$\bar{x}_1 = 4.1 \text{ minutes}$$

$$s_1 = 2.2 \text{ minutes}$$

System-B

$$n_2 = 100$$

$$\bar{x}_2 = 3.3 \text{ minutes}$$

$$s_2 = 1.5 \text{ minutes}$$

Check the above hypothesis at 1% level of significance?

Solution: (i) $H_0: \mu_1 > \mu_2$

$H_1: \mu_1 < \mu_2$ (One tailed test)

(ii) $\alpha = 0.01$, hence critical value is $Z_\alpha = 2.33$

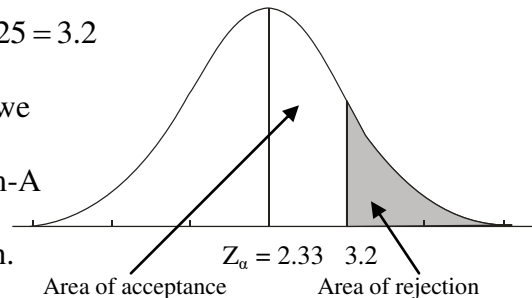
(iii) Standard error: $s_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{2.2^2}{120} + \frac{1.5^2}{100}} = \sqrt{0.0404 + 0.0225} = 0.25$

(iv) Test statistic: $z = \frac{(\bar{x}_1 - \bar{x}_2)}{s_{\bar{x}_1 - \bar{x}_2}} = (4.1 - 3.3) / 0.25 = 3.2$

Decision: Since $3.2 > \text{critical value } 2.33$ hence we may reject the null hypothesis.

This means that speedup customers with System-A Is not better than that of System-B

We should be 99% confident about this decision.



Tests of Difference between Two Population Means (Small Samples)

Suppose we have two normal populations with unknown but equal standard deviations σ_1 and σ_2 . Since, the independent samples used in testing means are small (less than 30) hence t-test is applied. The decision process is as under.

i. $H_0: \mu_1 - \mu_2 = 0$

$H_1: \mu_1 - \mu_2 \neq 0$

(Two tailed test)

ii. $H_0: \mu_1 - \mu_2 > 0$

$H_1: \mu_1 - \mu_2 < 0$

(One tailed test)

iii. $H_0: \mu_1 - \mu_2 < 0$

$H_1: \mu_1 - \mu_2 > 0$

(One tailed test)

The value of t-statistic is: $t_{v, \alpha/2} = \frac{(\bar{x}_1 - \bar{x}_2)}{s_{\bar{x}_1 - \bar{x}_2}}$

Here $v = n_1 + n_2 - 2$ is number of degrees of freedom. And a standard error

$$s_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \cdot \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

Example 9.1.6 A rancher wants to test two food mixtures 'Mixture-I' and 'Mixture-II' on random samples of steers to determine if there is a difference in their mean weight gains for the mixture. The sample data is as under:

Mixture-I: $n_1 = 12$ steers $\bar{x}_1 = 140$ pounds $s_1 = 6$ pounds

Mixture-II: $n_1 = 15$ steers $\bar{x}_2 = 124$ pounds $s_2 = 5$ pounds

Perform a hypothesis test at the 2% significant level. Interpret the results.

Solution: (i) $H_0: \mu_1 - \mu_2 = 0$ $H_1: \mu_1 - \mu_2 \neq 0$ (Two tailed test)

(ii) $\alpha = 0.02$, $v = 12 + 15 - 2 = 25$. Hence critical values of t-distribution are:

$$t_{v, \alpha/2} = t_{25, 0.01} = (-2.485, +2.485)$$

Standard error:

$$\begin{aligned} s_{\bar{x}_1 - \bar{x}_2} &= \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \cdot \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \\ &= \sqrt{\frac{11(36) + 14(25)}{25}} \sqrt{\frac{1}{12} + \frac{1}{15}} = 2.116 \end{aligned}$$

$$\text{Test statistic: } t = \frac{(\bar{x}_1 - \bar{x}_2)}{s_{\bar{x}_1 - \bar{x}_2}} = (140 - 124) / 2.116 = 7.56$$

Decision: Since $7.56 \notin (-2.485, +2.485)$ hence we reject the null hypothesis. We see from the data that sample mean of Mixture-I is greater than that of Mixture-II, hence we may conclude that steers should be fed Mixture-I.

Example 9.1.7 Two models of newly constructed roads were tested to see if mean mileage per gallon for Model-I is greater than Model-II. The data were as shown under.

Model-I: $n_1 = 8$ miles $\bar{x}_1 = 26.0$ miles per gallon $s_1 = 1.4$

Mixture-II: $n_1 = 10$ miles $\bar{x}_2 = 23.6$ miles per gallon $s_2 = 1.2$

Perform a hypothesis test at the 5% significant level. Interpret the results.

Solution: (i) $H_0: \mu_1 - \mu_2 > 0$ $H_1: \mu_1 - \mu_2 < 0$ (One tailed test)

(ii) $\alpha = 0.05$, $v = 8 + 10 - 2 = 16$. Hence critical values of t-distribution are

$$t_{v, \alpha} = t_{16, 0.05} = +1.746$$

$$\begin{aligned} s_{\bar{x}_1 - \bar{x}_2} &= \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \cdot \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \\ \text{(iii) Standard error:} &= \sqrt{\frac{7(1.4)^2 + 9(1.2)^2}{16}} \sqrt{\frac{1}{8} + \frac{1}{10}} = 2.45 \end{aligned}$$

$$\text{(iv) Test statistic: } t = \frac{(\bar{x}_1 - \bar{x}_2)}{s_{\bar{x}_1 - \bar{x}_2}} = (26 - 23.6) / 2.45 = 3.92$$

Decision: Since $1.746 < 3.92$ hence we accept the null hypothesis and conclude that mileage per gallon on newly constructed road Model-I is greater than that of Model-II and we say this with 95% confidence.

Test Concerning Population Variance σ^2

In this section we shall study testing the hypothesis concerning population variance. Let us consider the null hypothesis H_0 that the population variance σ^2 equals a specified value σ_0^2 against one of the usual alternatives $\sigma^2 < \sigma_0^2$, $\sigma^2 > \sigma_0^2$, or $\sigma^2 \neq \sigma_0^2$. An appropriate statistic is the chi-squared statistic to construct a confidence interval for σ^2 . Therefore, if we assume that the distribution of the population being sampled is normal, the chi-

squared value for testing $\sigma^2 = \sigma_0^2$ is given by $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$ where n is the sample size, s^2

is the sample variance and σ_0^2 is the value of σ^2 given by the null hypothesis. If H_0 is true, χ^2 is a value of chi-squared distribution with $(n - 1)$ degrees of freedom. Hence, for a two-tailed test at the α -level of significance, the critical region is $\chi^2 < \chi_{1-\alpha/2}^2$ and $\chi^2 > \chi_{\alpha/2}^2$. For the one tailed alternative $\sigma^2 < \sigma_0^2$, the critical region is $\chi^2 < \chi_{1-\alpha}^2$ and for the one tailed alternative $\sigma^2 > \sigma_0^2$, the critical region is $\chi^2 > \chi_{\alpha}^2$. Thus, for two tailed test:

$$\frac{(n-1)s^2}{\chi_{\alpha/2}^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_{1-\alpha/2}^2}$$

Example 9.1.8 A new drug for malaria fever is brought in the market with 99% confidence that average life of the drug is 4 years with variance of 1 year. If a sample of 15 is selected at random and is found that variance is 0.85 year. Should a wholesale dealer accept the claim of manufacturer?

Solution: Let us set up the hypothesis

- (i) $H_0 : \sigma^2 = \sigma_0^2 = 1, H_0 : \sigma^2 = \sigma_0^2 \neq 1$
- (ii) $\alpha = 0.01, n = 15, v = n - 1 = 14$ (degrees of freedom)
- (iii) $\chi_{v, 1-\alpha/2}^2 = \chi_{14, 0.995}^2 = 4.075, \chi_{v, \alpha/2}^2 = \chi_{14, 0.005}^2 = 31.319$
- (iv) $\frac{(n-1)s^2}{\chi_{\alpha/2}^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_{1-\alpha/2}^2}$
 or $\frac{14(0.85)^2}{31.319} < \sigma^2 < \frac{14(0.85)^2}{4.075}$ or $0.33 < \sigma^2 < 2.48$
- (v) You may observe that $1 \in (0.33, 2.48)$. Thus claim of drug manufacturer is correct and the wholesales dealer may accept this claim with 99% confidence.

Test Concerning the Ratio of Variances of Two Populations

We have discussed in the previous chapter that to compute the confidence interval for the ratio of two population variances we use F-distribution. In this section we shall use hypothesis testing to see that the claim made for this ratio based on sample information is valid or not to desire level of significance. The following assumptions are made

- (i) State the null hypothesis $H_0: \sigma_1^2 = \sigma_2^2$ and alternative hypothesis $H_1: \sigma_1^2 \neq \sigma_2^2$ for two tailed test. For one tailed test the alternative hypothesis will be $H_1: \sigma_1^2 > \sigma_2^2$ or $H_1: \sigma_1^2 < \sigma_2^2$.
- (ii) Decide the level of significance α

- (iii) Compute the test statistic: $F = s_1^2 / s_2^2$
 - (iv) Compute the critical values
 - (a) For two tailed test: Accept H_0 if $F_{1-\alpha/2}(v_1, v_2) < F < F_{\alpha/2}(v_1, v_2)$
 - (b) For right one tailed test: Accept if $F < F_{\alpha}(v_1, v_2)$
 - (c) For left one tailed test $F > F_{1-\alpha}(v_1, v_2)$
- Here v_1 and v_2 are degrees of freedoms given by $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$.
- (v) The null hypothesis is accepted if the value of F-statistic satisfies any of the conditions (a) to (c) otherwise reject H_0 .

It may be noted that, $F_{\alpha}(v_1, v_2) = \frac{1}{F_{1-\alpha}(v_2, v_1)}$

Example 9.1.9 A study was made to see the difference in production of a new wheat seed planted at two different agriculture farms. The sample from 12 different cites at Farm-I was taken and it was found that sample variance in weight per acre is 1.44 (100 kg²). Another sample from 10 different cites at Farm-II was taken and it was found that sample variance in weight per acre is 1.69 (100 kg²). Test the hypothesis that two population variances are equal using 10% level of significance.

Solution:

- (i) Null hypothesis is $H_0: \sigma_1^2 = \sigma_2^2$ and thus alternative hypothesis is $H_1: \sigma_1^2 \neq \sigma_2^2$
- (ii) Level of significance $\alpha = 0.10$
- (iii) Computation of statistic: $F = s_1^2 / s_2^2 = 1.44 / 1.69 = 0.85$
- (iv) Degrees of freedom: $v_1 = 12 - 1 = 11$ and $v_2 = 10 - 1 = 9$
- (v) Computation of critical values (For two tailed test)

$$F_{1-\alpha/2}(v_2, v_1) < F < F_{\alpha/2}(v_1, v_2) \quad \text{or} \quad F_{0.95}(11, 9) < F < F_{0.05}(11, 9)$$

$$\text{or } 0.34 < F < 3.11 \quad \text{NOTE: } F_{0.95}(11, 9) = 1 / F_{0.05}(9, 11) = 1 / 2.9 = 0.34$$
- (vi) Since value of F-statistic 0.85 lies in the interval (0.34, 3.11) hence we accept the null hypothesis and conclude that the variance of two populations is same that is variance in productions at Farms I and II is same.

9.2 TESTS OF GOODNESS OF FIT AND TEST OF INDEPENDENCE

In this section we shall deal with two important concepts in statistical analysis namely, *Test of Goodness of Fit* and *Test of Independence*. These tests are used in situations where study concerns *category variables* rather than *quantity variables*, that is when one is not interested in measurement of quantity for each observation in a sample but with counting how many observations fall into each category. For example, consider the case where management is interested in employing 300 people and is held responsible that employment is done on the base of sexual discrimination. The number of males and females employed is shown as under

Male	Female	Total
200	100	300

Here we have two categories, i.e., male and female. To deny the accusation, company has to prove that employment has been done without any discrimination. This means that company has to prove that employment is done according to the expectation of labour court. In other words one has to compare that *expected frequencies* with the actual number of *observed frequencies*. This gives rise to *Goodness of Fit Test* which means that there is a good fit between observed frequencies and expected frequencies. This is

also known as uniform distribution. It may be noted that this has to be done with certain level of confidence that there is no discrimination.

The chi-squared test can be extended to situations where cross-tabulations of variables can further be categorized. For example, the above data can further be categorized according to marital status. The company may now be accused of employment on the basis of marital status that is the employment of males and females is independent of their marital status. This is known as *Test of Independence*. We use chi-squared test to check the *goodness of fit* as well as *test of independence* which will be discussed shortly.

It may be noted that chi-squared test used here differs from that used in the previous chapter and the sections discussed above. The Chi-squared test used in the above sections is focused around the properties of normal distribution hence, it is known as *parametric* test. In contrast, the chi-squared test that we use in these sections is called *non-parametric* test because, it is applicable to situations where we do not make any assumption that population is normal.

Goodness of Fit Test

Goodness of fit test is used in different ways. Let us consider the problem of discrimination as stated above. The observed frequencies are 200 and 100 whereas expected frequencies are yet to be computed as stated above. Although there are different ways to find the expected frequencies, one way to find these is according to population ratio of males and females in the city where company is located. Let us suppose that this ratio is 55% and 45% respectively. Thus we have

	Male	Female
Observed frequencies (O)	200	100
Expected frequencies (E)	165	135

(Notice that 55% of 300 = 165 and 45% of 300 = 135).

If there is no difference between the observed and expected frequencies in each category, then this is a good indication of no discrimination.

The procedure to apply the chi-squared test in this case is as follows:

(i) Set the hypothesis

H_0 : The row and column categories are uniformly distributed.

H_1 : The row and column categories are not uniform.

(ii) Set the level of confidence: α

(iii) Set the degrees of freedom: $v = (k - 1)$ where k is number of categories.

(iv) Compute $\chi^2_{v, \alpha}$

(v) Compute χ^2 - statistic using the formula $\chi^2 = \sum \left(\frac{[O_i - E_i]^2}{E_i} \right)$

(vi) If $\chi^2 < \chi^2_{v, \alpha}$ accept the null hypothesis and conclude that row and columns categories are *uniformly distributed*, that is, a *goodness of fit* is followed at the given level of confidence.

Example 9.2.1 With reference to above table test the hypothesis whether or not the employment of male and female is uniform at 1% level of confidence. The table is reproduced below.

	Male	Female
Observed frequencies (O)	200	100
Expected frequencies (E)	165	135

Solution:

(i) Set the hypothesis H_0 : The row and column categories are uniformly distributed.

H_1 : The row and column categories are not uniform.

(ii) Set the level of confidence: $\alpha = 0.01$

(iii) Set the degrees of freedom: $v = k - 1 = (2 - 1) = 1$ where k is number of categories Here male and female).

(iv) Compute $\chi^2_{v, \alpha} = \chi^2_{1, 0.01} = 6.635$

(v) Compute χ^2 - statistic:

$$\chi^2 = \sum \left(\frac{[O_i - E_i]^2}{E_i} \right) = \frac{(200-165)^2}{165} + \frac{(100-135)^2}{135} = 16.49$$

(vi) We see that $\chi^2 > \chi^2_{1, 0.01}$, hence we reject the null hypothesis and conclude that employment is not uniformly distributed at 99% level of confidence.

Example 9.2.2 A television manufacturing company produces the same brand of televisions in six factories situated at six different cities. Each factory is employing same number of workers, engineers etc with same technology. The chief of these factories is considering enhancing production and so wishes to expand one of these factories rather than built a new factory. The expected production is 5000 television per week. The production level at these factories in a week is shown as under.

Factory Situated at	No. of Computers in (1000s)
Karachi	7
Sukkur	4
Quetta	6
Multan	5
Lahore	4
Peshawar	4
Total	30

What suggestions would you give to the chief if the level of significance is set at 5%.

Solution: Since the total production is 30 thousand a week (six working days) hence the average production that is 5000 televisions is achieved. Now the calculation of chi-squared test is shown in the following table.

Factory at	O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2$	$(O_i - E_i)^2/E_i$
Karachi	7	5	2	4	0.8
Sukkur	4	5	-1	1	0.2
Quetta	6	5	1	1	0.2
Multan	5	5	0	0	0.0
Lahore	4	5	-1	1	0.2
Peshawar	4	5	-1	1	0.2
Total	30	30			1.6

- (i) Set the hypothesis H_0 : Any factory may be enhanced for more production.
 H_1 : A particular factory may be enhanced for more production.
- (ii) Set the level of confidence: $\alpha = 0.05$
- (iii) Set the degrees of freedom: $v = k - 1 = 6 - 1 = 5$ (k being number of factories)
- (iv) Compute $\chi^2_{v, \alpha} = \chi^2_{5, 0.05} = 11.05$
- (v) Compute χ^2 - statistic:

$$\chi^2 = \sum \left(\frac{[O_i - E_i]^2}{E_i} \right) = 1.6$$

(vi) We see that $\chi^2 < \chi^2_{5, 0.05}$, hence we accept the null hypothesis and chief may be suggested to enhance any of the factories for more production instead of constructing a new factory with 95% level of confidence.

Goodness of Fit Test-Further Cases

Goodness of fit test is also applicable to test whether or not the given distribution is binomial or Poisson or normal distribution. We shall illustrate this by giving examples and which will show that how this test is used.

It may be noted that if any class contains frequency less than 5 it must be adjusted with adjacent class.

Example 9.2.3 (Goodness of Fit- Binomial Distribution) A company makes compressed gas cylinders. Twenty cylinders are packed in a box. Customers occasionally complain for the defectiveness of the cylinders (cylinder with low pressure). A sample of 100 groups was tested and it was found that 100 cylinders are defective. The following table shows the number of defective cylinders in this sample of 100 boxes.

No. of defective cylinders in a group of 20	0	1	2	3	4	5	≥ 6	Total
No. of boxes observed O_i	39	34	20	4	1	2	0	100

The complaint is accepted if it is found that distribution does not follow the binomial distribution. Test at 99% confidence level that complaint is rejected.

Solution: Here $p = 100/(100 \times 20) = 0.05$, $q = 0.95$ and $n = 20$. Thus

$P(X = 0) = b(20, 0, 0.05) = (0.95)^{20} = 0.3585$. [Using binomial distribution formula]

Thus expected frequency for no defective cylinder in the sample is

$$E_1 = 0.3538 (100) = 35.85.$$

Similarly, we may calculate the other expected frequencies. The following table now shows the complete computation for chi-squared values. It may be noted that frequencies of 3rd, 4th, 5th, and 6th classes are added because their frequencies are less than 5. The table is now as under:

No. of defectives	O_i	E_i	$(O_i - E_i)^2/E_i$
0	39	35.85	0.2768
1	34	37.73	0.3687
2	20	18.87	0.0677
3 or more	7	7.55	0.0401
			$\chi^2 = 0.7533$

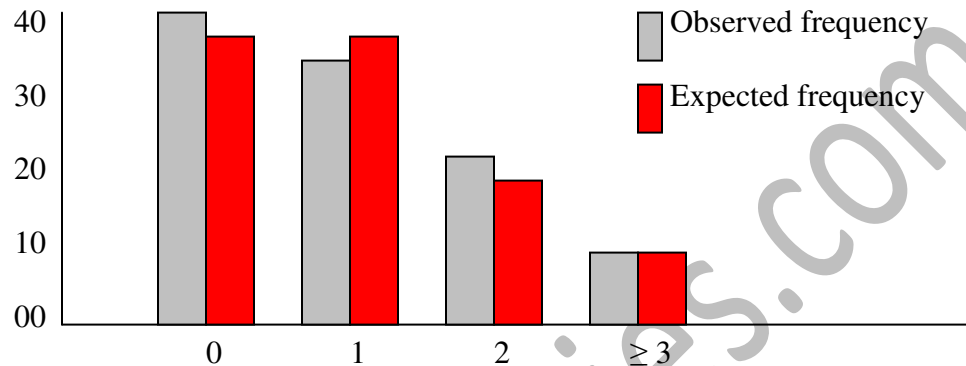
Now, H_0 = the distribution is binomial, H_1 = the distribution is not binomial
 Level of significance $\alpha = 0.01$ $v = df = k - p - 1 = 7 - 4 - 1 = 2$

Here k is total number of classes; p is number of classes merged.

$$\chi^2_{v,0.01} = \chi^2_{2,0.01} = 9.21$$

$$\text{Statistic: } \chi^2 = 0.7533$$

Since $\chi^2 < \chi^2_{v,0.01}$ hence we accept null hypothesis and decide that given distribution is binomial. We see from the figure below that expected frequencies fit the observed frequencies.



Example 9.2.4 Goodness of Fit- Poisson Distribution: A firm manufactures computer memory chips wants to see that number of defective memory chips follow Poisson distribution. The inspector of the firm takes a random sample of 60 rods and observes the following:

No. of defects	0	1	2	≥ 3
Observed frequencies	32	15	9	4

Fit a Poisson distribution and determine the goodness of fit.

Solution: We know that Poisson distribution is given by:

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!} \text{ where } \mu = \frac{\sum x.f}{\sum f} = 0.75$$

Using the above formula, we get Poisson probability distribution as

X	P(X = x)	Expected Frequency
0	$e^{-0.75} \cdot (0.75)^0 / 0! = 0.472$	$0.472 \times 60 = 28.32$
1	$e^{-0.75} \cdot (0.75)^1 / 1! = 0.354$	$0.354 \times 60 = 21.14$
2	$e^{-0.75} \cdot (0.75)^2 / 2! = 0.133$	$0.133 \times 60 = 7.98$
≥ 3	$1 - P(0) - P(1) - P(2) = 0.041$	$0.041 \times 60 = 2.46$

We see from the first table that observed frequency of the last class is less than 5 hence we combine the two classes. Thus new table for computing the χ^2 - statistic is as follows:

No. of defectives	O_i	E_i	$(O_i - E_i)^2 / E_i$
0	32	28.32	0.478
1	15	21.14	1.783
2 or more	13	10.44	0.63
			$\chi^2 = 2.891$

Now, H_0 = the distribution is Poisson,

H_1 = the distribution is not Poisson

Level of significance $\alpha = 0.05$

$v = df = k - p - 1 = 4 - 2 - 1 = 1$

Here k is total number of classes; p is number of classes merged.

$$\chi^2_{v,0.01} = \chi^2_{1,0.05} = 3.84$$

$$\text{Statistic: } \chi^2 = 2.891$$

Since $\chi^2 < \chi^2_{2,0.05}$ hence we accept null hypothesis and decide that given distribution is Poisson.

Example 9.2.5 Goodness of Fit- Normal Distribution: A company wishes to see that its sale of mineral water is normally distributed a random sale for 200 days is recorded and is produced as under. At 1% level of significance, perform a test of hypothesis that sales are normally distributed by using the values of parameters μ and σ as point estimates $\bar{x}$ and s .

Sales (Thousands of gallons)	No. of days
≤ 34	0
34 – 35.5	13
35.5 – 37.0	20
37.0 – 38.5	35
38.5 – 40.0	43
40.0 – 41.5	51
41.5 – 43.0	27
43.0 – 44.5	1
44.5 – 46.0	0
≥ 46.0	0
Total	200

Solution: Sample mean: $\bar{x} = \frac{\sum xf}{\sum f} = 40$ (Thousand gallons) = μ

$$\text{Sample S.D: } \sqrt{\frac{\sum (x - \bar{x})^2 f}{\sum f}} = 2.5 \text{ (Thousand gallons)} = \sigma$$

Now by Normal distribution:

$$Z = (X - \mu) / \sigma$$

For $X = 34$, $Z = (34 - 40) / 2.5 = -2.4$. Hence, $P(X \leq 34) = P(Z \leq -2.4) = 0.0082$

Thus expected frequency of this class is $0.0082 \times 200 = 1.64$. Expected frequencies for other classes may be calculated similarly. The following table shows the computation of χ^2 -statistic.

Sales	O_i	E_i	$(O_i - E_i)^2 / E_i$
≤ 34	0	1.64	---
34.0 – 35.5	$0 + 13 = 13$	$1.64 + 5.54 = 7.18$	4.7176
35.5 – 37.0	20	15.84	1.0925
37.0 – 38.5	35	31.84	0.3136
38.5 – 40.0	43	45.14	0.01015
40.0 – 41.5	51	45.14	0.7607
41.5 – 43.0	27	31.84	0.7357
43.0 – 44.5	10	15.84	2.1531
44.5 – 46.0	1	5.54	---
≥ 46	0	$5.54 + 1.64 = 7.18$	5.3193
			$\chi^2 = 15.1940$

Now, H_0 = the distribution is normal, H_1 = the distribution is not normal
 Level of significance $\alpha = 0.05$ $v = df = k - p - 1 = 10 - 4 - 1 = 5$
 Here k is total number of classes; p is number of classes merged.

$$\chi^2_{v,0.05} = \chi^2_{5,0.05} = 11.0705 \quad \text{Statistic: } \chi^2 = 15.1940$$

Since $\chi^2 > \chi^2_{5,0.05}$ hence we reject the null hypothesis and conclude that given distribution does not follow normal distribution.

Test of Independence

As mentioned in the introduction to this section, the χ^2 test can also be applied to situations in which observations are cross-classified among different variables known as *contingency table*. In such cases, the test involves a test of independence of the classificatory variables. Examples of the sorts of questions that may be addressed are

- ✓ Is there a relationship between the educational backgrounds of new employees and various salary scales on which they are placed?
- ✓ Is there a significant difference in the pattern of housing occupancy of people from social classes?
- ✓ Does the degree of customer satisfaction with three models of food-mixer depend on the type of mixer purchased?

A contingency table consists of data arranged in rows and columns. Actual sample counts are called *observed frequencies*. The *expected frequencies* are calculated by using the formula:

$$E_i = \frac{(\text{Row Total}) (\text{Column Total})}{\text{Grand Total}}$$

Degrees of freedoms are calculated as: $v = d.f = (r - 1) (c - 1)$, where 'r' is number of rows and 'c' is number of columns of contingency table.

Example 9.2.6 A firm manufactures apple juices where, apples are grown in three different regions of country wishes to collect opinions on quality of juice. The customers tested juices and they rated juice quality on a scale of 1 to 4 where 1 stands for the best, 2 for good, 3 for fair and 4 for not well at all. The sample data are given in the following table. Firm wants to know whether quality ratings are or are not independent of the apple-growing region. Perform the test for independence at the 1% level of significance.

Quality Rating	Growing Region			Row Total
	I	II	III	
1	15	10	6	31
2	7	13	12	32
3	11	12	8	31
4	3	8	15	26
Column Total	36	43	41	120

Solution: Using the formula $E_i = \frac{(\text{Row Total}) (\text{Column Total})}{\text{Grand Total}}$ the contingency table is

reproduced as under showing the observed frequencies and expected frequencies in braces.

Observed frequencies (O_i)	Expected frequencies (E_i)	$(O_i - E_i)$	$(O_i - E_i)^2 / E_i$
15	9.3	5.7	3.4935
10	11.11	-1.1	0.1090
6	10.6	-4.6	1.9962
7	9.6	-2.6	0.7042
13	11.5	1.5	0.1957
12	10.9	1.1	0.1110
11	9.3	1.7	0.3108
12	11.1	0.9	0.0730
8	10.6	-2.6	0.6377
3	7.8	-4.8	2.9538
8	9.3	-1.3	0.1817
15	8.9	6.1	4.1809
Total	---	---	$\chi^2 = 14.9475$

H_0 : Quality rating and growing areas of apples are independent.

H_1 : Quality rating and growing areas of apples are dependent.

$$v = d.f = (r - 1)(c - 1) = (4 - 1)(3 - 1) = 6$$

$$\alpha = 0.01, \text{ thus, } \chi^2_{v, 0.01} = \chi^2_{6, 0.01} = 16.8119$$

$$\text{Statistic } \chi^2 = 14.9475$$

Decision: Since $\chi^2 < \chi^2_{v, 0.01}$ hence we accept the null hypothesis and conclude that “Quality rating and growing areas of apples are independent”.

WORKSHEET 09

1. The output voltage for a certain electric circuit is specified to be 130 V. A sample of 30 independent readings on the voltage for this circuit had the following data:

130.1, 129.2, 129.4, 128.3, 128.1, 131.0, 132.2, 133.2, 129.3, 128.6, 128.5, 128.7, 130.1, 130.6, 128.2, 128.0, 128.1, 128.3, 129.1, 127.0, 128.0, 130.0, 130.1, 130.2, 129.2, 129.1, 128.2, 128.3, 128.2, 128.0.

Test the hypothesis that the average output voltage 130 against the alternative hypothesis that is less than 130. Use a 5% significance level.

2. The spot speeds at a particular location on an express way are known to be normally distributed with a mean of 80 km/h. A new radar speedometer was brought by traffic department and set of 100 observations of speeds was taken. The mean speed observed was 77.3 km/h and a standard deviation was 15 km/h. Is there any evidence to prove that:

- The new speedometer might have been faulty.
- The new speedometer is showing lesser speeds than actual.

3. The mean spot speeds of 15 vehicles observed on a Sunday at a particular location of a highway was 81.2 km/h and the standard deviation of 10.2 km/h. The mean speed of all vehicles at this location as per previous record was 75.5 km/h. Is there sufficient evidence to show that the speed of vehicles on the particular Sunday:

- i. Was higher than the average speed
 - ii. Was different from the average speed
4. A manufacturer of TV tube has for many years used a process giving a mean tube life of 4700 hours and standard deviation of 1460 hours. A new process is tried to see if it will increase the life significantly, a sample of 100 new tubes gave a mean life of 5000 hours. Is the new process better than the old at one percent level of significance?
5. The breaking strength of cables produced by a manufacturer has mean 2200 lbs. it is assumed that by a new technique, breaking strength can be increased. To test this assumption a sample of 16 cables, is obtained by a new technique was tested and found that mean breaking strength was 2320 lbs with a standard deviation of 220 lbs. Can this sample result support the assumption at 0.01 level of significance?
6. Study is being made of the failure of an electronic component. There are four possible types of failure and two mounting positions for the device. The following data have been taken. Would you conclude that the type of failure is independent of mounting position? Use $\alpha=0.01$.

Mounting Position	Failure Type				Total
	A	B	C	D	
1	22(21)	46(47)	18(17)	9(8)	95
2	4(3)	17(15)	6(5)	12(10)	39
Total	26	63	24	21	134

7. A sample of 200 hard disks is selected from the week output of hardware laboratory that employs three hardware engineers. The hard disks are inspected to determine whether or not they are defective and are categorized according to which hardware engineer did the work. The results are as follows:
Is the defective/non defective classification independent of hardware engineer classification? Conduct a test at one percent significance level.

	Hard Ware Engineer			Total
	A	B	C	
Defective	10(9)	8(10)	14(11)	32
Non-Defective	52(53)	60(57)	56(58)	168
Total	62	68	70	200

8. A computer scientist has developed an algorithm for generating pseudo random integer over the interval 0-9. He codes the algorithm and generates 1000 pseudo random digits. The data are shown as observed frequencies in the following table.
Is the evidence that the random number generator is working correctly, use $\alpha=0.05$.
9. The number of messages sent to computer bulletin board is hypothesized to follow a Poisson distribution. A random sample of 60 bulletin boards has been collected and the following number of messages observed
Fit a Poisson distribution and determine the goodness of fit.

Number of Messages	0	1	2	3
Observed Frequency	32	15	9	4

10. Union officials claim that mean salary per month paid by other companies is Rs. 6000/-. A sample of 49 workers was taken at random and it was found that their mean salary is Rs. 5400 with sample standard deviation Rs. 100/-. At 5% significance level, test the hypothesis that claims of union is true.

11. The environment protection officer of large industrial plant wants to determine whether the mean amount of carbon-mono-oxide emitted by the plant exceeds 7 tons per day. Suppose emission amounts are normally distributed. Perform a hypothesis test at 1% level if, for a random sample of 10 days, tons of emission per day are:

9 8 11 16 12 7 9 6 14 13

12. Chairman of a car selling company is of the opinion that profit on the sale of cars this year is less than the last year although the number of cars sold this year is more than the last year. The Manager claims that this year profit has increased. To test his claim, manager takes a sample of 35 sales for both years and finds that average profit on the sale last year was Rs. 32 millions with sample standard deviation of Rs. 1 million and profit on the sale of cars this year is Rs. 30 millions with sample standard deviation of Rs. 0.75 millions. Test the hypothesis at 2% level of significance that managers claim is correct.

13. Costly road testing of random samples of cars was carried out to determine if mean mileage (mile per gallon) is greater for road Model-I cars than for Model-II cars. The sample data is as under:

Model-I: $n_1 = 10$ sample mean = 25.9 sample S.D = 1.5

Model-II: $n_2 = 12$ sample mean = 24.5 sample S.D = 1.2

Perform a hypothesis test at 2% level. Interpret your results.

14. The capacitance of brand I and II split ACs are rated the same. The capacities of random samples of six units of each brand were determined. The sample data in thousands of BTU (British Thermal Units) were:

Brand-I: 6.1 6.5 5.7 6.3 6.5 5.9

Brand-II: 4.9 5.2 5.4 5.2 5.1 5.3

Perform a hypothesis test at 1% level to determine whether mean BTU consumption for the two brands differ.

15. A Structure Engineer is testing the strength of newly designed steel beam required in cantilever construction. As his null hypothesis assumes that the mean strength will be greater as 10,000 pounds per square inches (psi) for traditional beams. A test sample of 9 new beams provided mean 105,000 psi and S.D equal to 10,000 psi . At 1% level of significance should the engineer accept or reject the hypothesis.

16. An airline company conducting an analysis concerning load factors due to passenger and weights for plane to determine whether human and baggage are related. The airline selected random sample of 16 females and 14 males. The mean baggage weight for the female was 42 pounds with a variance of 9 pounds square while for males the mean weight was 39 pounds with variance of 10 squared. Is there a significance increase in the baggage weight for females than males? Use a 0.01 level of significance and assume the population to be approximately normal with equal variance.

17. From a central library situated at Meharn University, a random sample of size 50 days was taken and it was found that on the average 125 students came for reading with standard deviation of 36 students. From Sindh University a random sample of 40 days was taken and found that on the average 90 students came for reading with a standard deviation of 40 students. Has there a reason to believe that averagely more students at

Meharn come for study than Sindh University. Use the significance level equal to 0.05.

18. An advertising company claimed that the average advertising budget for small firms is more than Rs.45, 000 per annum. To test its claim a random sample of 125 small firms showed an advertising budget of Rs. 42,710 per annum with standard deviation of Rs. 1, 000/. At the level of 0.05 do we accept or reject the hypothesis?

19. A study of a sample of 198 bank accounts showed that average size of bank account is Rs.7, 542 with standard deviation of Rs.2, 984. Test the hypothesis that mean size of account is Rs. 8,000. Use a 0.01 level of significance.

20. The strength of steel wire made by an existing process is normally distributed with mean 1250 and standard deviation 150. A batch of wire is made by a new process and a sample of 25 measurements gives an average strength 1312 with standard deviation of 100. Has engineer strong evidence to justify changing to new process?

21. Two techniques for a certain laboratory test are compared to determine which one is more efficient. The average of 57 tests is 40 days with standard deviation of 6 days. The average of 65 tests is 50 days with standard deviation of 4 days. Would you be justified in concluding that the second method is preferable? Use significance 0.01.

22. A security analyst for four industries A, B, C and D classifies a random sample of stock as being high, average, or low in terms of safety to buyers of stock. The data is given below. Perform a test of independence of safety rating and industry classification (Take $\alpha = 0.01$).

Stock safety rating	Industry category			
	A	B	C	D
High	26	19	30	13
Average	33	31	41	20
Low	14	33	13	32

23. A hotel has rooms in the high, average, and low price levels. The owner advertises high-quality service in all rooms. Opinions of service obtained from a random sample of guests are given in the following table. Is guests' ratings independence of price? (Take $\alpha = 0.01$).

Guest service rating	Room price		
	High	Average	Low
Excellent	1400	2500	1100
Good	200	6600	1400
Poor	600	1900	500

24. The following table shows random sample data on the number of workers absent from their job. The sample contained same number of workers (10) for each day from Monday to Friday. Does it appear that number of workers absent is uniformly distributed over days of the week? (Perform a goodness of fit test at 1% level).

Day	Number of workers absent (O_i)	Expected value (E_i)
Monday	4	10
Tuesday	10	10
Wednesday	11	10
Thursday	12	10
Friday	15	10

25. To know if binomial distribution can be used in a statistical decision analysis that involves number of people who have seen the advertisement for the employment in automobile company. The company obtained 200 random samples where each sample contains 10 people. The number of people who saw the advertisement (Call them successors) is show below:

No. of successes	Frequency
0	14
1	31
2	49
3	55
4	30
5	15
6	6
7	2
8 to 10	0
TOTAL	200

At 5% level, perform the test oh hypothesis that the distribution of number of people who saw the advertisement is binomial with $n = 10$ and $p = 0.25$.

26. The following data shows life in hours of 200 bulbs chose at random. Do this data provide sufficient evidence that life of the bulbs is normally distributed with mean 500 hours and S.D equals to 100, perform a good-ness-of-fit test at 5% level.

Life in hours	Observed frequency
< 260	2
260 – 340	6
340 – 420	34
420 – 500	64
500 – 580	50
580 – 660	27
660 – 740	11
≥ 740	4
Total	200

27. The following table shows number of customers entering a departmental store in a random sample of 1-minute time interval. Does the number of customers entering appear to have a Poisson distribution with mean equal to 1 arrival per minute? Perform the goodness of fit test at 1% level.

Number of customers entering the store	Frequency
0	47
1	61
2	37
3	3
4	1
5	1
≥ 6	0
Total	150

CHAPTER TEN

LINEAR REGRESSION AND CORRELATION

10.1 INTRODUCTION

There are many situations in science, engineering, business, medical science etc; where observations on two or more variables appear to be related to one another. Simple regression analysis provides an equation that can be used to estimate or predict the value of a dependent variable for a given value of an independent variable. In a simple linear regression, the estimating equation has a graph which is a straight line. The estimating equation is determined by performing calculations on observed data. For example, breaking strength of iron bars produce in some factory may depend upon the material used, sale of new computer may depend on advertisement, number of workers who remain absent from their work may depend on the weather conditions; etc.

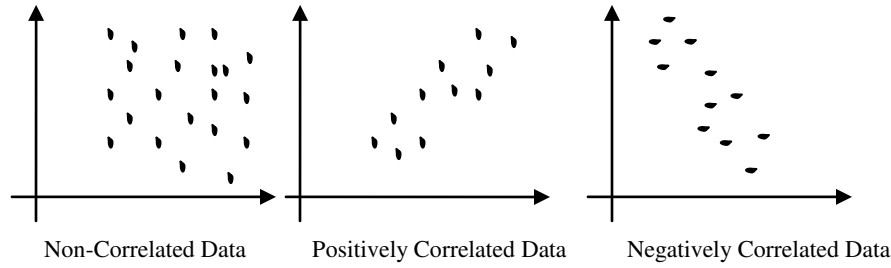
The purpose of simple linear correlation analysis is to determine whether or not two variables are related, that is, whether one variable tends to be larger/smaller as the other variable changes. The outcome of correlation analysis is not an estimating equation but a conclusion that there is, or is not, a relationship between the variables.

Often, two variables are correlated, but an estimating (regression) equation does a poor job of predicting the value of one variable from the known value of the other variable. However, the correlation between two variables aid in making a decisions.

Summarizing, we may say that '*Regression Analysis*' is concerned with measuring the way in which one variable is related to another, that is, how differences in one variable helps to explain differences in other variable. '*Correlation Analysis*' on the other hand, is simply concerned with providing a statistical measure of the strength of any relationship between variables. If this relationship only between two variables is being considered the term *simple* regression and correlation analysis is used. This book is only concerned with simple regression and correlation

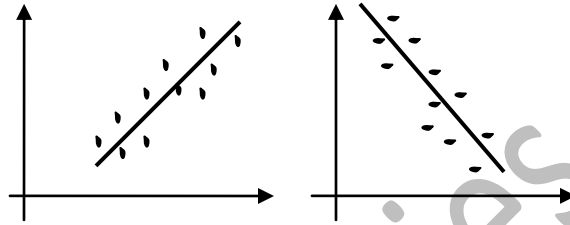
Scattered Diagram

In statistical studies, we collect pairs of data number, called observations. For example, let a person rent a car where fixed cost is Rs. 1, 000 and per kilometer charge is Rs. 50 then the total amount charged is given by $y = 1000 + 50x$, where y is the total amount and x is the distance traveled. Thus if $x = 100$ the $y = 6000$ hence the ordered pair is $(100, 5000)$. Now if the car is rented with the condition that if it is used for more than 200 kilometers then extra Rs. 500 would be charged. In this case we get an ordered pair that will not satisfy above equation. It may be noted that first component of ordered pair represents the independent variable and the second component the dependent variable. If such pairs of data are plotted in xy -plane using dot for each pair, the diagram so obtained is called *Scattered Diagram*. It may be noted that these ordered pairs may not lie on the same line. But there is a tendency for the points to rise or drop as we move away to the right. Some of the scattered diagrams are shown below indicating that whether the data points are not correlated, positively correlated or negatively correlated.



Simple Regression Analysis

We have introduced the notion of simple regression and correlation in the introduction of this chapter. It may be noted that best fit in regression is a straight line. This is shown below.



It should be noted that regression analysis is not restricted to such linear relationship, but can be extended to non-linear relationships as well; however, the principles are same. The central problem of simple regression analysis is to employ the criteria which will allow us to identify the best-fitting line passing through any set of paired observations on x and y , and hence to determine its equation in mathematical terms. In general equation of any straight line is given by an expression

$$y = a + b x$$

Here y is dependent and x is an independent variables, ' b ' is the slope and ' a ' the y -intercept which are to be determined. The method used to determine the values of ' a ' and ' b ' is known as method of least squares.

Method of Least-Squares

Consider the data points (x_i, y_i) , $i = 1, 2, 3, \dots, n$. If the line $y = a + b x$ passes through these points then method of least squares suggests that sum of the squares of deviation/difference between y and y_i should be as small as possible. That is,

$$S = \sum_i d_i^2 = \sum_i (y - y_i)^2 = \sum_i (a - bx_i - y_i)^2$$

is minimum. Using mathematical analysis for maxima/minima criteria, we have:

$$\begin{aligned} \frac{\partial S}{\partial a} &= 2 \sum_i (a - bx_i - y_i) = 0 & \text{and} & & \frac{\partial S}{\partial b} &= 2 \sum_i (a - bx_i - y_i)x_i = 0 \\ &= na - b \sum x_i - \sum y_i = 0 & \text{and} & & &= a \sum x_i - b \sum x_i^2 - \sum x_i y_i = 0 \end{aligned}$$

Solving for ' a ' and ' b ' we obtain,

$$a = \frac{\sum y_i \sum x_i^2 - \sum x_i \sum x_i y_i}{n \sum x_i^2 - (\sum x_i)^2} \quad \text{and} \quad b = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

Example 10.1.1 The following data show monthly expenditures in (Lacs of Rs.) and sales in (Millions of Rs.) over a period of six months. Estimate the relationship between sales (y) and advertising expenditures (x) by using least square regression line. Also predict y for $x = 7$.

Advertisement Expenditures:	1	2	3	4	5	6
Sales:	3	15	6	20	9	25

Solution: The following table shows the computations of various terms that are necessary in calculations of 'a' and 'b'.

x_i	y_i	x_i^2	$x_i y_i$	y_i^2
1	3	1	3	9
2	15	4	30	225
3	6	9	18	36
4	20	16	80	400
5	9	25	45	81
6	25	36	150	625
$\sum x_i = 21$	$\sum y_i = 78$	$\sum x_i^2 = 91$	$\sum x_i y_i = 326$	$\sum y_i^2 = 1376$

$$\text{Now, } a = \frac{\sum y_i \sum x_i^2 - \sum x_i \sum x_i y_i}{n \sum x_i^2 - (\sum x_i)^2} = \frac{78(91) - 21(326)}{6(91) - (21)^2} = 2.40$$

$$\text{and } b = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2} = \frac{6(326) - 21(78)}{6(91) - (21)^2} = 3.026$$

Hence the estimated relationship (regression line) between sales and advertisement expenditure is:

$$y = 2.40 + 3.026 x$$

Now if $x = 7$ then $y = 23.58$. This means that if expenditures on advertisement are made up to Rs. 7 lacs then it is estimated that sale may go to Rs. 23.58 millions.

Inference Regarding Slope of Population Regression Line

In chapters 8 and 9 we discussed inference of two kinds, confidence intervals and hypothesis testing. In this section we shall first test the hypothesis about the population slope say B; then we shall construct confidence intervals for B.

Testing Hypothesis for Slope of Population Regression Line

While dealing with regression equations, we have not considered their statistical significance. The significance of the regression equations depends on 'a' and 'b', where 'a' and 'b' represent estimates population intercept and population slope. Mostly we test the significance of 'b' because the intercept 'a' is of little interest. It may be noted that population regression line is denoted by $y = A + B x$.

We know that different samples can be taken from a population. For the regression purpose, different samples will give different values of slope 'b'. Each sample slope 'b' is an estimate of population slope 'B'. The sample slopes will vary around the value of 'B', so we need a measure of slope variation. This measure is denoted by S_b , and is called the sample standard error of the slope; and is given by:

$$S_b = \frac{S_e}{\sqrt{\sum x_i^2 - (\sum x_i)^2 / n}}, \text{ where } S_e = \sqrt{\frac{\sum y_i^2 - a \sum y_i - b \sum x_i y_i}{(n-2)}}$$

Here S_e is known as *sample standard error of estimate*.

Assumptions:

1. State the null and alternative hypothesis:

$H_0: B = 0$, where B is the slope of population regression line.

$H_1: B \neq 0$ (For two tailed test); $H_1 < 0$ or $H_1 > 0$ (for one tailed test).

This means that if $H_1 > 0$ then we accept the alternative otherwise reject H_1 and accept the null hypothesis which means that slope of population is not in increasing mode.

2. Decide the level of significance α

3. Compute degrees of freedom $v = n - 2$, where n is sample size (Data values).

4. (i) Compute the critical values for two tailed test $\pm t_{\alpha/2}$

(ii) Compute the critical values for one tailed test $\pm t_{\alpha}$

5. Find the value of t-statistic, $t = b/S_b$.

6. Accept H_0 if the t-value lies in the area of acceptance otherwise, reject H_0 .

Example 10.1.2 Using the data of Example 01 estimate the value of 'b' with 1% level of significance and decide that slope of population B is greater than 0.

Solution: $v = 6 - 2 = 4$ and $\alpha = 0.01$ then critical value $t_{v, \alpha} = t_{4, 0.01} = 2.132$

$$S_e = \sqrt{\frac{\sum y_i^2 - a \sum y_i - b \sum x_i y_i}{(n-2)}} = 17.11 \quad \text{or} \quad S_b = \frac{S_e}{\sqrt{\sum x_i^2 - (\sum x_i)^2 / n}} = 4.183$$

Thus $t = b/S_b = 3.026/4.183 = 0.723$.

Since statistical $t = 0.723$ is less than the critical value $t_{v, \alpha} = t_{4, 0.01} = 2.132$, hence we accept the alternative hypothesis and reject the null hypothesis and conclude that sale will not increase with the increase in expenditures on advertisement.

Confidence Interval for Slope of Population Regression Line

A test of hypothesis about B starts with hypothesized value of the population slope; $H_1: B > 0$ ($B < 0$). On the other hand, the confidence interval starts with b , the sample estimate of B . We then develop an inference which states that B is between two values calculated from the sample data. Remember that, in Chapter 8, the confidence interval for the population mean is expressed as:

$$\bar{x} - t_{\alpha/2, v} s_{\bar{x}} \leq \mu \leq \bar{x} + t_{\alpha/2, v} s_{\bar{x}}$$

A confidence interval for B has the same form, where $\bar{x}$ is replaced by b and $s_{\bar{x}}$ is replaced by S_b . Thus, we have:

$$b - t_{\alpha/2, v} S_b \leq B \leq b + t_{\alpha/2, v} S_b$$

Example 10.1.3 Find the confidence interval for B with 1% significance level using the data of Example 01.

Solution: Here $v = 6 - 2 = 4$ and $\alpha = 0.01$ thus, critical value $t_{v, \alpha/2} = t_{4, 0.005} = 4.604$, $S_b = 4.183$ and $b = 3.026$. Thus confidence interval for population slope B is

$$b - t_{\alpha/2, v} S_b \leq B \leq b + t_{\alpha/2, v} S_b \quad \text{or} \quad 3.026 - 4.604(4.183) \leq B \leq 3.026 + 4.604(4.183)$$

Or,

$$-16.23 \leq B \leq 19.26$$

We conclude that population slope lies in the interval $(-16.23, 22.28)$ and we say this with 99% confidence.

Correlation Analysis

Regression analysis is carried out to obtain an estimating equation. But it is also important to know whether or not two variables are correlated. Thus statistician is interested in knowing the measure of correlation instead of estimating equations. The two commonly computed measures are the *coefficient of correlation* and *coefficient of determination*.

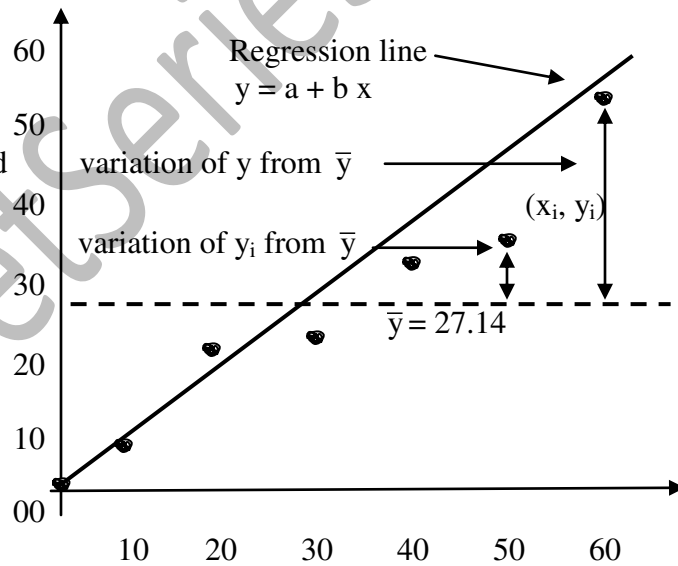
The Coefficient of Determination

To understand the coefficient of determination, let us consider the example of amount of pollution at different distances in a river. We consider the point where the flow starts as zero point where no pollution occurs. Then we measure the pollution at equal distances where the result is as follows.

Distance (x)	0	10	20	30	40	50	60
Amount of Pollution (y)	0	8	21	23	34	36	53

The scattered diagram is also shown below along with the regression line.

Certainly, the variation in the pollution occurs. This is also called expected pollution. The sample coefficient of determination, denoted by r^2 , is the proportion of total variation in y . If r^2 is zero then no variation in y is expected. If r^2 is equal to 1 then certainly there is a variation in y and if r^2 lies in between 0 and 1 then there is a part of variation in y . Hence, $0 \leq r^2 \leq 1$. It may be noted that mean value of y values is: $\bar{y} = 27.14$.



The coefficient of determination can be computed without finding the equation of regression line. But reference to regression line will help to understand the number r^2 . The coefficient of determination is defined as

$$r^2 = \frac{\text{expected variation}}{\text{total variation}} = \frac{\sum (y - \bar{y})^2}{\sum (y_i - \bar{y})^2}$$

To use this formula, we need to compute the regression line. Instead, we may use an alternative formula that can be derived from the above formula which is given by:

$$r^2 = \frac{[n \sum x_i y_i - (\sum x_i)(\sum y_i)]^2}{[n \sum x_i^2 - (\sum x_i)^2][n \sum y_i^2 - (\sum y_i)^2]}$$

Although this formula looks complex, but it permits you to use the same column sums used in computation of regression line.

Example 10.1.4 Find the coefficient of determination using the data given in the above illustration.

Solution: The necessary calculations are shown in the table below. It may be noted that we do not consider the first value (0, 0) as it won't disturb the entire computation. Thus we take $n = 6$.

x	y	xy	x^2	y^2
10	7	70	100	49
20	21	420	400	441
30	23	690	900	529
40	34	1360	1600	1156
50	36	1800	2500	1296
60	53	3180	3600	2809
$\sum x_i = 210$	$\sum y_i = 174$	$\sum x_i y_i = 7520$	$\sum x_i^2 = 9100$	$\sum y_i^2 = 6280$

Using the formula, $r^2 = \frac{[n \sum x_i y_i - (\sum x_i)(\sum y_i)]^2}{[n \sum x_i^2 - (\sum x_i)^2][n \sum y_i^2 - (\sum y_i)^2]}$ and the above values we obtain,

$$r^2 = 0.95.$$

This shows that there is 95% variation in pollution with respect to the distance from the point of origin that is; from where the river starts.

The Correlation Coefficient

The simple linear coefficient of correlation is the square root of the simple linear coefficient of determination. In other words,

$$r = \frac{[n \sum x_i y_i - (\sum x_i)(\sum y_i)]}{\sqrt{[n \sum x_i^2 - (\sum x_i)^2][n \sum y_i^2 - (\sum y_i)^2]}}$$

Since the coefficient of determination r^2 is a square, so its value is positive or zero. The largest possible value of r^2 is 1, therefore,

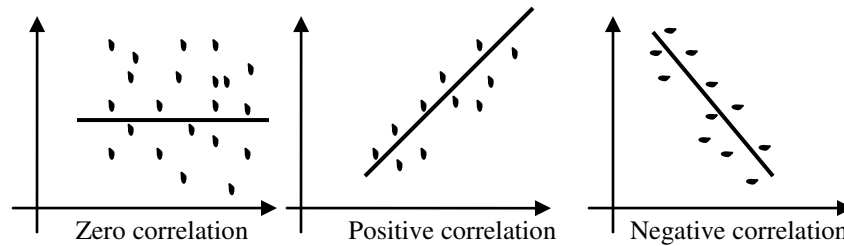
$$r = \pm\sqrt{1} = \pm 1.$$

In other words, $-1 \leq r \leq 1$

If we look at the expression for b in the equation of regression line, we see that

$$b = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

You may observe that numerator of both b and r is same. We know that slope of regression line may be positive or negative hence so the value of r. Thus if b is positive then r is also positive and if b is negative then so is r. Thus we say that correlation between x and y is positive if r is positive and vice versa. If $b = 0$ then regression line is parallel to the x-axis. In this case, we deduce that there is no correlation between x and y.



In the above example, $r = \sqrt{0.95} = 0.975$. This shows that there is an excellent correlation between *Pollution and the Distance* from the point from where the river starts. This means that as the distance increases the pollution also increases.

Probable Error (P.E)

In the calculation of correlation we use sample data. Thus the result that we get may not be true representation of the population correlation. Probable error is used to see the conformity of the result obtained from the sample data for the population correlation. It is also used to guard against false conclusion for the population correlation. Probable error is defined as:

$$P.E = \pm 0.6745 \left[\frac{1-r^2}{\sqrt{n}} \right]$$

1. If r is less than P.E, there is no correlation between x and y .
2. If r is greater than P.E there is a correlation between x and y . However, if r is less than 0.20, the correlation is considered as very poor.
3. If r is six times more than the size of P.E the correlation is highly noteworthy.

For the example discussed above:

$$P.E = \pm 0.6745 \left[\frac{1-r^2}{\sqrt{n}} \right] = \pm 0.6745 \left[\frac{1-0.95}{\sqrt{6}} \right] = \pm 0.014..$$

Now $6(P.E) = 0.082$. We observe that $r > 6(P.E)$ hence, we conclude that correlation is highly significant or noteworthy.

Testing the Significance of r

The sample correlation coefficient r is computed from a random sample of size n . Different samples of size n from the same population will generally produce different values of r . If ρ represents the population correlation coefficient then r is estimate of ρ then in testing the significance of r we in fact attempt to test whether there is no difference between the values of r and ρ . We test the hypothesis by considering that $\rho = 0$ that is, there is no correlation exists between the two variables x and y . For this we have to test the hypothesis for linear regression.

Test of Hypothesis for Linear Correlation

Assumptions:

Step I: State the null and alternative hypotheses:

$H_0 : \rho = 0$ that is, variables x and y are not linearly related.

$H_1: \rho \neq 0$ that is, the variables x and y are linearly correlated.

$H_1: \rho > 0$ that is, the variables x and y are positively correlated.

$H_1: \rho < 0$ that is, the variables x and y are negatively correlated.

Step II: Decide the level of significance α .

Step III: Find the critical values using t-test with $df \ v = n - 2$.

- (i) Critical values for two tailed test are $\pm t_{v, \alpha/2}$
- (ii) Critical value for right tailed test is $t_{v, \alpha}$
- (iii) Critical value for left tailed test is $-t_{v, \alpha}$

Step-IV: Compute t-statistic by using the formula $t = \frac{\sqrt{n-2}}{\sqrt{1-r^2}} \cdot r$

Step V: Accept the null hypothesis if value of t lies in the region of acceptance otherwise reject it.

Example 10.1.5 Use the data of Example 04, find at 1% level of significance to conclude that pollution in river and distance are linearly correlated.

Solution:

- I. $H_0: \rho = 0, H_1: \rho \neq 0$
- II. $\alpha = 0.01$ hence $\alpha/2 = 0.005$
- III. d.f: $v = n - 2 = 6 - 2 = 4$
- IV. $t_{v, \alpha/2} = t_{4, 0.005} = 4.604$. Thus critical interval is $(-4.604, 4.604)$
- V. Computation of t-statistic

$$t = \frac{\sqrt{n-2}}{\sqrt{1-r^2}} \cdot r = \frac{\sqrt{4}}{\sqrt{1-0.95}} \cdot 0.975 = 8.72$$

Decision: Since the value of t lies outside the critical interval $(-4.604, 4.604)$ hence we reject the null hypothesis and conclude that two variables x and y, that is, the distance and the pollution are linearly correlated. We conclude this with 99% confidence.

WORKSHEET 10

1. An experiment was conducted on a new model of a particular make of automobile to determine the stopping distance at various speeds. The following data's are recorded as,

Speed	34	49	64	79	94	109
Stopping distance	15	25	40	61	87	118

- (a) Estimate the stopping distance when car is traveling at a speed of 70KM/h.
- (b) Compute the coefficient of correlation and interpret your results.
- (c) At 1% significance level, do the data provide sufficient evidence to conclude that stopping distance is useful as a predictor of speed?
- (d) At 5% significance level, do the data provide sufficient evidence to conclude that stopping distance and speed are linearly correlated?
- (e) At 2% level, find the confidence interval for the slope of population regression line. Interpret your results.

2. Following table shows the data on rainfall and discharge in a certain river. Find the equation of regression line to predict the discharge from the rainfall. Estimate the discharge when the rainfall is 2.8. At 4% level, test the hypothesis for slope and coefficient of correlation. Also compute the coefficient of determination and probable error and confidence interval for the slope of population regression line. Interpret your results.

Rainfall (inches)	1.5	1.8	2.6	3.0	3.5
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Discharge (1000 cc)	33	36	40	46	54
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3. The following data shows the cost of using CPU time verses time. Compute the coefficient of correlation between the time and cost. Also predict time for the cost of 350 and cost for using CPU time of 18 hours. At 5% level, test the hypothesis for slope and coefficient of correlation. Also compute the coefficient of determination and probable error and confidence interval for the slope of population regression line. Interpret your results.

CPU Time	5	8	15	20	23
Cost	125	175	350	460	520

4. The Voltage drop across a resistor for a number of different values of a current, the results are

I (Amperes)	0.25	0.75	1.25	1.50	2.00
V (Volts)	0.23	0.38	0.76	1.88	6.00

(a) Estimate the voltage drop for $I = 0.9$ and Interpret your result.

(b) Compute the coefficient of correlation between voltage and current.

At 1% level, test the hypothesis for slope and coefficient of correlation. Also compute the coefficient of determination and probable error and confidence interval for the slope of population regression line. Interpret your results.

5. A sample of six days is produced in the following table showing the relation between temperature and sale of deep freezers during a month. Compute the regression line and coefficient of correlation. Estimate the sale when temperature reaches at 45°C . At 2% level, test the hypothesis for slope and coefficient of correlation. Also compute the coefficient of determination and probable error and confidence interval for the slope of population regression line. Interpret your results.

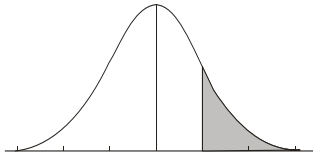
Temperature $^{\circ}\text{C}$	48	46	42	47	52	51
Sale of Deep-Freezers	26	28	20	25	28	29

A normal distribution curve is shown. A vertical line is drawn at a point on the x-axis to the right of the mean. The area under the curve to the left of this line is shaded gray, representing the cumulative probability $P(Z \leq z)$.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0017	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0352	0.0344	0.0339	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0603	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0828	0.0793	0.0778	0.0764	0.0749	0.0735	0.06722	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2149	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2627	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3288	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4201	0.4168	0.4129	0.4090	0.4054	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4582	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8108	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9278	0.9292	0.9308	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9460	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9990	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

APPENDIX “B”
Table of Random Numbers

65 23 68 00	77 82 58 14	10 85 11 85	57 11 73 74	45 25 60 46
06 56 76 51	04 73 94 30	16 74 69 59	04 38 83 98	30 20 87 85
55 99 98 60	01 33 06 93	85 13 23 17	25 51 92 04	52 31 38 70
72 82 45 44	09 53 04 83	03 83 98 41	67 41 01 38	66 83 11 99
04 21 28 72	73 25 02 74	35 81 78 49	52 67 61 40	60 50 47 50
87 01 80 59	89 36 41 59	60 27 64 89	47 45 18 21	69 84 76 06
31 62 46 53	84 40 56 31	74 96 52 23	72 95 96 06	56 83 85 22
29 81 57 94	35 91 90 70	94 24 19 35	50 22 23 72	87 34 83 15
39 98 74 22	77 19 12 81	29 42 04 50	62 34 36 81	43 07 97 92
56 14 80 10	76 52 38 54	84 13 99 90	22 55 41 04	72 37 89 33
29 56 62 74	12 67 09 35	89 33 04 28	44 75 01 57	87 45 52 21
93 32 57 38	39 36 87 42	72 55 73 97	98 36 57 41	76 09 11 68
95 69 51 54	43 19 20 49	57 25 90 55	26 20 70 98	43 73 56 45
65 71 32 43	64 67 22 55	65 65 48 86	10 88 20 12	40 18 49 25
90 27 33 43	97 84 20 57	49 91 41 20	17 64 29 60	66 87 55 97
95 29 42 45	61 34 30 13	30 39 21 52	59 28 64 98	08 76 09 27
99 74 06 29	20 55 72 70	11 43 95 82	75 37 90 24	77 43 63 21
87 87 56 91	16 97 51 50	61 36 96 47	76 68 49 11	50 56 51 06
46 24 17 74	97 37 39 03	54 83 34 00	74 61 77 51	43 63 15 67
66 79 81 43	40 92 84 72	88 32 83 24	67 01 41 34	70 19 26 93
36 42 94 58	83 30 92 39	18 40 03 00	12 90 32 27	91 65 48 15
07 66 25 08	99 27 69 48	85 32 16 46	19 31 85 02	86 36 22 96
93 10 05 72	18 26 36 67	68 48 31 69	68 58 93 49	45 86 99 29
49 50 63 99	26 71 47 94	32 71 72 91	34 18 74 06	32 14 40 80
20 75 58 89	39 04 42 73	37 93 11 07	28 77 91 36	60 47 82 62
02 40 62 09	00 71 09 37	80 44 50 37	32 70 20 38	71 86 75 34
59 87 21 38	29 78 72 67	42 83 65 21	54 79 66 42	47 86 31 15
48 08 99 66	43 38 28 13	50 25 47 93	11 15 07 84	28 30 19 07
54 26 86 75	44 15 20 39	20 03 58 54	80 29 62 53	06 97 71 51
35 35 58 45	23 58 63 66	09 62 80 92	14 55 81 41	21 48 87 34
73 84 90 49	01 21 90 29	57 06 68 73	51 10 51 95	63 08 57 99
34 64 78 00	92 59 67 74	58 48 92 09	42 20 40 37	63 80 58 93
68 56 87 47	63 06 24 71	41 98 79 06	07 18 58 29	16 49 67 37
72 47 05 42	88 07 27 55	58 74 82 08	42 28 26 48	25 32 00 31
44 44 96 75	89 57 12 60	42 38 77 36	45 69 21 68	32 70 04 96
28 11 57 47	61 57 89 88	62 18 93 67	57 32 96 72	21 17 13 54
87 22 38 88	91 99 16 08	17 76 27 47	52 14 98 86	35 68 23 85
44 93 14 59	67 40 24 10	11 63 40 47	07 56 14 22	62 74 93 39
81 84 37 25	90 43 56 62	94 58 49 03	84 22 57 22	47 98 86 37
09 75 35 21	04 47 54 08	98 44 08 16	44 86 69 71	20 52 64 94
77 65 05 04	22 18 20 10	81 87 05 69	43 70 96 76	42 05 21 10
19 06 51 61	34 03 61 55	98 58 83 50	01 48 99 85	08 67 15 91
52 91 87 07	19 62 32 28	04 91 42 48	65 24 86 09	87 68 55 51
52 47 25 14	93 91 75 51	49 26 49 41	20 83 30 30	43 22 69 08
52 67 87 40	63 41 91 86	10 47 80 70	56 87 25 86	89 94 21 42
65 25 71 73	78 60 50 62	91 04 95 97	64 16 71 31	32 80 19 61
29 97 56 42	56 90 16 75	74 95 99 26	01 63 25 16	54 18 54 46
15 25 03 68	92 45 53 00	26 29 46 43	46 66 27 12	85 05 22 44
82 08 65 67	64 13 51 14	38 28 24 30	39 62 20 35	23 90 57 36
81 35 03 25	87 24 83 59	04 67 51 52	26 21 69 75	87 28 61 50



APPENDIX “C”
Critical Values of the t -Distribution

V	A				
	0.10	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.169
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

APPENDIX “D”
Critical Values of the Chi-Squared Distribution

df: ν	α							
	0.995	0.99	0.975	0.95	0.05	0.025	0.01	0.005
1	0.0393	0.0157	0.0982	0.0393	3.841	5.024	6.635	7.879
2	0.0100	0.0201	0.0506	0.103	5.991	7.378	9.210	10.597
3	0.0717	0.115	0.216	0.352	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	11.070	12.832	15.086	16.750
6	0.676	0.872	1.237	1.635	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	36.415	39.364	42.980	45.558
25	10.520	11.524	13.120	14.611	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	40.113	43.194	46.963	49.645
28	12.461	13.565	15.308	16.928	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	43.773	46.979	50.892	53.672

APPENDIX “E”
Critical Values of the F-Distribution

$F_{0.05}(v_1, v_2)$

v_2	v_1								
	1	2	3	4	5	6	7	8	9'
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
4	7071	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88

APPENDIX “E” (Continued)
Critical Value of the F-Distribution
 $F_{0.05}(v_1, v_2)$

v ₂	v ₁									
	10	12	15	20	24	30	40	60	120	∞
1	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
2	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

APPENDIX “E” (Continued)
Critical Value of the F-Distribution
 $F_{0.01}(\nu_1, \nu_2)$

ν_2	ν_1								
	1	2	3	4	5	6	7	8	9
1	4052	4999.5	5403	5625	5764	5859	5928	5981	6022
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41

APPENDIX “E” (Continued)
Critical Values of the F-Distribution
 $F_{0.01}(v_1, v_2)$

v_2	v_1									
	10	12	15	20	24	30	40	60	120	∞
1	6056	6106	6157	6209	6235	6261	6237	6313	6339	6366
2	99.40	99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.49	99.50
3	27.23	27.05	26.87	26.69	26.60	26.50	26.41	26.32	26.22	26.13
4	14.55	14.37	14.20	14.02	13.93	13.84	13.75	13.65	13.56	13.46
5	10.05	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02
6	7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88
7	6.62	6.67	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65
8	5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.86
9	5.26	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31
10	4.85	4.71	4.56	4.41	4.33	4.25	4.17	4.04	4.00	4.91
11	4.54	4.40	4.25	4.10	4.02	3.94	3.86	3.78	3.69	3.60
12	4.30	4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45	3.36
13	4.10	3.96	3.82	3.66	3.59	3.51	3.43	3.34	3.25	3.17
14	3.94	3.80	3.66	3.51	3.43	3.35	3.27	3.18	3.09	3.00
15	3.80	3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96	2.87
16	3.69	3.55	3.41	3.26	3.18	3.10	3.02	2.93	2.84	2.75
17	3.59	3.46	3.31	3.16	3.08	3.00	2.92	2.83	2.75	2.65
18	3.51	3.37	3.23	3.08	3.00	2.92	2.84	2.75	2.66	2.57
19	3.43	3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58	2.49
20	3.37	3.23	3.09	2.94	2.85	2.78	2.69	2.61	2.52	2.42
21	3.31	3.17	3.03	2.88	2.80	2.72	2.64	2.55	2.64	2.36
22	3.26	3.12	2.98	2.83	2.75	2.67	2.58	2.50	2.40	2.31
23	3.21	3.07	2.93	2.78	2.70	2.62	2.54	2.45	2.35	2.26
24	3.17	3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31	2.21
25	3.13	2.99	2.85	2.70	2.62	2.54	2.45	2.36	2.27	2.17
26	3.09	2.96	2.81	2.66	2.58	2.50	2.42	2.33	2.23	2.13
27	3.06	2.93	2.78	2.63	2.55	2.47	2.38	2.29	2.20	2.10
28	3.03	2.90	2.75	2.60	2.52	2.44	2.35	2.26	2.17	2.06
29	3.00	2.87	2.73	2.57	2.49	2.41	2.33	2.23	2.14	2.03
30	2.98	2.84	2.70	2.55	2.47	2.39	2.30	2.21	2.11	2.01
40	2.80	2.66	2.52	2.37	2.29	2.20	2.11	2.02	1.92	1.80
60	2.63	2.50	2.35	2.20	2.12	2.03	1.94	1.84	1.73	1.60
120	2.47	2.34	2.19	2.03	1.92	1.86	1.76	1.66	1.53	1.38
∞	2.32	2.18	2.04	1.88	1.79	1.70	1.59	1.47	1.32	1.00

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